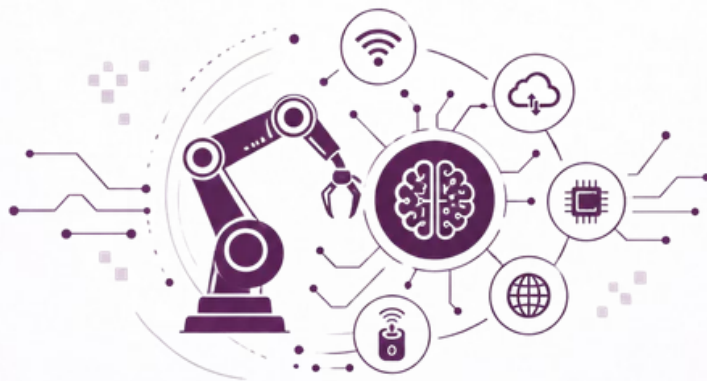




IEEE 5TH INTERNATIONAL CONFERENCE ON ROBOTICS, AUTOMATION, ARTIFICIAL-INTELLIGENCE AND INTERNET-OF-THINGS (RAAICON) 2026

CONFERENCE BOOKLET

Conference Date: 18-19 September, 2026



Jointly Organized By:



Hosted By:



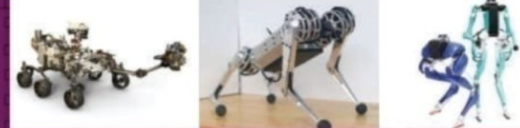
Jashore University of
Science and Technology (JUST)



2026 5th IEEE

International Conference on
Robotics, Automation, Artificial-Intelligence
and Internet-of-Things (RAAICON) 2026

www.raaicon.org



The abstracts of full papers presented at RAAICON 2026 are included in this book. All papers have been peer-reviewed by at least two independent reviewers. The short biographies and abstracts of keynote and invited speakers are also included; the contents herein do not necessarily reflect the opinion or endorsement of editors, organizers, or sponsors, and any liability cannot be imposed upon them.

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CONTENTS

Section	Page No.
Front & Title Page	1
Copyright & Editorial	2
Contents	3
Messages	4
Conference Committee	12
Program Schedule	18
Technical Program Schedule – At a Glance	19
Keynote Speakers	21
Technical Sessions	30
Presented Abstracts	43
Co-located Events	110
HTA Project Exhibition – Participating Projects	111
Workshop on Underwater Robotics	129

MESSAGE

**Chief Patron**

IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things 2026

It is a great privilege and an immense pleasure for me to welcome you to the 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things—RAAICON 2026, hosted by Jashore University of Science and Technology.

RAAICON 2026 brings together four transformative areas of contemporary science and engineering—robotics, automation, artificial intelligence, and the Internet of Things. These fields are no longer isolated domains of research. Rather, their convergence is fundamentally reshaping industries, and transforming healthcare, agriculture, transportation, education, energy, manufacturing, and many other aspects of our daily lives.

We are living in an era of unprecedented technological transformation. The pace of innovation is accelerating rapidly, creating both extraordinary opportunities and new challenges for humanity.

Artificial intelligence is transforming the way we process information, solve complex problems, and make decisions. Robotics is changing the nature of manufacturing, healthcare, exploration, and human assistance. Automation is enhancing productivity, efficiency, precision, and safety. Meanwhile, the Internet of Things is connecting physical systems, machines, people, and environments, creating intelligent and interconnected ecosystems that were unimaginable only a few decades ago.

However, technological progress must ultimately serve humanity. As researchers, educators, and innovators, our responsibility is not merely to develop sophisticated technologies, but also to ensure that these technologies are ethical, inclusive, sustainable, accessible, and responsive to the needs of society.

This is precisely why conferences such as RAAICON are so important. They provide a valuable platform not only for presenting research findings, but also for exchanging ideas, challenging established assumptions, fostering collaborations, and exploring new directions in scientific research and technological innovation.

The scope of RAAICON 2026 reflects this broad and forward-looking vision. The conference covers a wide range of areas, including robot design and control, robotic vision, artificial intelligence, machine learning, autonomous systems, swarm robotics, human–robot collaboration, mechatronics, intelligent systems, IoT architectures, smart automation, cybersecurity, energy and sustainability, healthcare automation, climate monitoring, and many other emerging fields.

I am particularly pleased that this conference brings together researchers and professionals at different stages of their academic and professional careers. For young researchers and students, participation in an international conference is much more than an opportunity to present a research paper. It is an opportunity to build confidence, receive constructive feedback, interact with leading researchers, establish professional networks, and discover new avenues for research and collaboration.

At the same time, we must recognize that the successful adoption of advanced technologies in developing societies requires careful consideration of local realities and needs. Technologies developed for one context cannot always be transferred directly to another. We therefore need solutions that are locally appropriate, economically viable, environmentally sustainable, and socially acceptable.

In this regard, universities have a vital role to play. Universities must serve not only as centers of knowledge generation but also as engines of innovation, entrepreneurship, and national development. We need to strengthen university–

industry collaboration, promote interdisciplinary research, encourage entrepreneurship, and create opportunities for young researchers to transform innovative ideas into practical solutions to real-world problems.

I am particularly proud that Jashore University of Science and Technology is hosting an international conference of this stature. Hosting RAAICON 2026 reflects our University's commitment to excellence in teaching, research, innovation, technological advancement, and international collaboration.

At this point, I would like to express my sincere appreciation to the IEEE Bangladesh Section and the IEEE Robotics and Automation Society Bangladesh Chapter for jointly organizing this important international conference. I also extend my heartfelt gratitude to the organizing and technical committees, reviewers, keynote speakers, researchers, and participants whose dedication and hard work have made RAAICON 2026 possible.

I wish the conference every success and hope that it will make a significant contribution to the advancement of robotics, automation, artificial intelligence, and the Internet of Things in Bangladesh and beyond.

Once again, I warmly welcome you all to Jashore University of Science and Technology. I wish you a productive, inspiring, and memorable conference and a pleasant stay in Jashore.

Thank you all.

Professor Dr. Yearul Kabir
Chief Patron, RAAICON 2026
Vice Chancellor
Jashore University of Science and Technology



Advisor

IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things 2026

Welcome to the 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things, hosted by Jashore University of Science and Technology.

I would like to congratulate the IEEE Bangladesh Section, the IEEE Robotics and Automation Society Bangladesh Chapter, and the organizing committee of RAAICON 2026 for bringing together such a distinguished gathering of researchers, academics, professionals, and young innovators.

We are currently witnessing a remarkable transformation in science and technology. Technologies that were once considered part of science fiction are now becoming an integral part of our daily lives. Artificial intelligence is changing the way we analyze information and make decisions; robotics is redefining human-machine interaction; automation is transforming industrial processes; and the Internet of Things is creating increasingly connected and intelligent physical environments.

More importantly, the convergence of these technologies is opening new possibilities that extend far beyond individual technological disciplines.

This makes the theme and scope of RAAICON particularly relevant to the present time. The conference provides an important platform for researchers to exchange knowledge, present emerging ideas, discuss technological challenges, and explore innovative solutions to real-world problems.

As universities, our responsibility goes beyond transmitting existing knowledge. We have a responsibility to create new knowledge, nurture innovation, and prepare the next generation to address challenges that may not yet have clearly defined solutions.

In this regard, research conferences play an essential role. They provide researchers with opportunities to receive constructive feedback, refine their ideas, establish professional networks, and develop collaborations across institutions and national boundaries.

Another important aspect that I would like to emphasize is interdisciplinary research. For example, combining artificial intelligence with IoT can create intelligent monitoring systems; integrating robotics with machine learning can enable more autonomous systems; and combining automation, sensing, data analytics, and intelligent decision-making can fundamentally transform agriculture, healthcare, manufacturing, and other sectors. For Bangladesh, these technological opportunities are particularly significant.

As a developing and densely populated country, Bangladesh faces challenges that require innovative and affordable technological solutions. We need to think carefully about how advanced technologies can be adapted to our local context and used to improve productivity, efficiency, safety, environmental sustainability, and the quality of life of our people.

However, technological advancement should not be measured solely by technological sophistication. A successful innovation should also be accessible, affordable, sustainable, ethical, and socially beneficial.

This is where universities have a particularly important responsibility. We must encourage research that moves beyond theoretical contributions and addresses practical problems. At the same time, we need to strengthen partnerships among universities, industries, government organizations, and research institutions so that promising research can move from laboratories and publications toward practical implementation.

I believe IEEE has an important role in facilitating this transition. Through its global professional network, technical communities, conferences, publications, and standards, IEEE provides researchers and professionals with opportunities to exchange knowledge and contribute to the development of emerging technologies.

RAAICON 2026 is therefore not only a conference; it is also an opportunity to strengthen the research and innovation ecosystem in Bangladesh.

I am proud that Jashore University of Science and Technology is hosting this international event. Such an event strengthens our University's academic environment and creates opportunities for our faculty members and students to interact with researchers and professionals from diverse backgrounds.

Let us remember that the ultimate purpose of research is not merely to increase the number of publications or citations. Research should contribute to knowledge, innovation, human welfare, and sustainable development.

I hope RAAICON 2026 will generate ideas that go beyond the conference rooms and laboratories—that it will inspire new research projects, interdisciplinary collaborations, innovative technologies, and solutions to some of the pressing challenges facing Bangladesh and the world.

I wish the conference great success and hope that it will become a memorable milestone in the advancement of robotics, automation, artificial intelligence, and the Internet of Things in Bangladesh and internationally.

Professor Sk. Mahmudul Hasan

Advisor, RAAICON 2026

Pro-Vice Chancellor

Jashore University of Science and Technology

MESSAGE



Advisor

IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things 2026

It is my great pleasure to extend my warmest greetings to all participants, researchers, academics, professionals, and young innovators attending the 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON 2026).

RAAICON has emerged as an important platform for bringing together researchers and professionals to exchange knowledge, present innovative ideas, and explore emerging developments in robotics, automation, artificial intelligence, and the Internet of Things. These rapidly advancing technologies are transforming industries, reshaping our daily lives, and creating new possibilities for addressing complex challenges in society.

The organization of RAAICON 2026 at Jashore University of Science and Technology (JUST) is a significant opportunity to strengthen academic collaboration, promote research and innovation, and connect our researchers and students with the wider international scientific community. I am confident that RAAICON 2026 will foster meaningful intellectual exchange, new collaborations, and innovative solutions with lasting impact.

As Treasurer, I recognize that the successful organization of an international conference requires not only strong academic leadership but also careful financial planning, transparency, accountability, and effective coordination. I sincerely appreciate the dedication of the organizing committees, technical teams, volunteers, sponsors, partners, and all those who have contributed their time and effort to making RAAICON 2026 possible.

I am particularly pleased to welcome the distinguished keynote speakers, invited speakers, authors, delegates, and participants joining us from home and abroad. I hope their participation will enrich the conference and create opportunities for sustained academic and professional collaboration. I also encourage our young researchers and students to take full advantage of this platform. Their curiosity, creativity, and innovative thinking will be instrumental in shaping the technological future of Bangladesh and the world. I firmly believe that the true success of RAAICON will be measured not only by the quality of the research presented, but also by the collaborations initiated, ideas exchanged, and innovations inspired through this gathering.

May RAAICON 2026 transcend the boundaries of a conference and become a catalyst for transformative ideas, enduring collaborations, and impactful innovations. Let us unite knowledge, technology, and human ingenuity to shape a smarter, more connected, sustainable, and inclusive future. May the ideas we share today become the innovations that define tomorrow. I extend my sincere gratitude to everyone whose dedication and contributions have made RAAICON 2026 possible, and I wish this prestigious conference every success.

With profound regards and best wishes for a distinguished and inspiring RAAICON 2026.

Professor Dr. Md. Faruq Hasan

Advisor, RAAICON 2026

Treasurer

Jashore University of Science and Technology



Organizing Chair

IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things 2026

It is my great pleasure and privilege, as the Dean of the Faculty of Engineering and Technology and Organizing Chair of RAAICON 2026, to welcome you to the 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things, hosted by Jashore University of Science and Technology.

RAAICON 2026 represents a significant academic occasion for our University. Hosting an IEEE international conference of this scale is both an honor and a responsibility. It provides us with an opportunity to bring together researchers and professionals working at the forefront of some of the most rapidly evolving areas of science and engineering.

The fields represented by RAAICON—robotics, automation, artificial intelligence, and the Internet of Things—are transforming the technological landscape at an extraordinary pace. Their convergence is creating intelligent, connected, and increasingly autonomous systems that are influencing virtually every sector of modern society.

From smart manufacturing and autonomous vehicles to intelligent healthcare, precision agriculture, energy management, environmental monitoring, and smart cities, the applications of these technologies are expanding rapidly.

This conference has been designed as a platform where researchers can present their latest findings, exchange knowledge, discuss emerging challenges, and explore new possibilities for collaboration. More importantly, we hope that the discussions taking place during RAAICON 2026 will lead to research partnerships and innovative solutions that extend well beyond the duration of this conference.

As Organizing Chair, I am particularly pleased to see participation from researchers and professionals from diverse institutions and backgrounds. Such diversity is essential for meaningful scientific exchange. Different perspectives often allow us to look at familiar problems in new ways and identify solutions that may otherwise remain unexplored.

I would also like to highlight the importance of young researchers and students in this conference. They represent the future of our research and innovation ecosystem. Conferences such as RAAICON provide them with a valuable opportunity to present their work, interact with experienced researchers, receive feedback, and develop international academic networks.

The organization of an international conference is itself a collaborative endeavor. RAAICON 2026 would not have been possible without the collective efforts of numerous individuals and organizations. In this regard, I would like to express my deepest gratitude to the **IEEE Bangladesh Section** and the **IEEE Robotics and Automation Society Bangladesh Chapter** for their valuable partnership and support in organizing this conference.

I would like to acknowledge the authors and presenters whose research contributions form the intellectual foundation of RAAICON 2026. Without your research, ideas, and willingness to share your findings with the wider community, this conference would not have its true academic significance.

As Dean, I am particularly pleased that our Faculty has the opportunity to host such an event. We believe that excellence in engineering and technology education must be supported by a strong culture of research and innovation. International academic events provide an important bridge between classroom learning, cutting-edge research, professional practice, and global collaboration.

I also hope that this conference will strengthen the international academic profile of Jashore University of Science and Technology. We aspire to develop our University as a center of excellence where teaching, research, innovation, entrepreneurship, and international collaboration reinforce one another.

Hosting RAAICON 2026 is an important step in that direction.

Finally, on behalf of the organizing committee, I once again extend my warmest welcome to all national and international participants. I hope your participation in RAAICON 2026 will be academically enriching, professionally rewarding, and personally memorable.

It is my sincere hope that the **5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things—RAAICON 2026** will serve as a catalyst for new ideas, stronger collaborations, meaningful innovations, and impactful research for Bangladesh and the global community.

With these words, I wish RAAICON 2026 every success.

Thank you very much.

Professor Dr. Engr. Imran Khan
Organizing Chair, RAAICON 2026
Dean, Faculty of Engineering and Technology
Chairman, Dept. of Electrical and Electronic Engineering
Jashore University of Science and Technology

MESSAGE



Organizing Secretary

IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things 2026

It is with immense pleasure and a deep sense of gratitude that I welcome all participants to the 2026 IEEE 5th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026), hosted by the Department of Computer Science and Engineering at Jashore University of Science and Technology (JUST). This conference marks a significant milestone in our collective journey to advance research, innovation, and collaboration in the field of computational intelligence.

As the Organizing Secretary of RAAICON 2026, I have had the privilege of witnessing the tireless dedication and teamwork that have gone into making this event possible. I would like to extend my heartfelt appreciation to the organizing committee and sub-committee members of the conference, IEEE RAS BD Chapter members and IEEE ExCom members for their unwavering support, leadership, and commitment throughout the planning and execution of this conference. Their efforts have been crucial in upholding the standards of this conference and fostering meaningful scholarly exchange.

In an era characterized by rapid technological evolution, computational intelligence stands as a pivotal force driving transformative progress. The disciplines explored in this conference—spanning artificial intelligence, machine learning, Internet-of-Things, data science, robotics, and beyond—are redefining the paradigms of human engagement with technology. RAAICON 2026 holds particular significance in its dedication to harnessing rigorous academic inquiry to address pressing real-world challenges, including those in healthcare, power, sustainability, security, and education.

My sincere thanks also go to my colleagues, students and volunteers whose enthusiasm and hard work have been instrumental in managing every aspect of the event—from logistics and hospitality to technical support and promotion. Their contributions reflect the vibrant spirit of Jashore University of Science and Technology and the passion of our next generation of researchers and professionals.

RAAICON 2026 serves not only as a platform for sharing innovative research but also as a celebration of collaborative spirit and academic excellence. I am confident that the ideas and interactions emerging from this forum will lead to new directions in research and continued growth in Bangladesh's computer and software engineering ecosystem.

Thank you all for being a part of this important journey. I wish everyone a productive, engaging, and memorable conference.

Dr. Md. Kamrul Islam

Organizing Secretary, RAAICON 2026
Chairman, Department of Computer Science and Engineering
Jashore University of Science and Technology

**IEEE 5TH INTERNATIONAL CONFERENCE ON
ROBOTICS, AUTOMATION, ARTIFICIAL-INTELLIGENCE
AND INTERNET-OF-THINGS (RAAICON) 2026**



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Hosted By:



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Vice Chancellor

Jashore University of Science and Technology



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Professor Sk. Mahmudul Hasan
Pro-Vice Chancellor

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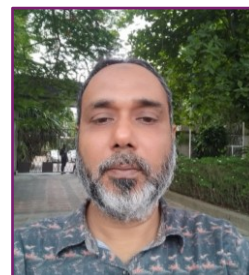
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P R O G R A M S C H E D U L E

Day 1: Friday, 18th September 2026

Venue: Sir Jagadish Chandra Bose Academic Building, JUST

TIME	PROGRAM DETAILS
08:30 AM – 09:00 AM	Registration <i>[Location: Ground Floor]</i>
09:00 AM – 10:30 AM	Parallel Technical Sessions TS 1 -Hall 1 (Room #219): MIDL TS 10 (VR-1): DMIA TS 2 -Hall 2 (Room #220): AACI TS 11 (VR-2): PSEM TS 3 -Hall 3 (Room #205): AAFR TS 12 (VR-3): RMHI
10:30 AM – 10:45 AM	Tea Break <i>[Location: 1st Floor]</i>
10:45 AM – 12:00 PM	Inaugural Ceremony <i>[Location: FET Gallery, Room 106]</i>
12:00 PM – 01:00 PM	Keynote 1 [12:00 PM]: Predictive Maintenance in Semiconductor Manufacturing – AI Driven Automation for Tool Downtime Reduction by Dr. Ram Chandra Palsaniya Keynote 2 [12:30 PM]: Soft Robotics and the Embodied Side of Intelligence by Dr. Cecilia Laschi <i>[Location: FET Gallery, Room 106]</i>
01:00 PM – 02:30 PM	Lunch and Prayer Break
02:30 PM – 04:00 PM	Parallel Technical Sessions TS 4 -Hall 1 (Room #219): BNWH TS 13 (VR-1): ECRP TS 5 -Hall 2 (Room #220): PREG TS 14 (VR-2): CFDF TS 6 -Hall 3 (Room #205): CTMF TS 15 (VR-3): RDKM
04:00 PM – 04:30 PM	Tea Break <i>[Location: 1st Floor]</i>
04:30 PM – 06:00 PM	Parallel Technical Sessions TS 7 -Hall 1 (Room #219): ECAH TS 16 (VR-1): BNLP TS 8 -Hall 2 (Room #220): ITUI TS 17(VR-2): ACPA TS 9 -Hall 3 (Room #205): RDKV TS 18 (VR-3): IADP
07:30 PM – 09:30 PM	Gala Dinner <i>[Location: TSC 2nd Floor, JUST]</i>

[All virtual sessions will be conducted at Room #233]

Day 2: Saturday, 19th September 2026**Venue:** Sir Jagadish Chandra Bose Academic Building, JUST

TIME	PROGRAM DETAILS
08:00 AM – 08:30 AM	Reporting & Registration <i>[Location: Ground Floor]</i>
08:30 AM – 10:00 AM	IEEE R10 Humanitarian Technology Activity Volunteer Training Program HTA Training Session I: Humanitarian Project Design by Prof. Shaikh A Fattah HTA Training Session II: Technology-Driven Solutions for Underserved Communities by Prof. Celia Shahnaz <i>[Location: FET Gallery, Room 106]</i>
09:00 AM – 11:00 AM	Parallel Technical Sessions TS 19 (VR-1): TFMA TS 20 (VR-2): NSWL TS 21 (VR-3): EIHS Keynote 3 [09:00 AM]: Bomb Disposal Robot for Human Security by Elmer P. Dadios, PhD Keynote 4 [10:00 AM]: Artificial Intelligence and robotics for obtaining an IEEE Milestone recognition for the first brain-robot interface, 1988 by Dr. Stevo Bozinovski <i>[Location: Room 205]</i>
09:30 AM – 10:00 AM	Tea Break <i>[Location: 1st Floor]</i>
10:00 AM – 11:00 AM	Pre-screening of HTA Project Exhibition
10:00 AM – 12:00 PM	Technical Expert Forum
11:00 AM – 11:45 AM	Keynote 5 [11:00 AM]: Fundamental Challenges in Robotics and Embodied AI by Dr. Steven M. LaValle <i>[Location: Room 205]</i> HTA Keynote Talk: Empowering IEEE Volunteers by Amira Ouerfelli <i>[Location: FET Gallery, Room 106]</i>
11:45 AM – 01:00 PM	Final Evaluation of Selected Projects Workshop: Engineering the Deep: A Workshop on Underwater Robotics (by IEEE MIST Student Branch) Speakers: Areeb Ayman Haq, Sheikh Tanvir Islam Uday, Rokonzaman Shaad <i>[Location: FET Gallery, Room 106]</i> Keynote 6 [12:00 PM]: Artificial Intelligence Approaches for Integrating Cooperative, Connected and Automated Mobility in the Urban Traffic by Dr. Maria Pia Fanti <i>[Location: Room 205]</i>
01:00 PM – 02:00 PM	Lunch and Prayer Break <i>[Location: TSC 2nd Floor, JUST]</i>
02:00 PM – 03:00 PM	Closing & Award Ceremony
03:00 PM	Conference Tour

[All virtual sessions will be conducted at Room #233]

TECHNICAL PROGRAM SCHEDULE – AT A GLANCE

DAY 1 • 18 SEPTEMBER 2026, FRIDAY

PHYSICAL (ON-SITE)

TS-ID	SHORT NAME	TIME	ROOM	PAPER IDs
1	MIDL	9:00 AM - 10:30 AM	Hall 1 (Room #219)	237, 301, 371, 441, 471, 545 [6]
2	AACI	9:00 AM - 10:30 AM	Hall 2 (Room #220)	183, 310, 474, 527 [4]
3	AAFR	9:00 AM - 10:30 AM	Hall 3 (Room #205)	128, 255, 262, 387, 449 [5]
4	BNWH	2:30 PM - 4:00 PM	Hall 1 (Room #219)	263, 398, 500, 502, 505, 512 [6]
5	PREG	2:30 PM - 4:00 PM	Hall 2 (Room #220)	232, 259, 260, 305, 362, 447 [6]
6	CTMF	2:30 PM - 4:00 PM	Hall 3 (Room #205)	224, 240, 385, 437, 452, 515 [6]
7	ECAH	4:30 PM - 6:00 PM	Hall 1 (Room #219)	1, 326, 443, 459, 521, 539 [6]
8	ITUI	4:30 PM - 6:00 PM	Hall 2 (Room #220)	101, 315, 357, 359, 361 [5]
9	RDKV	4:30 PM - 6:00 PM	Hall 3 (Room #205)	238, 325, 328, 369, 492, 543 [6]

ONLINE (VIRTUAL)

TS-ID	SHORT NAME	TIME	ROOM	PAPER IDs
10	DMIA	9:00 AM - 10:30 AM	VR-1	207, 209, 358, 433, 508, 518, 552 [7]
11	PSEM	9:00 AM - 10:30 AM	VR-2	69, 187, 210, 266, 334, 413 [6]
12	RMHI	9:00 AM - 10:30 AM	VR-3	49, 277, 281, 288, 456, 479 [6]
13	ECRP	2:30 PM - 4:00 PM	VR-1	254, 268, 276, 395, 523, 538, 566 [7]
14	CFDF	2:30 PM - 4:00 PM	VR-2	219, 236, 278, 283, 312, 384 [6]
15	RDKM	2:30 PM - 4:00 PM	VR-3	225, 294, 360, 445, 495, 542 [6]
16	BNLP	4:30 PM - 6:00 PM	VR-1	203, 213, 264, 352, 516, 553 [6]
17	ACPA	4:30 PM - 6:00 PM	VR-2	63, 145, 157, 293, 544, 550, 562 [7]
18	IADP	4:30 PM - 6:00 PM	VR-3	94, 308, 416, 525, 551 [5]

DAY 2 • 19 SEPTEMBER 2026, SATURDAY

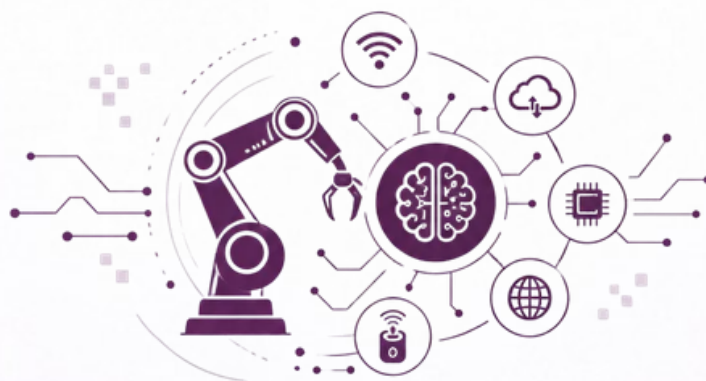
ONLINE (VIRTUAL)

TS-ID	SHORT NAME	TIME	ROOM	PAPER IDs
19	TFMA	9:00 AM - 11:00 AM	VR-1	108, 257, 487, 489, 496, 514, 558 [7]
20	NSWH	9:00 AM - 11:00 AM	VR-2	252, 258, 274, 465, 468, 534 [6]
21	EIHS	9:00 AM - 11:00 AM	VR-3	333, 349, 353, 462, 499, 503, 524 [7]

**IEEE 5TH INTERNATIONAL CONFERENCE ON
ROBOTICS, AUTOMATION, ARTIFICIAL-INTELLIGENCE
AND INTERNET-OF-THINGS (RAAICON) 2026**



KEYNOTE SPEAKERS



Jointly Organized By:



Hosted By:



Jashore University of
Science and Technology (JUST)

KEYNOTE SPEAKER 1

**Ram Chandra Palsaniya**

Co-Chair, IEEE RAS Semiconductor Manufacturing Automation Technical Activities Board

TALK TITLE

Predictive Maintenance in Semiconductor Manufacturing – AI Driven Automation for Tool Downtime Reduction**Abstract**

Semiconductor fabrication relies on high equipment reliability, where unexpected tool failures can cause major yield and productivity losses. This talk examines AI machine learning–based predictive maintenance approaches designed for semiconductor manufacturing automation environments. We review key methods – including anomaly detection, condition monitoring, and system-level fault diagnosis and compare supervised, unsupervised, and hybrid models using fab-specific data characteristics. Industry challenges such as sparse failure events, high-dimensional sensor streams, and tool variability are highlighted. The session concludes with practical guidance for deploying scalable PdM solutions and outlines future directions for intelligent maintenance systems aligned with human-centered Industry 5.0 semiconductor fabs.

Biography

Mr. Ram Chandra Palsaniya, IEEE Senior Member, is a featured technology leader with over 20 years of experience in the semiconductor manufacturing industry. He currently serves as a Senior Product Manager at Applied Materials Inc. California, where he drives innovation at the intersection of advanced manufacturing with AI, Robotics and Automation.

Mr. Palsaniya has demonstrated sustained leadership and service within IEEE. He served as Chair of the IEEE Robotics and Automation Society (RAS) Technical Chapter and Vice Chair of the IEEE Control Systems Society Technical Chapter in Silicon Valley, California, and contributes as TC Co-Chair IEEE RAS Technical Activities Board. He has organized several technical talks and panel discussions for chapter events, for example recent RAS Chapter Fireside chat at Stanford University with 200+ participants from industry and academics.

Mr. Palsaniya has presented in multiple IEEE international conferences including IEEE ISIE, IEEE CAI, and IEEE NMDC. His work focuses on AI driven predictive maintenance, semiconductor manufacturing robotics and automation systems.

KEYNOTE SPEAKER 2

**Dr. Cecilia Laschi**

Provost's Chair Professor at the National University of Singapore, Dept. Mechanical Engineering, College of Design and Engineering, Advanced Robotics Centre, Director

TALK TITLE

Soft Robotics and the Embodied Side of Intelligence**Abstract**

Observing living beings means observing complex systems, in complex dynamic environments. Obtaining similar behaviours has been challenging roboticists since long. What we can learn from nature is how to simplify perception-action loops, without simplifying the systems, the environment, or the task, but taking advantage of their interplay. According to this embodied intelligence paradigm, control is simplified and efficiency is increased, as we see in a few examples of octopus-inspired soft robots. Octopus locomotion uses soft legs that shorten and elongate to walk underwater, avoiding the need to move rigid limbs against water drag. Octopus arm movements also reduce water drag by avoiding rigid translations and instead leveraging the arm softness and extreme deformability to unfold in water. Among other simplifying principles, those strategies can help roboticist develop effective and efficient robots, for marine operations, and for many more application fields.

Biography

Cecilia Laschi is best-known for her research in soft robotics, an area that she pioneered internationally. She investigates how to build robots with soft materials, studying the octopus as a model for robotics. She explores marine applications of soft robots and their use in the biomedical field, with a focus on eldercare. She has worked in humanoid and neuro-robotics, applying brain models to humanoid robots. Cecilia Laschi is Provost's Chair Professor at the National University of Singapore, in the College of Design and Engineering, and the Director of the Advanced Robotics Centre.

She is IEEE Fellow and AAAS member. She is Editor-in-Chief of IEEE Transactions on Soft Robotics, Scientific Advisory Board member of Science Robotics, IEEE Transactions on Bionics and Medical Robotics, and Editorial Board member of the Intelligent Robotics and Autonomous Agents (IRAA) Series of MIT Press.

She founded the IEEE-RAS International Conference on Soft Robotics (RoboSoft) in 2018 and co-founded the IEEE RAS Technical Committee in Soft Robotics.

She co-founded the start-up company RoboTech, in edutainment robotics.

KEYNOTE SPEAKER 3

**Elmer P. Dadios, PhD**

Academician, National Academy of Science and Technology, Philippines Distinguished Professor and University Fellow, DLSU Manila.

TALK TITLE

Bomb Disposal Robot for Human Security**Abstract**

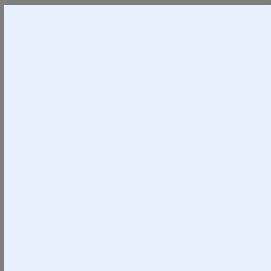
Robotics and Artificial Intelligence are considered to be among the best and most significant discoveries in the 20th century. The goal of Robotics is to develop machines that can perform tasks, automate processes, and interact with the environment, mimicking human actions or surpassing them in speed, endurance, and accuracy. This presentation will discuss the role and impact of robotics for human security. The bomb removal automated vehicle (BRAVE) is developed and designed based on the expertise from the Military Personnel of the Philippine National Police EOD/K9 team. Knowledge and Information from the military experts on disposing explosives were utilized in creating BRAVE. Aside from disposing bombs, BRAVE is also use for rescue operations. The accuracy, reliability, and robustness of the capability of BRAVE will be shown in this presentation.

Biography

Dr. Elmer P. Dadios or “Elmer” to colleagues and friends, obtained his PhD for an unprecedented two years from the Manufacturing Engineering, Loughborough University, United Kingdom in the area of Robotics and Intelligent Systems. A multi-awarded educator and scholar, Elmer was awarded over 20 recognitions and distinctions from various international and national scientific award-giving bodies and professional organizations. Among the awards he garnered include: The 2023 National Academy of Science and Technology Academician award. The ScholarGPS Highly Ranked Scholar – Both Lifetime and Prior Five Years in the Specialties of Environment (Systems), and Nanotechnologies. The 2022 Severino & Paz Koh Lectureship Awardees in Engineering given by the Philippine American Academy of Science and Engineering (PAASE); The top 100 scientists listed in Asian Scientist Magazine; The 2019 Leaders in Innovation Fellowship “Fellow” given by the United Kingdom Royal Academy of Engineering; The 2018 Philippine Association for the Advancement of Science and Technology D. M. CONSUNJI AWARD for Engineering Research; Lifetime Achievement Award from the National Research Council of the Philippines (NRCP); The Department of Science and Technology (DOST) 50 Men and Women of Science and Technology; The Department of Science and Technology (DOST) Scholar Achievers; De La Salle University Miguel Febres Cordero Research Award. Currently, Elmer serves as editor of the Journal of

Advanced Computational Intelligence and Intelligent Informatics (JACIII) published by the Fuji Technology Press, Ltd. (Tokyo, Japan). He is an editor of Journal of AI, Computer Science and Robotics Technology published by IntechOpen London and Croatia. He is an Editorial Board Member of IntechOpen Series “Artificial Intelligence” for Evolutionary Computation topic. He is the Editor in Chief of the Journal of Computational Intelligence in Engineering Applications (JCIEA). He published five books on fuzzy logic systems and a book on Control and Automation with IntechOpen (Croatia/London). On top of these, Elmer has published over 670 Scopus indexed technical papers in highly reputable journals and IEEE Xplore. Elmer joined DLSU in May 1992 and currently holds the title of Distinguished Research Fellow. In May 2008, Elmer was conferred the title of University Fellow, the highest award and recognition given by DLSU to faculty members who exhibit consistent scholarly pursuit through research and publication, effective teaching performance, and exemplary modeling of the Lasallian core values. He is the founder and Chairman of the Board of the Intelligent Systems Innovation Corporation. He was the past Chair of the IEEE Philippines Section; past EXCOM member of the IEEE Region 10; past Chair of the IEEE Region 10 Awards and Recognition Committee; the Founder and Chair of the IEEE Computational Intelligence Society Philippines Chapter, and IEEE Robotics and Automation Society Philippines Chapter. His current research interests include: robotics, artificial intelligence and its applications to smart cities, transportation management, smart farms, human security, biomedical engineering, computational intelligence systems, mechatronics, and manufacturing processes.

KEYNOTE SPEAKER 4



Dr. Stevo Bozinovski

Professor of Computer Science at the Computer Science and Mathematics Department, South Carolina State University, USA (BSc in Computer Science, M.Sc in Electronics, and PhD in Computer Science from Faculty of Electrical Engineering, University of Zagreb)

TALK TITLE

Artificial Intelligence and robotics for obtaining a IEEE Milestone recognition for the first brain-robot interface, 1988

Abstract

In 2023 on recommendation of the IEEE History Committee, the IEEE Board of Directors approved a designation of IEEE Milestone to the achievement First control of a physical object (a robot) using signals emanating from a human brain, which happened in 1988 in Skopje, Macedonia. The paper was published at a IEEE conference in New Orleans, USA, in 1988.

This invited talk describes the pioneering events that led to this achievement. The pioneering events described here are divided into three categories: artificial intelligence events, robotics events, and brain-

computer interface events. The Artificial Intelligence part includes the first solution of a challenge of reinforcement learning with delayed rewards, the introduction of emotion in neural network learning, and self-learning, learning without external rewards. The robotics part includes first control of a robot using concurrent (multitasking) programming and first control of a movement of a robot using spoken commands. The brain-computer interface part includes contingent negative variation (CNV) signal processing and control of a computer buzzer, and alpha rhythm processing including Machine Learning for control of a robot. All that solved the challenge of psychokinesis, moving a physical object with energy emanating from a human brain, which before 1988 was in the realm of the science fiction.

Biography

Stevo Bozinovski. Obtained BSc in Computer Science, M.Sc in Electronics, and PhD in Computer Science from Faculty of Electrical Engineering, University of Zagreb, Croatia. He studied and worked in various places including: IBM in Bayreuth, Germany; the Institute for Communications and Informatics at the Technical University in Munich, Germany; Tsukuba University and the Kanazawa Institute of Technology in Japan; the Computer Science Department of the University of Massachusetts at Amherst, USA; and the German National Center for Information Technologies in Sankt Augustin, Bonn, in Germany. He was contracted as an Expert from Macedonia by the European Commission in Brussels, Belgium. Since 2001 he is a professor of Computer Science at the Computer Science and Mathematics Department, South Carolina State University, USA. His research interest is in Artificial Intelligence, Robotics, and Biocybernetics.

Some of his most important pioneering achievements in science are: Introducing transfer learning in neural networks, 1976; Solving challenge of reinforcement learning with delayed rewards, 1981; and of learning without external rewards, 1981, Introducing emotion and self learning in neural networks, 1981; Introducing parallel programming in neural networks, 1981; First control of a robot using concurrent (multitasking) programming, 1983; First recognition that tRNA is a mobile robot, 1985; First control of movements of a robot using speech commands, 1986; Revision of the Central Dogma in molecular genetics using Flexible Manufacturing metaphor, 1987; First control of a robot using EEG signals, 1988; First control of a robot using EOG signals, 1989; First mathematical model of motivation, emotion, and happiness, 2003; EEG emulation of control circuits, 2015; First mathematical explanation of EEG rhythms including alpha rhythm, 2022. In 2023 he received an IEEE Milestone recognition for the achievement of the first control of a robot using brain signals.

KEYNOTE SPEAKER 5

**Dr. Steven M. LaValle**

Professor of Robotics and Virtual Reality, University of Oulu & University of Illinois

TALK TITLE

Fundamental Challenges in Robotics and Embodied AI**Abstract**

The field of robotics is wildly exciting and rapidly gaining worldwide attention, yet it is often an enigma in terms of its scope and scientific foundations. Robotics involves the design, programming, and analysis of movable machines that accomplish useful work through sensing and manipulation of the surrounding world. Throughout the decades it has been varyingly viewed as an application field of more mature disciplines such as computer science (AI, algorithms, machine learning) and mechanical engineering (kinematics, dynamics, nonlinear control). This talk will argue that robotics has its own unique and growing scientific core, with deep questions and modeling challenges that should inspire new directions in computer science, engineering, and even pure mathematics.

We will start with a Turing-inspired way to view robotics or embodied AI, leading to some of our recent results that characterize minimally sufficient amounts of sensing, actuation, or computation that are required to solve physical tasks. Questions addressed include: How are learning, planning, and control related? How do we know when it is impossible to solve a task? When will learning fail, even with an infinite amount of data? Does a universal action sequence exist that would cause a robot to solve any possible task without modification? How important are semantics and representations? Interspersed throughout the talk will be results and perspective from my research in the field over three decades, produced with many inspiring students, mentors, and collaborators.

Biography

Steven M. LaValle has been Professor of Computer Science and Engineering, in Robotics and Virtual Reality, at the University of Oulu, Finland since 2018. Since 2001, he has been a professor in the Department of Computer Science at the University of Illinois. He has also held positions at Stanford University and Iowa State University. His research interests include robotics, virtual reality, sensor fusion, planning algorithms, computational geometry, and control theory. In research, he is mostly known for his introduction of the Rapidly exploring Random Tree (RRT) algorithm, which is widely used in robotics and other engineering fields. He also authored the books Planning Algorithms, Sensing and Filtering, and Virtual Reality. He currently leads an Advanced Grant project from the European Research

Council on the Foundations of Perception Engineering. With regard to industry, he was an early founder and chief scientist of Oculus VR, acquired by Facebook for \$3 billion in 2014, where he developed patented tracking technology for consumer virtual reality and led a team of perceptual psychologists to provide principled approaches to virtual reality system calibration, health and safety, and the design of comfortable user experiences. From 2016 to 2017, he was a Vice President and Chief Scientist of VR/AR/MR at Huawei Technologies, where he was a leader in mobile product development on a global scale. He has worked as an angel investor and adviser to startups in robotics and virtual reality.

KEYNOTE SPEAKER 6

**Dr. Maria Pia Fanti**

Professor of Control System Engineering, Polytechnic University of Bari

TALK TITLE

Artificial Intelligence Approaches for Integrating Cooperative, Connected and Automated Mobility in the Urban Traffic.

Abstract

Cooperative, Connected and Automated Mobility (CCAM) is expected to reshape the way of travelling and moving in Europe and around the world. By the CCAM, the automated vehicles have to be integrated into the mobility and transport system by designing and implementing infrastructures, new services, platforms, cooperation and governance models. The smart traffic management and the deployment of CCAM innovative technologies and services is increasing efficiency and reducing congestion.

The talk will present some approaches developed to accelerate the integration of innovative CCAM technologies and systems for passengers and goods in terms of design and implementation of enhanced physical, digital and operational infrastructures. In particular, the talk will describe some recent results obtained in case studies implemented in a current European Project by simulation environments and in the field.

Biography

Maria Pia Fanti (M'92-SM'02-F'17) received the Laurea degree in electronic engineering from the University of Pisa, Pisa, Italy, in 1983. She was a visiting researcher at the Rensselaer Polytechnic Institute of Troy, New York, in 1999. Since 1983, she has been with the Department of Electrical and Information Engineering of the Polytechnic of Bari, Italy, where she is currently a Full Professor of system and control engineering and Chair of the Laboratory of Automation and Control. Her research interests include

management and modeling of complex systems, such as transportation, logistics and manufacturing systems; discrete event systems; Petri nets; consensus protocols; fault detection. Prof. Fanti has published more than 370 papers and two textbooks on her research topics. Prof. Fanti received from the IEEE SMCS in 2008 the best Associate editor award of the Trans. on SM, the best technical committee chair award and the 2020 IEEE SMC outstanding Contribution Award.

She is senior editor of the IEEE Trans. on Automation Science and Engineering and of the Transaction on Intelligent Vehicles. She is member at large of the Board of Governors of the IEEE Systems, Man, and Cybernetics Society, and was member of the AdCom of the IEEE Robotics and Automaton Society, founder and chair of the Technical Committee on Automation in Logistics and of Machine learning for Automation of the IEEE Robotics and Automation Society. Prof. Fanti was involved in the organizing committee of a large number of International conference and in particular she was General Chair of the 2011 IEEE Conference on Automation Science and Engineering, the 2017 IEEE International Conference on Service Operations and logistics, and Informatics and the 2019 IEEE Systems, Man, and Cybernetics Conference, Program co-Chair of the IEEE SMC 2023, Automation Chair of the IEEE ICRA

Technical Session 1 | 9:00 AM - 10:30 AM**TRACK : MEDICAL IMAGING AND DIAGNOSTIC DEEP LEARNING (MIDL)**

Hall 1 (Room #219), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Md. Tofael Ahmed, Professor, ICT, CoU Session Chair-2: Dr. Muhibul Haque Bhuyan, Professor, AIUB Session Moderator: Dr. Md. Mehedi Hasan, Assistant Professor, EEE, JUST Volunteer(s): MD Shoriful Islam (EEE), Mohammad Basel (EEE)	
#	Paper ID	Paper Title	Author(s)
1	237	Automated PET to Fused PET-CT Mapping Using PCGAN for Lung Cancer Diagnosis	Md. Abu Sayed, Ashiqur Rahman, Sourav Basak Shuvo, Muhammad Abdul Kadir, Md. Ibrahim Al Imran, Md. Sharjis Ibne Wadud, Md. Nahid Hossain
2	301	CoNeViG: Interpretable Breast Lesion Classification on BUS-CoT Using a ConvNeXtTiny and ViT-B/16 Gated Fusion Network	Iftekhar Ahmmed, Shahriar Kabir, Rafiqul Islam, Md Mahbub Saroar, Md. Rifat, Sharmin Alam
3	371	EMFNet: An Explainable Dual-Backbone Feature Fusion Network for Multi-Class Colorectal Image Classification	Md. Sadman Hafiz Shaon, Al Arian Ahmad, Khalid Hossen, Mahabub Hasan Biplop, Md. Ashikuzz Zaman, Md. Ibn Masud
4	441	EffViT-LCNet: A Gated EfficientNetV2M-ViT Fusion Network with Explainable AI for Lung Cancer Classification from CT Images	Roman Mia, Shahriar Kabir, Iftekhar Ahmmed, Md. Nadim Khan, Atiar Rahman Ritu, Md Saiful Alam, Shamima Akter
5	471	WSV-ViT: A Weighted Soft Voting Ensemble of Regularized Vision Transformers with LIME and Grad-CAM for Accurate Cervical Cancer Diagnosis	Jihadul Islam, Tanvir Anjum Rahat, MD Sifat, Md Abdul Masud
6	545	GAD-Net: An Interpretable Deep Learning Framework for Automated Diabetic Retinopathy Screening in Smart Healthcare	Abrar Rakin, Obaidul Haque, Md Istiak Ahamed, Md. Ashrafur Islam, Ferdaus Anam Jibon, Abu Bakar Rakib

Technical Session 2 | 9:00 AM - 10:30 AM**TRACK : AGRICULTURAL, AQUACULTURE AND CROP INTELLIGENCE (AACI)**

Hall 2 (Room #220), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Lasker Ershad Ali, Professor, KU Session Chair-2: Dr. Md. Sarwar Hosain, Professor, ICE, PUST Session Moderator: Md. Tarequzzaman, Assistant Professor, EEE, JUST Volunteer(s): Mehedi Hasan (CSE), Md. Al Muzahid Pranto (EEE)	
#	Paper ID	Paper Title	Author(s)
1	183	Physics-Guided Feature Embedded BiLSTM Framework for Precision Irrigation and Crop Water Requirement Forecasting	Zonaid Hossain Nafim, Sabiha Suraiya, Swapnil Tarafdar, Mahbubur Rahman, Fahad Mahmud, Jul Jalal Al-Mamur Sayer
2	310	A Multi-Stage Deep Learning Framework for Automated Detection and Localization of Shrimp Disease	Pritam Sarkar, Mohammad Tasneem Hasan, Arjun Saha, Md. Muntasir Abdullah
3	474	Agroclimatic Crop Yield and Production Prediction in Bangladesh Using Machine Learning and Explainable AI	Jannatul Maa Binta Rahim, Susmita Sejuti Saha, Rashik Reza Fahim, Aishwarja Paul Sourav, Dola Das
4	527	A Cloud-Integrated AIoT System for Automated Small-Scale Aquaculture Toward Urban Food Security	Md. Rahman, Mohammad Ali Khan, Anita Tasnim, Md. Abdur Razzak,

			Shaem Hasan Shakib, Md. Mahmudul Hasan
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Technical Session 3 | 9:00 AM - 10:30 AM

TRACK : AUTONOMOUS AERIAL AND FIELD ROBOTICS FOR SEARCH AND RESCUE (AAFR)

Hall 3 (Room #205), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Md. Rubel Basar, Associate Professor & Head, EEE, BAUET Session Chair-2: Dr. Sujan Chandra Roy, Assistant Professor, IICT, CUET Session Moderator: Sk. Shalauddin Kabir, Lecturer, CSE, JUST Volunteer(s): Md Tanvir Sikder (EEE), Maimuna Rahman Mysha (CSE)	
#	Paper ID	Paper Title	Author(s)
1	128	Benchmarking Vision-Language Models for Real-Time UAV Obstacle Avoidance in Controlled Simulation	Samiha Tarannum Noor, Abyead Zarif Hassan, Dr Md Mehedi Hasan
2	255	Quantitative Evaluation of Embedded ROS2 Navigation and Sensor Fusion for a Differential Drive Autonomous Mobile Robot	Md. Abu-Talha Roni, Alok Kumar Paul
3	262	FedSwarm-Attn: Federated Attention-Based Deep Reinforcement Learning for Energy-Efficient Multi-Robot Swarm Navigation in IoT Environments	Musavvir Haque, Masud Rana, Saklain Mustak, Mirza Mahadir Hasan, Jahedul Islam
4	387	A Collaborative UAV-UGV Heterogeneous System For Enhanced Search And Rescue In Complex Disaster Environments	Md. Nafis Mahmud, Atia Shahnaz Ipa, Zibon Nesa Zim
5	449	HEXAGUARD: An IoT-Based Search and Rescue Rover for Hazard Monitoring and Emergency Response	Md Zahid Hasan Zihad, Imtiaz Ahmed, Md Toha Bin Asad Dip, Dr. Md. Mahfuzur Rahman

Technical Session 4 | 2:30 PM - 4:00 PM

TRACK : BIOSIGNAL PROCESSING, NEURAL INTERFACES AND WEARABLE HEALTH SYSTEMS (BNWH)

Hall 1 (Room #219), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Md. Tofael Ahmed, Professor, ICT, CoU Session Chair-2: Dr. Md. Sarwar Hosain, Professor, ICE, PUST Session Moderator: Md. Mahmudul Haque Milu, Assistant Professor, BME, JUST Volunteer(s): MD Shoriful Islam (EEE), Mohammad Basel (EEE)	
#	Paper ID	Paper Title	Author(s)
1	263	DS-SleepNet: An Explainable Dual-Stream Meta Fusion Architecture for Sleep Disorder Diagnosis	Fazle Azim Pathan, Shihab Hossain, Md Masum, Md. Rakib Hasan, Asaduzzaman Noor, Md. Miraz Hossain
2	398	A Wearable Smart Diabetic Shoe with Hybrid Adaptive Multi-Index Classification for Early Detection and Prevention of Diabetic Foot Ulcers	Md. Rakibuzzaman Rabin, Sheikh Md. Rabiul Islam, Noshin Akhter, Maisha Mehnaz
3	500	Explainable Low Back Pain Classification Using Multifidus sEMG and Multi-Domain Feature Selection	Shuvo Ronjon, Tanmoy Datta, Ritu Mondal, Iftiar Rahman Emon, Md Moznuzzaman, Asraf Mridha Limon
4	502	Adapting a Lightweight Time-Series Foundation Model for sEMG Hand Gesture Classification	Rifat Tanbhir, Md. Sabbir Reza, Tanmoy Chandra Datta, Wakee Arefin, Md Mehedi Hasan, Md. Moznuzzaman
5	505	NeuroFocal-Net: A Multi-Scale CNN–Vision Transformer Hybrid with Focal Attention for Robust Five-Class EEG Seizure Detection	Md Minhazul Islam Minar, Imam Abu Zubaer, Prof. Dr. Mohammad Shamsul Arefin, Dr. Mahfuzulhoq Chowdhury

6	512	Adaptive Topology-Switched Wireless Power Transfer for Implantable Neural Stimulators: SAR-Constrained Optimisation and Misalignment Robustness Analysis	Anit Barua, Md. Ifthakhar Khan Sagor, Ayesha Joha Purnota, Swaran Kumar Pal Pal, Mehedi Hasan, Shah Mohammed Shahrir
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Technical Session 5 | 2:30 PM - 4:00 PM

TRACK : POWER ELECTRONICS, RENEWABLE ENERGY AND GRID AUTOMATION (PREG)

Hall 2 (Room #220), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Abdullah Al Nahid, Professor, ECE, KU Session Chair-2: Dr. Md. Rubel Basar, Associate Professor & Head, EEE, BAUET Session Moderator: Anika Anjum, Lecturer, BME, JUST Volunteer(s): Mehedi Hasan (CSE), Md. Al Muzahid Pranto (EEE)	
#	Paper ID	Paper Title	Author(s)
1	232	Explainable Hybrid ARIMA–ML Forecasting of Bangladesh’s GDP: Sectoral Energy Consumption and Carbon Emission Dynamics	Md Kaif Afran Khan, S.M. Sadab Shakib, Rafsat Wairan, Mohammad Sakib Mahmood, Md Al-Amin, Md Ajmin Shuvo, Md. Tareq Hasan
2	259	Continuation-Accelerated Multi-Start Newton Solution of Selective Harmonic Elimination for a Seven-Level Inverter	Rubell Sen Goopta, Md. Musfiqur Rahman Akib, Abdullah Al Nayem
3	260	A Constraint-Aware Neural Warm-Start with Newton Refinement for Selective Harmonic Elimination PWM: A Reproducible Study on the Three-Angle Fifth-Seventh Elimination Problem	Rubell Sen Goopta, Md. Musfiqur Rahman Akib, Abdullah Al Nayem
4	305	A Venturi Wind Concentrator for Enhancing 16 kW PMSG Turbine Output and Capacity Factor at Low Wind Sites in Bangladesh	Md. Abu Salman, Mehedi Hassan, Md. Mehedi Hasan, Subratto Biswas, Md. Sanwar Faraji, Md. Nazmul Hossain, Krishnapada Mondal
5	362	Fault-Ride-Through and Power Quality Enhancement of a Grid-Connected PV/PMSG-Wind Hybrid System Using DC-Link Chopper and LCL-Filtered Three-Level VSC	Nazmul Hossain Siam, Arizit Chaki Artha, Minan Mahmud Munna, Md. Bayzid Rahaman, Tareq Abir Tonmoy, Md Mortuza Ahmmed
6	447	A Multi-Paradigm Meta-Ensemble Framework for Short-Term Electricity Load Forecasting	Siyam Azad, Mahfuze Alam Shahin, Md. Shahiduzzaman, Mostafa Galib Jami, Imran Khan

Technical Session 6 | 2:30 PM - 4:00 PM

TRACK : CYBERSECURITY, TRUSTWORTHY AI AND MEDIA FORENSICS (CTMF)

Hall 3 (Room #205), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Md. Arshad Ali, Professor, CSE, HSTU Session Chair-2: Dr. Lasker Ershad Ali, Professor, KU Session Moderator: Md. Faisal Mina, Lecturer, BME, JUST Volunteer(s): Md Tanvir Sikder (EEE), Maimuna Rahman Mysha (CSE)	
#	Paper ID	Paper Title	Author(s)
1	224	SpatSeqFake: A Hybrid Spatial-Sequential Framework for Cross-Dataset Deepfake Image Detection	Sadia Afroz Oishi, Ariful Islam, Md Sarwar Hosain, Ohidujjaman, Abul Fazal Mohammad Zainul Abadin, Liton Chandra Paul
2	240	Towards Intelligent SOC Automation: An Open-Source Analyst-in-the-Loop Framework with Dual-LLM Alert Triage and MITRE ATT&CK RAG	Saif Ahmed Shuvo, Sarfin Uddin Bhuiyan Anik, Md Saniul Basir Saz, Md Tuhin Reza, Dr. Md. Kamrul Islam, Monishanker Halder, Sk. Shalauddin Kabir, Dr. Mohammad Nowsin Sheikh

3	385	Aligning the Unseen: MMD-Regularized Autoencoders for Cross-Device IoT Botnet Detection with Statistically Significant False Positive Reduction	Abdul Hasib, Waladur Rahman, Nasif Shahriar Alif, Md. Shakil Ahmed, Md. Rakibul Hasan Nishat, Arjina Nasrin Khan
4	437	Dialect-Aware Real vs. Telephone-Processed Voice Classification Using AASIST-Lite for Bangladeshi Regional Speech	Md. Sadman Hafiz Shaon, Al Arian Ahmed, Rotno Chandra Das, Mahabub Hasan Biplob, Khalid Hossen, Md. Abu Rayhan
5	452	Compact and Calibrated: Multi-Component Knowledge Distillation for Lightweight LLM Jailbreak Prompt Detection	Mowmita Parvin Hera, Syed Azmul Hasan, Ziyi Chen, Hazrat Ali, Mohammad Farhad Bulbul
6	515	Privacy Preserving Federated Learning for Early Student Dropout Prediction: Combining FedProx, Differential Privacy and Institutional Non-IID Partitioning	Md Shohedozzaman Basunia, Syed Shakib Shahriar Islam, Tridib Banik, Abdullah Al Rafi, Md. Naimul Pathan

Technical Session 7 4:30 PM - 6:00 PM			
TRACK : EXPLAINABLE CLINICAL PREDICTION AND ASSISTIVE HEALTH MONITORING (ECAH)			
Hall 1 (Room #219), Sir Jagadish Chandra Bose Academic Building, JUST		Session Chair-1: Dr. Md. Abdullah Al Humayun, Professor & Head, EEE, Eastern University Session Chair-2: Dr. Md. Abdulla Al Mamun, Professor, CSE, HSTU Session Moderator: Dr. Md. Mehedi Hasan, Assistant Professor, EEE, JUST Volunteer(s): MD Shoriful Islam (EEE), Mohammad Basel (EEE)	
#	Paper ID	Paper Title	Author(s)
1	1	NightCare-VideoMAE: A Framework for Low-Light Patient Activity Recognition	Md Rafiqul Islam, Md. Liakot Ali, Joydeb Kumar Sana
2	326	A Comparative Study of Federated XGBoost for Stroke Prediction under IID and Non-IID Healthcare Data Distributions	Raisa Jerin Sristy, Sajjad Waheed
3	443	An Explainable Ensemble Learning Framework for IoT-Based Elderly Healthcare Monitoring and Fall Detection Using Multi-Dataset Validation	Nusrat Mim, Sanjida Nuha, Aishwariya Sardar, Afrin Jahan, Md Abdul Masud
4	459	Lightweight Two-Stage CNN-to-LLM Pipeline for Automated Chest X-Ray Report Generation	Tanzin Ahamed Shohag, Md.Abu Raihan, Jahanara Perven Juthy, F M Javed Mehedi Shamrat, Md. Asraf Ali
5	521	From Perception to Reconstruction: Type-Aware Semantic Color Priors for Color Vision Deficiency	Abu Bakar Rakib, Obaidul Haque, Abrar Rakin, Ferdous Anam Jibon, Rafat Ara, Esrat Nourin
6	539	Enhancing Explainable Diabetes Prediction with CTGAN Based Data Augmentation: A Comparative Study with SMOTE	Shaykh Yunus, Abdullah Al Rafi, Md Shohedozzaman Basunia, Md. Shihab Prodhan, S. M. Rakibul Islam

Technical Session 8 | 4:30 PM - 6:00 PM**TRACK : INTELLIGENT TRANSPORTATION AND URBAN INFRASTRUCTURE AUTOMATION (ITUI)**

Hall 2 (Room #220), Sir Jagadish Chandra Bose
Academic Building, JUST

Session Chair-1: Dr. Md. Arshad Ali, Professor, CSE, HSTU
Session Chair-2: Dr. Mohammad Nurul Huda, Professor, CSE, UIU
Session Moderator: Md. Tarequzzaman, Assistant Professor, EEE, JUST
Volunteer(s): Mehedi Hasan (CSE), Md. Al Muzahid Pranto (EEE)

#	Paper ID	Paper Title	Author(s)
1	101	Design and Implementation of a Hybrid IoT-Based Emergency SOS Alert System Using GSM and MQTT	<i>Md. Al Riad, MD. Rakib Hassan Dipu, Wasif Azan Nafiz, Fazle Rabbi Rifat, Sadia Enam, Saurav Chandra Das</i>
2	315	PDRL-TSC: Predictive Deep Reinforcement Learning for Hardware-Constrained Adaptive Traffic Signal Control	<i>S. M. Sadikuzzaman Abir, Most. Kaniz Fatema Isha, Md. Nazirul Hasan Shawon</i>
3	357	Multi-Objective Adaptive Signal Control of Urban Roundabouts Using Genetic Algorithms	<i>Turjoy Das Turjo, Mahbub Hassan, Arpita Saha Tithi, Touhid Bhuiyan, Md Maruf Hassan</i>
4	359	A Multi-Paradigm Forecasting Framework for Urban Traffic Volume Prediction Using Data-Driven Time Series Modeling	<i>Turjoy Das Turjo, Mahbub Hassan, Arpita Saha Tithi, Abubokor Hanip, Md Maruf Hassan</i>
5	361	Agentic Vision-Language Framework for Resilient Railway Track Defect Detection and Maintenance Orchestration	<i>Sumon Chandra Ray, Mahbub Hassan, Md Maruf Hassan, Cynthia I. Teran Valadez</i>

Technical Session 9 | 4:30 PM - 6:00 PM**TRACK : ROBOT DESIGN, KINEMATICS AND MACHINE VISION (RDKV)**

Hall 3 (Room #205), Sir Jagadish Chandra
Bose Academic Building, JUST

Session Chair-1: Dr. Md. Golam Rashed, Professor, ICE, RU
Session Chair-2: Dr. Sujon Chandra Roy, Assistant Professor, IICT, CUET
Session Moderator: Sk. Shalauddin Kabir, Lecturer, CSE, JUST
Volunteer(s): Md Tanvir Sikder (EEE), Maimuna Rahman Mysha (CSE)

#	Paper ID	Paper Title	Author(s)
1	238	Design of a Low-Cost Humanoid Robot for STEM Education in Resource-Constrained Classrooms	<i>Tanbir Hossain, A A M Muzahid, Md Awsaf Iqbal Chowdhury, Rasheduzzaman Md, Osama Ibn Mahfuz, Jiadian Wang, Tafhimul Islam</i>
2	325	Latent Geometry-Guided Gaussian Splatting for Depth-Free Novel View Synthesis	<i>Mst. Jeba Shazida, Md. Mahafuj Anam Murad, Dr. Md. Mahfuzur Rahman, Shariful Haque</i>
3	328	Comparative Analysis of 6-DOF and 7-DOF Robotic Manipulators Through Structural, Kinematic, and Motion Planning Evaluation.	<i>Arnob Chakraborty, Zephyr Uddin, Sadman Saif Brinto</i>
4	369	A Comparative Study of Skeleton and Silhouette Features for Drone-Based Gait Person Identification Using Deep Learning	<i>Zahin Daiyan, Samin Yesar Tousib, Md Abdullah Al Mahim, Nakib Aman, Mohammad Ali Sakib, Md. Zillur Rahaman</i>
5	492	A Low-Cost IMU-Based Wearable Device for Real-Time Elbow Range-of-Motion Monitoring with Clinically Acceptable Accuracy	<i>Hasin Abrar, Ahad Hossain Nibir, Tonmoy Chandra Datta, Ritu Mondal, Md Moznuzzaman</i>
6	543	VISTA RT: Heterogeneous Local Global Residual Fusion for State-Invariant Object Recognition and Retrieval	<i>Md Ibrahim Shikder Mahin, Muhammad Aminur Rahaman</i>

Technical Session 10 | 9:00 AM - 10:30 AM**TRACK : DEEP LEARNING FOR MEDICAL IMAGE ANALYSIS (DMIA)**

Room: VR-1 (Online Session)		Session Chair-1: Dr. Taufique Sayeed, Fulbright Visiting Research Scholar, ESU, USA Session Chair-2: Dr. Ashis Kumar Mandal, Professor, CSE, HSTU Session Moderator: Dr. Md. Abdul Halim, Associate Professor, ChE, JUST Volunteer: Saklain Mustak (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/92948435824?pwd=vz9P2qag5HNQmfnLEps1yeOzCwOc3c.1 Meeting ID: 929 4843 5824	
#	Paper ID	Paper Title	Author(s)
1	207	A Cascaded Pipeline for Melanoma Detection, AJCC Staging, and Skin-Lesion Classification with Grad-CAM Interpretability	Hasan Mahmud, Farhana Tazmim Pinki
2	209	CaneScan-XAI: An Explainable Ensemble Deep Learning Framework for Automated Sugarcane Leaf Disease Classification	Mahfuz Uddin Ahmed, Shawna Akter, Rafid Bin Taher, Moin Uddin Ahmed
3	358	Explainable AI-Enhanced Deep Learning Models for Automated White Blood Cell Subtype Classification in Blood Smear Images	Sagor Kumar Pandit, Rotno Chandra Das, Swapnomoy Biswas, Md. Ashikuzz Zaman, Nitun Kumar Podder
4	433	Benchmarking Attention-Enhanced Encoder–Decoder Models for Polyp Segmentation in Colonoscopy Images	Md. Nur Alam, Eric Mithun Bonik, Tamim Mahmud, Md. Monir Hossain Shimul, Kulsom Akter Sinthia, Md Raj Kabir
5	508	Hierarchical Attention and Dynamic Gating Network for Medical Image Classification	Sifat Rahman Ahona, Hussain Mohammad Ricon, Tabassum Alam, Amatur Rahman Oshin
6	518	HCSNet: Hybrid CNN–Swin Transformers with Diffusion Augmentation for Explainable Breast Ultrasound Cancer Classification	Nisha Jaman Khan, Adnan Sakib, Muhammad Aminur Rahaman
7	552	Whose Cuisine? Cultural Bias in Digital Eating Support Systems	Sharmin Sultana Lincoln, Soham Biren Katlariwala, Fahima Hossain Muna

Technical Session 11 | 9:00 AM - 10:30 AM**TRACK : POWER SYSTEMS, SOLAR ENERGY AND ELECTRICAL MACHINE AUTOMATION (PSEM)**

Room: VR-2 (Online Session)		Session Chair-1: Dr. Afroza Nahar, Professor, CS, AIUB Session Chair-2: Dr. Shruti Jain, Professor & Dean (A), JUIT, India Session Moderator: Dr. Md. Sumon Rahman, Associate Professor, IPE, JUST Volunteer: Md. Mustakim Islam (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/98461036539?pwd=u3YuMUbaA268nEHxUaQHOIUbaIIlBVq.1 Meeting ID: 984 6103 6539	
#	Paper ID	Paper Title	Author(s)
1	69	Leakage-Resistant Physics-Informed Learning for Multiclass Transmission-Line Fault Diagnosis	Sa-Ad Ahmmed, Md Maniruz Zaman, Md. Mizanur Rahman, Md. Jannat Khan, Jahangir Alam
2	187	ESP32-Based Outage-Resilient Solar Tracking and IoT Energy Analytics	Md. Walid Hossian, Abdul Mumen Masum, Sohel Mia, Robiul Alam Riad, Rubel Paloan, Md Maniruz Zaman
3	210	High-Efficiency Lead-Free Cs ₂ PdBr ₆ Perovskite Solar Cells: A SCAPS-1D Numerical Study of Transport Layer Optimization	Fatematug Johara Mim, Md. Mizanur Rahman, Md. Rafiur Rahman Rowdra, Md. Tanvir Ahamed, Md. Motiur Rahman Tareq, Md. Sazedur Rahman

4	266	Field-Oriented Control of Enhancing Torque Ripple and Flux Estimation Accuracy of Induction Motor based on Multi-Level Inverter	Harith Nashwan, Harith Al-Badrani
5	334	PowerMamba-BD: A Selective State-Space Forecaster with Inference-Time Conservation Enforcement for Demand and Load Shedding on the Bangladesh National Grid	Rashid Tanzil, Shehab Ahamed, Mahinul Islam, Sanjan Habib
6	413	Data-Driven Techno-Economic and Environmental Assessment of Rooftop Solar PV	Md Maniruz Zaman, Sheikh Farhana Binte Ahmed, Md. Mominur Rahman, Tahsin Abrar Nabil, Umma Saima Jisha, Md Jisan

Technical Session 12 | 9:00 AM - 10:30 AM

TRACK : ROBOT MANIPULATION, NAVIGATION AND HUMAN-ROBOT INTERACTION (RMHI)

Room: VR-3 (Online Session)		Session Chair-1: Dr. M. Jayalakshmi, Associate Professor, ECE, RCEW, India Session Chair-2: Dr. Monjur Morshed, Professor, AUST Session Moderator: Dr. A F M Shahab Uddin, Assistant Professor, CSE, JUST Volunteer: Milton Mistry (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/91241594090?pwd=LjJw7QEexj36imNqJTG4YPZxI5aPrW.1 Meeting ID: 912 4159 4090	
#	Paper ID	Paper Title	Author(s)
1	49	A Voice-Based Semantic Task Interface for Sequential Navigation in ROS 2-Based Systems	Amal CN, Blesson Mathews Luke, Abraham Sampson S
2	277	Hand-Gesture-Guided One-Demo-to-Many Robot Skill Generation with Teacher-Weighted Residual Correction in MuJoCo	Fahim Shahriar, M A Hafiz, Tasfia Noor, Wasima Noor Iqra
3	281	IRIS: Intelligent Robotic Interactive System with Multimodal Sensing for Autonomous Object Manipulation	Joezac Joseph Sunil, Nithya S, Rahul Krishnan, Shon Mathews, Mary Catherine V G
4	288	Human-Demonstration-Guided Warehouse Navigation with Safety-Aware Gap Reasoning for Mobile Robots	Wasima Noor Iqra, M A Hafiz, Fahim Shahriar, Tasfia Noor Chowdhury
5	456	An Automated Scan-to-Simulation Pipeline for Indoor Digital Twin Construction Using BLK360 and NVIDIA Isaac Sim	Sangmin Kim, Byeongjun Kim, Tae-Yong Kuc
6	479	Low-Cost 3D Environment Reconstruction from 2D LiDAR Data Using the OMRON TM5-900 Robotic Manipulator	Genito Pedro, Kleber Roberto Da Silva Santos

Technical Session 13 | 2:30 PM - 4:00 PM

TRACK : EXPLAINABLE MACHINE LEARNING FOR CLINICAL RISK PREDICTION (ECRP)

Room: VR-1 (Online Session)		Session Chair-1: Dr. Shaik Mahaboob Basha, Professor, NEC Gudur, India Session Chair-2: Dr. Hadaate Ullah, Associate Professor, EEE, USTC Session Moderator: Dr. Md. Abdul Halim, Associate Professor, ChE, JUST Volunteer: Saklain Mustak (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/98837687965?pwd=9qLidKFRaiJQLaQ75HhbmNrhP1o9XG.1 Meeting ID: 988 3768 7965	
#	Paper ID	Paper Title	Author(s)
1	254	Machine Learning Based Survival Prediction in Lung Cancer: A Comparative Study of TNM and UICC Staging Systems	Md Yusuf Hossain Ador, Samrat Kumar Dev Sharma, Saem Mohammad Nirob, Rumana Islam Riya, Iqramul Haq, Mohammed Islam

2	268	HepaPredict: A Web-Driven Hybrid LightGBM-CatBoost Framework for Early Liver Disease Prediction	Mossaddak Hossain, Mim Bin Hossain, Md. Rizwan Chowdhury, Md Ashikur Rahman Anik, Meherun Nessa Jerin, Farhana Hoque Nishi, Dipto Kumar Das, Ragib Mahatab
3	276	How Little Is Enough for CPU-Only Surface-Defect Classification? A Controlled Comparison of Classical and Compact Convolutional Models	Md Aftab Uddin, Jaafar Gaber, Pierre Petitjean, Francois Cortinavis
4	395	EDRP-XAI: Explainable Diabetes Risk Prediction Using Optimized Ensemble Learning for Smart Healthcare	Afroza Sultana, Kaniz Mashruhra Nusrat, Nafisa Prity, Nazia Binte Salam, Subarna Yeasmin Sheemu
5	523	Explainable and Uncertainty-Aware Machine Learning for Predicting Cryotherapy Treatment Success in Cutaneous Warts	Md Tanzim Hossain, Tanvir Rahman, Rakibul Hassan Rejon, Tasdik Hasan, Mohammad Anwar Hossain, Nasirul Mumenin
6	538	A Machine Learning-Based Clinical Decision Support System for Early Detection of Chronic Kidney Disease	Nahid Islam, Ananya Kundu, Md. Romjan Ali, Md. Harun Or Rashid, Md. Ashiqul Tareq Titu, Md. Delowar Hossain, Tangina Sultana
7	566	HealthCal-XAI: An Explainable Multi-Endpoint Machine Learning Framework for Preventive Disease Risk Screening	Hrithik Shibu, Kanika Chakraborty

Technical Session 14 | 2:30 PM - 4:00 PM

TRACK : CYBERSECURITY, FRAUD DETECTION AND DIGITAL FORENSICS (CFDF)

Room: VR-2 (Online Session)		Session Chair-1: Dr. Kazi Newaj Faisal, Associate Professor, EECE, MIST Session Chair-2: Dr. Mohebbanaaz, Dean (R&D), ECE, RCEW, India Session Moderator: Dr. Engr. Md. Arif Rahman, Assistant Professor, CSE, JUST Volunteer: Md. Mustakim Islam (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/93159611809?pwd=cWkGuDfsbo9nJnQtsfNqVhU7ZdoC.1 Meeting ID: 931 5961 1809	
#	Paper ID	Paper Title	Author(s)
1	219	Real-Time Browser Threat Detection Security Agent via Local VLM–LLM	Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, Tanvir Ahmed, Md Mizanur Rahman
2	236	A Comparative Deep Learning Framework for Phishing Detection Using Transformer and Reconstruction-Based Models	Md. Impreeaj Hossain, Musaib Ibn Habib Mikdad, Samir Uddin Ahmed, Israt Jahan Esha, Muhammad Iqbal Hossain
3	278	Beyond Learned Fusion: Modality-Aware Max-OR for Asymmetric Deepfake Detection in Short Audio-Visual Clips.	Abu Hayat Mohammad Nibras, Tasnova Islam, Md. Jahedul Islam
4	283	VulPatchNet: A Dual-Encoder Cross-Attention Framework for Multi-Language Vulnerability Patch Generation	S. M Imran Reza, Iffath Tanjim Moon, Md. Forhan Shahriar Fahim, Nafiu Rahman, Adrita Alam
5	312	An In-Depth Comparative Analysis of Credit Card Fraud Detection Using A Hybrid Machine Learning Framework and Explainable AI	Divyashree Dutta, Md. Nazrul Islam Mondal
6	384	Intrusion Detection System Using Optimized Machine Learning with Explainable AI	Sabbir Ahmed, Nafisa Prity, Sharupya Barua, Nazia Binte Salam, Sadia Ahmed Eva, Afroza Sultana

Technical Session 15 | 2:30 PM - 4:00 PM**TRACK : ROBOT DESIGN, KINEMATICS AND MECHATRONIC SYSTEMS (RDKM)**

Room: VR-3 (Online Session)		Session Chair-1: Dr. Lachit Dutta, Assistant Professor, ECE, Gauhati University, India Session Chair-2: Dr. Taufique Sayeed, Fulbright Visiting Research Scholar, ESU, USA Session Moderator: Md. Tariquzzaman, Assistant Professor, EEE, JUST Volunteer: Milton Mistry (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/99256115536?pwd=SygxFU2F5sH0WmazFQpC0Xq2ad3FKq.1 Meeting ID: 992 5611 5536	
#	Paper ID	Paper Title	Author(s)
1	225	Design and Experimental Evaluation of a Low-Cost Autonomous Solar Panel Cleaning Robot Using Line-Following Navigation	Shuvodip Dey, Md. Faiead Imtyaz, K.B.M. Miyad Hasan Shamippo, Affan Sabith, Md. Maruf Hasan Mahi
2	294	Kinematic Analysis, Trajectory Generation, and Low-Cost Hardware Evaluation for a Quadruped Robot Leg	Md. Shahariar Hossain, Rafin Md Abrar, Shadid Al Akib, Tazwar Ahmed, Shahed Alam, Atib Mohammad Oni
3	360	A Low-Complexity 3-DOF Collaborative Robot with Vision-Based Expiry Date Recognition for Retail Automation	Sayma Jhara, Alireza Sanaei
4	445	Design, Development, and Experimental Demonstration of a Low-Cost Additively Manufactured Morphing Quadcopter	Tawhidul Islam Tahsif, Tasnim Kibria, Anik Datta, Tanzibul Islam
5	495	Design, Development, and Workspace Evaluation of a Ceiling-Mounted Four-Axis SCARA Robotic Arm for Light-Duty Automation	Md Ridwanul Islam Antu, Md Hasibur Rahman
6	542	Inverse Kinematics of 6-DOF IRB 2400 Robots with Customized MLPs and Explainable Joint Feature Analysis	Md. Abid Hasan Rafi, Mohima Binte Rasel, Mst. Fatematuj Johora, Mst. Sharmin Ara, Md. Mahabub Hossain

Technical Session 16 | 4:30 PM - 6:00 PM**TRACK : BANGLA NATURAL LANGUAGE PROCESSING AND LANGUAGE TECHNOLOGIES (BNLP)**

Room: VR-1 (Online Session)		Session Chair-1: Dr. Mohammad Nurul Huda, Professor, CSE, UIU Session Chair-2: Dr. Kazi Newaj Faisal, Associate Professor, EECE, MIST Session Moderator: Dr. Engr. Md. Arif Rahman, Assistant Professor, CSE, JUST Volunteer: Saklain Mustak (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/92910153405?pwd=EFpmdZ3C54CP8Menv7CgQaFgATdhC.1 Meeting ID: 929 1015 3405	
#	Paper ID	Paper Title	Author(s)
1	203	Bangla Sentence Function Classification: Corpus Development, Model Benchmarking, and Interpretability	Abdullah Al Shafi, Swapnil Kundu Argha, Rowzatul Zannat, Shoumik Barman Polok, M.A. Moyeen, Abdul Muntakim, Jannatul Ferdousi
2	213	Leakage-Aware Sentiment Analysis of Bangla-English E-Commerce Reviews Using RoBERTa with Rigorous Cross-Validation	Md. Impreeaj Hossain, Musaib Ibn Habib Mikdad, Israt Jahan Esha, Archana Das, Muhammad Iqbal Hossain
3	264	Fine-Grained Sentiment Classification of Bangla Book Reviews: A Large-Scale Benchmark Dataset and Ensemble Learning Approach	Mithila Akter, Sadman Ahmed Sakin, Rafsan Hasan Pronay
4	352	Multi-Modal Stacking Ensemble with SHAP Explainability for Bangla Clickbait Detection	Chowdhury Mohammad Mehedi, Rimon Kanthi Devnath, Md Akash Rahman, Masum Ahmed Mozumder,

			Nusrat Jahan, Fariha Tasnim, Mohammed Borhan Uddin
5	516	Real-Time Isolated Bangla Sign Language Recognition Using MediaPipe Hand Landmarks and LSTM	Moinul Hossain, Md Kishor Morol
6	553	Bengali Dialect Classification via Feature-Enriched RAG and Multi-Strategy Prompting Ensembles	Mahjabin Rahman, Khadeja Islam, Mozdahaer Abdul Quader, Fateha Jannat Ayrin, Md Mizanur Rahman, Montaser Abdul Quader, Sithi Mozumder, Madhusodan Chakraborty, Md. Ashikur Rahman Api, Sadia Afroze

Technical Session 17 4:30 PM - 6:00 PM			
TRACK : ARTIFICIAL INTELLIGENCE FOR CROP DISEASE AND PRECISION AGRICULTURE (ACPA)			
Room: VR-2 (Online Session)		Session Chair-1: Dr. M. Jayalakshmi, Associate Professor, ECE, RCEW, India Session Chair-2: Dr. Afroza Nahar, Professor, CS, AIUB Session Moderator: Md. Tariquzzaman, Assistant Professor, EEE, JUST Volunteer: Md. Mustakim Islam (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/91747671444?pwd=NboWCa3Dr7OgsRTtoabrI4bHOAZbQFG.11 Meeting ID: 917 4767 1444	
#	Paper ID	Paper Title	Author(s)
1	63	Performance Analysis of Sequential CNN Architectures for Regional Fruit Leaf Disease Detection	Jannatul Ferdouse, Sumaiya Yeasmin, Md. Rasel Ahmed, Md. Abid Hasan Roni Bokshi, Aslam Mahmud Siam
2	145	Comparative Analysis of CNN Architectures for Hog Plum Leaf Disease and Maturity Stage Classification	Bishwa Das, Md. Asiful Islam, Ifath Rahman Tushar, Subroto Sarker, Bijoy Sarkar, Ajoy Shutra Dhar
3	157	Deep Learning-Based Morphological Segment Classification of Piper Chaba: A Benchmark Study with Class-Imbalance Mitigation	Manam Khan, Moonsia Alifa, Manzima Rahman, Siprian Gain, Raufu Fahad, Fatin Nabil, Rohul Amin
4	293	Bridging Local and Global Feature Learning: An Explainable Ensemble Framework for Intelligent Crop Disease Diagnosis	Rafsan Hasan Pronay, Tamzid Raihan, Nusrat Sinthiya
5	544	An Explainable Machine Learning Framework for Crop Recommendation Using Soil and Environmental Features	Maruf Kabir Refat, Baivab Das, Afridi Jubair, Yamin Hossain, Abul kalam Azad, Sumya Akter
6	550	TriAttenNet: A Lightweight Multi-Branch Separable-Attention Network for Explainable Ash Gourd Leaf Disease Classification	Mst Lutfun Nahar, Bibi Aysha Raha, Md. Nahidul Islam Rabbi Sheikh, Riazul Islam Rahat, Shahariar Hossain, Shaikh Ahmed
7	562	A Robust Weighted Ensemble Framework with Explainable AI for High-Precision Cotton Pest Identification	Md. Toha Hayder, Ibrahim Khalil Farsim, Sazzad Islam, Sazzad Hossain, Md. Faysal Hossan, Saifuddin Sagor

Technical Session 18 | 4:30 PM - 6:00 PM**TRACK : INDUSTRIAL AUTOMATION, DEFECT INSPECTION AND PREDICTIVE MAINTENANCE (IADP)**

Room: VR-3 (Online Session)		Session Chair-1: Dr. Shruti Jain, Professor & Dean (A), JUIT, India Session Chair-2: Dr. Mohebbanaaz, Dean (R&D), ECE, RCEW, India Session Moderator: Dr. A F M Shahab Uddin, Assistant Professor, CSE, JUST Volunteer: Milton Mistry (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/93948597193?pwd=QgWF59gUJtz8kphQRbmpiGMHp6OWFb.1 Meeting ID: 939 4859 7193	
#	Paper ID	Paper Title	Author(s)
1	94	Fatigue-Aware Break Schedule Optimization for Garment Assembly Lines Using Discrete Event Simulation and NSGA-II	Rashid Tanzil, Fawjia Islam, Niloy Roy
2	308	Adjacency-Aware Graph Transformers for Learning Mesh Dynamics	Md Yasir Arafat, Wissem Inoubli, Said Jabbour
3	416	TRUST-RUL: Reliability Auditing of Turbofan Remaining Useful Life Models Under Critical-Region, Uncertainty, and Domain-Shift Stress Tests	Tahiatul Siddique, MD Tahsin Hoque, Mofakkarul Islam Turjo
4	525	A Multi-Task U-Net Model with ASPP and Attention for Joint Segmentation and Classification of Industrial Defects	Arafath Fahim, Dr. Mohammad Mizanur Rahman, Kazi Naimur Rahman, Nusaiba Ibnath, Mohammad Ekram Ullah
5	551	RoadDG-Lite: Source-Only Cross-Country Generalization and Confidence Calibration for Lightweight Road Damage Detection	Moinul Hossain, Md Kishor Morol

Technical Session 19 | 9:00 AM - 11:00 AM**TRACK : TRUSTWORTHY, FEDERATED AND PRIVACY-PRESERVING MEDICAL AI (TFMA)**

Room: VR-1 (Online Session)		Session Chair-1: Dr. Afroza Nahar, Professor, CS, AIUB Session Chair-2: Dr. Hadaate Ullah, Associate Professor, EEE, USTC Session Moderator: Dr. Shakhawat Hossain, Associate Professor, IPE, JUST Volunteer: Saklain Mustak (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/94885677585?pwd=v9u1g52PbYFbflkvZQVNxWsgYwi2YD.1 Meeting ID: 948 8567 7585	
#	Paper ID	Paper Title	Author(s)
1	108	Voice-Guided Traffic Sign Classification Using Progressive Deep CNN with Batch Normalization for Driver Assistance	Mohammad Noimul Islam, Md. Momtasir Mohtad Noman, Tanvinur Rahman Siam, Mohammad Ashraf Ali
2	257	Fed-Physio-Tunnel: Secure Physiological Gating for Reversible Memory Editing in Federated Cognitive Digital Twins	Mir Tahmid Hossain, Sadik Al Jarif, Md Nawab Yousuf Ali, Md. Akil Hossain
3	487	RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI	Md Abu Sayeb Rayhan, Mosammat Fahimah, Md Tarek Mahmud, Hustagna Hiya, Ajoy Shutra Dhar
4	489	Failure-Aware Brain Tumor Segmentation with Tumor Type Conditioned Reliability Estimation and Explainability Guided Refinement	Md. Masud Rana Redoy, Tanvir Ahmed Tamim, Sk. Md. Tashfin Sami, Md Abdul Masud, Md. Samsuzzaman
5	496	A Triple-Layer Privacy-Preserving Federated Learning Framework for Secure Medical Image Classification	Avishek Debnath, Md. Nazirul Hasan Shawon
6	514	Beyond Data Mirage: Hallucination Detection and Mitigation in Small Language Models	Fariha Rahman Toha, Md Shohidul Islam, Saiful Islam

7	558	IB-Net : An Explainable Hybrid CNN-Transformer Model for Brain Tumor Classification	<i>Sad Ibne Musa Nabil, Saifuddin Sagor, Fuad Hossain, Shahariar Hossain, Md. Shafiul Alam Chowdhury, Md Masum Billah</i>
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Technical Session 20 | 9:00 AM - 11:00 AM**TRACK : NEUROLOGICAL SIGNAL ANALYSIS AND WEARABLE HEALTH MONITORING (NSWH)**

Room: VR-2 (Online Session)		Session Chair-1: Dr. Lachit Dutta, Assistant Professor, ECE, Gauhati University, India Session Chair-2: Dr. Ashis Kumar Mandal, Professor, CSE, HSTU Session Moderator: Dr. Mohammad Nowsin Amin Sheikh, Associate Professor, CSE, JUST Volunteer: Md. Mustakim Islam (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/91780710570?pwd=Kq8moU5G3bDLEDcK9LjVVpeXWbPuvf.1 Meeting ID: 917 8071 0570	
#	Paper ID	Paper Title	Author(s)
1	252	An Ordinal Deep Learning Framework with FT-Transformer and CORAL for Assessing Parental Perception of Smartphone-Related Harm in Bangladeshi Children	Saimuna Akter Shefa, Pritom Kumar Paul, Nuraiya Jannat Labiba, Most. Najmun Nahar Mim, Oli Ahmed Khan, Shah Md Abdur Razzak
2	258	BrainGATv2-Mamba: Spatial-Temporal Learning for Schizophrenia	Nuzhat Tabassum, Md. Shoabe Akter, Riadul Islam Rabbi
3	465	Prediction Epileptic seizures using deep learning	Shahad Abd, Hussein K. Ibrahim
4	468	Explainable Tinnitus Detection Using Audiograms: An Image-Based Deep Learning Approach	Md. Ajwad Hossain, Md. Farhad Hossain, Md. Azad Hossain
5	274	Bidirectional LSTM with Temporal Pooling for Cross-Day Robust Intrusion Detection on CIC-IDS2017	Toyeer-E- Ferdoush, Md. Rifat Hassan
6	534	SMOTE-Augmented Gradient Boosting for Epileptic Seizure Detection from Imbalanced Scalp EEG	Md. Sayed Parvej, Tanmoy Chandra Datta, Md Moznuzzaman

Technical Session 21 | 9:00 AM - 11:00 AM**TRACK : EMBEDDED IOT, HARDWARE SECURITY AND NETWORKED SENSING SYSTEMS (EIHS)**

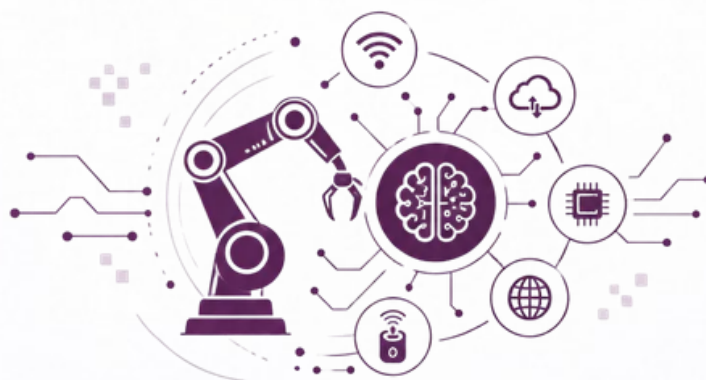
Room: VR-3 (Online Session)		Session Chair-1: Dr. Shaik Mahaboob Basha, Professor, NEC Gudur, India Session Chair-2: Dr. Md. Abdulla Al Mamun, Professor, CSE, HSTU Session Moderator: Md. Mahmudul Haque Milu, Assistant Professor, BME, JUST Volunteer: Milton Mistry (EEE)	
Joining Information		Zoom: https://bdren.zoom.us/j/98894633954?pwd=823Q4ohlX0umu0wXApC3nZLZQkz9r.1 Meeting ID: 988 9463 3954	
#	Paper ID	Paper Title	Author(s)
1	333	Lightweight CNNs Under Resource Constraints: A Controlled Multigenerational Evaluation of Architecture, Initialization, Training Budget, and Efficiency Tradeoffs	Tasnim Shahriar
2	349	A Decentralized P2P Offline Mesh Networking System Using Drones	Khan Ar-Rafi, Shihab Ankur, Mohammad Hossain, Md. Kafi, Sheikh Sohan, Moinul Alam, Md. Khan, Md. Roman
3	353	Design and Evaluation of Lightweight Adiabatic CMOS PUF for IoT Device Authentication	Afif Choudhury, Pijush Majumder
4	462	High-Isolation Orthogonally Oriented 2-Port MIMO Vivaldi Antenna for Ku/Ka-Band Satellite Applications	Mohammad Habibur Rahman Razib, Md. Arif Islam, Debprosad Das, Md. Azad Hossain

5	499	Adaptive Replanning for Multi-UAV Swarm Search-and-Rescue Missions Under Failure and Priority Constraints	<i>Farha Priti</i>
6	503	A Dual-Controller Internet of Things Programmable Electronic Load for Autonomous Multimode Battery Characterization	<i>Zubayer Jahin, Turja Majumder, A.S.M Al Mosabbir</i>
7	524	ULTRAFLOAT: An ESP32-S3-Based Non-Contact Water Level Sensing and Pump Control Platform with Dynamic Speed-of-Sound Compensation	<i>Shourov Paul, Md Samiul Islam Mobin, Abu Sayem, Mohammad Rezaul Karim</i>

**IEEE 5TH INTERNATIONAL CONFERENCE ON
ROBOTICS, AUTOMATION, ARTIFICIAL-INTELLIGENCE
AND INTERNET-OF-THINGS (RAAICON) 2026**



**PRESENTED
ABSTRACTS**



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Jashore University of
Science and Technology (JUST)

TRACK 1: MEDICAL IMAGING AND DIAGNOSTIC DEEP LEARNING (MIDL)

Paper ID: 237

Automated PET to Fused PET-CT Mapping Using PCGAN for Lung Cancer Diagnosis

¹MD. ABU SAYED*, ¹Ashiqur Rahman, ¹Sourav Basak Shuvo, ¹Muhammad Abdul Kadir, ¹Md. Ibrahim Al Imran, ¹Md. Sharjis Ibne Wadud and ²Md. Nahid Hossain

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Abstract

Accurate lung cancer diagnosis relies on a combined assessment of the metabolic activity and anatomical structure jointly captured by PET and CT imaging. PET imaging alone, however, presents diagnostic limitations in settings where CT is unavailable. This work presents a deep learning pipeline for generating synthetic CT images from PET scans using a PCGAN, with the resulting CT subsequently fused with the original PET image. The dataset comprised 2,725 paired PET CT slices collected from 50 lung cancer patients at a nuclear medicine center in Bangladesh. Thorax localization and field-of view (FOV) correction were performed using YOLOv11, while PET images were enhanced with CLAHE to improve anatomical visibility prior to synthesis. The synthesized CT images were fused with their corresponding PET scans to generate color enhanced, diagnostically useful PET-CT images. For the CT generation task, the proposed system achieved a PSNR of 13.31 dB, mean SSIM of 55.03%, FID of 120.61, and MAE of 0.1115, while the detection model attained mAP50 scores of 99.50% for CT and 98.50% for PET, outperforming CycleGAN and Pix2Pix baselines under identical evaluation conditions. Collectively, these findings indicate the system's utility as a diagnostic aid for lung cancer, particularly in clinical settings lacking CT imaging capability.

Paper ID: 301

CoNeViG: Interpretable Breast Lesion Classification on BUS-CoT Using a ConvNeXtTiny and ViT-B/16 Gated Fusion Network

Iftekhar Ahmmed*, Shahriar Kabir, Rafiqul Islam, Md Mahbub Saroar, Md. Rifat and Sharmin Alam

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Abstract

Breast lesion classification in ultrasound images is still difficult because of the speckle variation, indistinct boundary and different shape of the lesions. This paper proposes CoNeViG, a gated fusion network combining ConvNeXtTiny and ViT-B/16 for binary breast lesion classification on the BUS-CoT BUS-Lesion benchmark. ConvNeXtTiny extracts local spatial features through four hierarchical convolutional stages, while ViT-B/16 captures global context via patch-based self-attention. A learned sigmoid gate combines their 768-dimensional outputs, adaptively weighting each feature dimension before classification. Grad-CAM visualizations identify the lesion regions that drive each prediction. Evaluated on 869 test images, the model achieves 89.76% accuracy, 0.8976 F1-score, 0.9445 AUC, and 92.79% malignant sensitivity, outperforming all published baselines on This benchmark, including Qwen2.5-VL with chain-of-thought prompting (80.64% accuracy, AUC 0.8354) and a dual-stage visual-textual alignment model (81.77% accuracy, AUC 0.8852).

Paper ID: 371

EMFNet: An Explainable Dual-Backbone Feature Fusion Network for Multi-Class Colorectal Image Classification

Md. Sadman Hafiz Shaon*, Al Arian Ahmad, Khalid Hossen, Mahabub Hasan Biplop, Md. Ashikuzz Zaman and Md. Ibn Masud

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Abstract

Automated colorectal image classification is an important area in medical image analysis because accurate identification of colorectal conditions can support early diagnosis and clinical decision-making. However, this task remains challenging due to the high visual similarity among different colorectal pathological classes, complex image patterns, and the limited interpretability of deep learning-based diagnostic systems. To address these challenges, this study proposes EMFNet, a hybrid dual-backbone feature-fusion deep learning framework for accurate and interpretable multi-class colorectal image classification. A balanced dataset of 12,000 colorectal medical images from 12 distinct classes was used with separate training, validation, and testing subsets. Initially, EfficientNetB0 and MobileNetV2 were trained and evaluated individually as baseline models. EfficientNetB0 achieved a test accuracy of 97.05% with an F1-score of 0.9705, while MobileNetV2 achieved 97.02% accuracy with an F1-score of 0.9701. EMFNet integrates EfficientNetB0 and MobileNetV2 as parallel feature extractors, fuses their complementary visual and semantic representations, and passes them through dense classification layers. EMFNet achieved 98.72% accuracy, 0.9873 precision, 0.9872 recall, 0.9872 F1-score, and 0.9997 ROC-AUC. Furthermore, Grad-CAM and LIME were applied to interpret classification decisions, making EMFNet an effective, reliable, and interpretable framework for colorectal medical image classification. Index Terms—Colorectal image classification, medical image analysis, deep learning, EMFNet, hybrid feature fusion, GradCAM.

Paper ID: 441

EffViT-LCNet: A Gated EfficientNetV2M-ViT Fusion Network with Explainable AI for Lung Cancer Classification from CT Images

Roman Mia, Shahriar Kabir*, Iftekhar Ahmmed, Md. Nadim Khan, Atiar Rahman Ritu, Md Saiful Alam and Shamima Akter
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Abstract

Lung cancer is a predominant cause of cancer-related mortality globally, and early precise detection with CT scans is essential for prompt intervention. This manuscript presents a hybrid deep learning framework combining EfficientNetV2M with a Vision Transformer (ViT) through a gated feature-fusion mechanism for automated lung cancer classification. The EfficientNetV2M branch captures local texture patterns, while the ViT branch models global contextual dependencies across CT slices; a learnable gate then adaptively weights each branch's contribution before classification. Instructed and assessed on the IQ-OTH/NCCD dataset with preprocessing and enhancement to tackle limited, imbalanced data, the proposed model achieves 96.38% accuracy, 96.36% F1-score, and an AUC of 0.9985, surpassing VGG16, ResNet50V2, DenseNet121, ConvNeXtTiny, and ViT used individually. Grad-CAM visualization further confirms that the projections are influenced by clinically significant lung areas. These results indicate that the proposed EfficientNetV2M-ViT fusion framework offers an effective, interpretable approach for computer-aided lung cancer screening.

Paper ID: 471

WSV-ViT: A Weighted Soft Voting Ensemble of Regularized Vision Transformers with LIME and Grad-CAM for Accurate Cervical Cancer Diagnosis

Jihadul Islam*, Tanvir Anjum Rahat, MD Sifat and Md Abdul Masud
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Abstract

One of the most significant causes of death among women in the world is cervical cancer and proper cytological screening. While traditional manual Pap smear analysis is still very time-consuming and prone to human error. Although the outcomes are promising, DL models, especially CNNs, have not been reported to be able to capture the overall morphological context and lack transparency in decision-making, which is central in clinical applications.

To overcome the limitations, the paper presents a new framework which is a Weighted Soft Voting (WSV) Ensemble of state-of-the-art Vision Transformers (ViTs), such as ViT-B/16, Swin-T, DeiT-B, and MobileViT-v2, on the SIPaKMeD benchmark dataset. In order to avoid overfitting and improve the generalization, more sophisticated regularization methods like CutMix, Mixup, and Label Smoothing were added to the training pipeline. The ensemble weights were dynamically determined according to the validation accuracy of each. The experimental performance on the independent test set indicates that the proposed WSV Ensemble has a remarkable Test Accuracy of 96.81%, a macro F1-score of 0.9694, and a peak AUC-ROC of 0.9988, indicating absolute determinism (100% accuracy and recall) in recognizing benign Parabasal cells. Moreover, we use Explainable AI (XAI) diagnostics with Grad-CAM and LIME to visualize that the network attention is specific to clinically relevant nuclear abnormalities to avoid the black-box nature of neural networks. The strong results and transparency that can be interpreted support the potential of the framework as a highly reliable computer-aided diagnostic (CAD) tool in screening cervical cancer.

Paper ID: 545

GAD-Net: An Interpretable Deep Learning Framework for Automated Diabetic Retinopathy Screening in Smart Healthcare

¹Abrar Rakin*, ¹Obaidul Haque, ¹Md Istiak Ahamed, ²Md. Ashraful Islam, ¹Ferdous Anam Jibon and ¹Abu Bakar Rakib

¹International University of Business Agriculture and Technology

²Military Institute of Science and Technology

*Corresponding author: rakinabrar151@gmail.com

Abstract

The latest developments in AI and intelligent healthcare systems have made possible the automated detection of retinal diseases, facilitating the diagnosis of diabetic retinopathy (DR) under limited healthcare conditions. The current paper introduces GAD-Net (Gaussian Attention DenseNet), which is an interpretable deep learning model for automated classification of diabetic retinopathy using retinal fundus images. This work involves using the Gaussian filter in image processing, data augmentation through GAN to reduce class imbalance, and the improvement of the DenseNet121 network by utilizing channel attention and GeM Pooling. The experiments were performed on the APTOS 2019 Blindness Detection dataset that consists of five stages of DR. The accuracy rate of GAD-Net is 84.74% with a ROC-AUC value of 96.83%, which performs better than many other baseline CNN architectures such as ResNet50, VGG16, MobileNetV3, and even the common DenseNet121. In addition to that, Grad-CAM gives explainable predictions by indicating clinical meaningful areas in the retina. The suggested architecture shows the power of using artificial intelligence in the interpretation of retinal images for future intelligent screening processes.

TRACK 2: AGRICULTURAL, AQUACULTURE AND CROP INTELLIGENCE (AACI)

Paper ID: 183

Physics-Guided Feature Embedded BiLSTM Framework for Precision Irrigation and Crop Water Requirement Forecasting

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Abstract

A hydro-meteorological forecasting system needs to be physically dependable and should be accurate enough to be used in real time to plan irrigation operations and reduce losses while improving the sustainability of the agricultural sector. Even though the present irrigation models are physically representative and they are empirical models, there are some limits, and empirical models cannot be applied at extreme microclimatic conditions. On the other hand, there are physically unimplementable anomalies in data-driven black box models like machine learning, such as under extreme microclimatic conditions. This work proposes a lightweight physics-guided hybrid machine learning approach, termed BiLSTM (PG-FE BiLSTM), to predict daily high-resolution temperature under closed-loop irrigation decision-making and crop water requirements. In contrast to hard PDE loss functions, which implement physics, we employ basic agro-meteorological principles in a nonlinear feature space regularised by bidirectional deep recurrent attention networks. Using a thermodynamic equation, temporal cyclic codings, autoregressive thermal memory and meteorologic predictors, proxies of ET are combined together and predictions are generated with significantly decreased statistical error bounds and, at the same time, are physically admissible. The proposed framework exhibits chronological experimental results with the meteorological data of ERA5 and NASA POWER due to its high performance with an MAE of 0.1369°C, RMSE of 0.2274°C, R2 of 0.9977, and a continuous Physical Consistency (PC) score of 100% compared to baseline models such as XGBoost and LightGBM. Lastly, an irrigation decision-support layer is added (lightweight) to enable edge actuator deployment.

Paper ID: 310

A Multi-Stage Deep Learning Framework for Automated Detection and Localization of Shrimp Disease

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Abstract

This work presents an extensive Multi-Stage Deep Learning Framework to reduce the significant financial effects of White Spot Syndrome Virus (WSSV) and Black Gill (BG) disease on the \$64 billion global shrimp aquaculture sector by fixing the interpretive problems and restricted ability to diagnose current automated systems. The study develops an ordered detection pipeline starting with disease detection using a dataset of 5,498 high-resolution images. Following a comparison of five state-of-the-art architectures, ConvNeXt-Tiny proved to be the best classifier. It accomplished a test accuracy of 96.8% and a Macro F1-score of 0.97. To solve baseline weaknesses, Projected Gradient Descent (PGD) adversarial training was executed to restore robust accuracy to 55.5% against environmental disturbances. After classification of disease, a conditional disease specific U-Net++ segmentation module had been developed for precise lesion identification, giving a Dice Score of 0.8217 for Black Gill and 0.71 for WSSV, which enabled the development of a novel pixel-level Severity Staging system (Mild, Moderate, Severe) for actionable intervention. This research successfully bridges the gap between theoretical computation and dependable, practical application by combining Explainable AI (XAI) with Grad-CAM visualizations for assessing histological features and optimizing for edge-deployment efficiency. This has been achieved by both of these techniques.

Paper ID: 474

Agroclimatic Crop Yield and Production Prediction in Bangladesh Using Machine Learning and Explainable AI

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Abstract

Forecasting crop yields and production is essential to ensure food security in Bangladesh, as agricultural production here is affected by agro-climatic conditions. This study proposes a machine learning-based framework for the simultaneous prediction of crop yield and total production using the Bangladesh Agroclimatic Crop Yield dataset (2000–2024) and applies a logarithmic transformation to one of the target variables, production, for reducing skewness. Four regression models, such as eXtreme Gradient Boosting (XGBoost), Random Forest (RF), Gradient Boosting (GB), and Linear Regression (LR), are trained and evaluated using Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) metrics. Among the individual models, GB achieves the best predictive capability, and a stacking-based meta-feature approach employing out-of-fold XGBoost predictions is implemented, achieving an R^2 of 0.9117 for crop yield and 0.9820 for production. Here, we not only predicted crop yields, but also used SHAP to find out which weather and climate factors have the greatest impact on crops. The results show that the model we created is very effective for agricultural planning and future crop cultivation in Bangladesh.

Paper ID: 527

A Cloud-Integrated AIoT System for Automated Small-Scale Aquaculture Toward Urban Food Security

¹Md. Rahman*, ¹Mohammad Ali Khan, ²Anita Tasnim, ¹Md. Abdur Razzak, ¹Shaem Hasan Shakib and ¹Md. Mahmudul Hasan

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Abstract

The increasing global demand for protein has created a growing gap between nutritional needs and the production capacity of traditional agriculture. As land and water resources become more limited, small scale aquaculture is emerging as a promising approach for supporting urban food security. However, maintaining proper water quality, fish feeding and monitoring fish health can be challenging, especially those without the knowledge about aquaculture. To address this issue, this study proposed a cloud-integrated Artificial Intelligence and Internet of Things (AIoT) system designed to automate small-scale aquaculture operations. The proposed system utilizes a publicly available dataset Freshwater Fish Disease Aquaculture in South Asia and integrates an ESP32-based IoT module for automating and cloud connectivity for remote monitoring and management. In addition, a pretrained transfer learning model is used to identify the common fish pathogens, enabling early disease detection. To investigate the efficiency of the proposed system several experiment was done. The study is further extended with the analysis of System Usability Scale (SUS) to track the user satisfaction and gained the SUS score of 84%. The experimental results also demonstrate that the proposed model achieved the accuracy of 98.85%. The proposed system provides a scalable solution for small-scale aquaculture reducing manual effort, improve fish health management, and support the global food safety.

TRACK 3: AUTONOMOUS AERIAL AND FIELD ROBOTICS FOR SEARCH AND RESCUE (AAFR)

Paper ID: 128

Benchmarking Vision-Language Models for Real-Time UAV Obstacle Avoidance in Controlled Simulation

Samiha Tarannum Noor, Abyeadd Zarif Hassan and Dr Md Mehedi Hasan*

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Abstract

Vision-Language Models (VLMs) are increasingly being explored as high-level reasoning engines for autonomous unmanned aerial vehicle (UAV) navigation. This paper presents a simulation-based benchmark evaluating six contemporary VLMs for monocular UAV obstacle avoidance in a high-fidelity Unity environment integrated with Robot Operating System 2 (ROS 2). The benchmark compares three proprietary cloud-based models (GPT-4o, GPT-5, and Gemini 2.0 Flash) with three locally deployed open-source models (LLaVA-7B, Llama3.2-Vision, and Qwen2.5VL-3B) across nine factorially designed scenes varying in obstacle density, lighting, weather, and obstacle behavior, resulting in 265 evaluation episodes and 478 shared decision points. Each model receives identical multimodal input comprising an RGB frame, a DepthAnything-V2 depth map, and a structured prompt based on a 3 times spatial grid. Performance is assessed using seven metrics: obstacle avoidance accuracy, false positive rate, path efficiency, safe-action alignment, collision rate, decision latency, and trajectory complexity. GPT-5 achieves the best overall performance with 73.3% accuracy, 7.3% false positive rate, and 1.34 s decision latency, followed by GPT-4o as a consistent second-tier model. Locally deployed open-source models exhibit systematic decision failures and high per-decision latencies ranging from 143 s to 366 s, making them unsuitable for real-time closed-loop navigation on consumer-grade hardware. These results provide a reproducible basis for selecting and screening VLMs for autonomous aerial navigation.

Paper ID: 255

Quantitative Evaluation of Embedded ROS2 Navigation and Sensor Fusion for a Differential Drive Autonomous Mobile Robot

Md. Abu-Talha Roni* and Alok Kumar Paul

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Abstract

Low-cost autonomous mobile robots frequently fail to translate simulation-level navigation performance into reliable real-world deployment because odometry drift, heading error, embedded resource limits, and planner behavior are often evaluated separately or only qualitatively. A ROS2 Jazzy based simulation-to-real evaluation framework is developed for a low cost differential-drive indoor AMR. Unlike prior low-cost AMR studies that mainly report platform design or navigation success, this work quantitatively evaluates both physical robot performance and simulation-only planner behavior while keeping the two experimental domains separate. Real-world experiments achieved a combined linear odometry error of 5.5 cm, and EKF fusion of wheel odometry with Mahony-filtered IMU yaw reduced rotational error by 71.8%. The robot completed six sequential Nav2 goals over a 7.0 m planned route in 28.3 s while maintaining 43.1% CPU and 12.4% RAM usage on the embedded platform. Navigation was observed to be more stable in larger arenas, while confined spaces increased scan and pose clutter. In a separate simulation benchmark, NavfnDijkstra, NavfnAStar, Smac2D, and ThetaStar were evaluated across three occupancy-grid maps using path and execution metrics. Smac2D achieved the best overall balance, with the highest average coverage (97.81%) and lowest path-pose density (18.92 poses/m), while ThetaStar produced the shortest paths and NavfnAStar the lowest execution time.

Paper ID: 262

FedSwarm-Attn: Federated Attention-Based Deep Reinforcement Learning for Energy-Efficient Multi-Robot Swarm Navigation in IoT Environments

Musavvir Haque*, Masud Rana, Saklain Mustak, Mirza Mahadir Hasan and Jahedul Islam

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Abstract

Swarm robotics in Internet-of-Things (IoT) environments must balance reliable goal-directed navigation, energy-efficient execution, and privacy-preserving distributed learning. Multi-agent reinforcement learning (MARL) improves coordination but scales poorly and can compromise data privacy, while fully decentralised approaches protect privacy at the cost of slower convergence and no knowledge-sharing across agents. We introduce FedSwarm-Attn, a framework that combines federated deep Q-network (DQN) learning with a four-head attention communication module. Each robot trains a local DQN; a shared encoder produces latent state embeddings, which multi-head attention aggregates into context-aware action representations. Periodic federated averaging (FedAvg) synchronises model parameters across robots without sharing raw sensor data. Over 350 training episodes and 5 random seeds in a 20x20 grid with 4 robots and 8 IoT sensors, FedSwarm-Attn reaches a 99.6% goal success rate, converges within 102 episodes, and cuts the energy cost per successful navigation by up to 32% compared with non-federated baselines. These gains are supported by paired t-tests ($p < 0.0001$), and the attention mechanism further reduces variation across seeds. The comparisons against non-federated baselines also show large effect sizes (Cohen's $d = 9.13$ and 4.77), not just statistical significance. The results indicate that federated aggregation and attention-based coordination are complementary: federation drives the convergence and success-rate gains, while attention improves consistency across seeds, together supporting scalable, energy-efficient, and privacy-preserving swarm intelligence for IoT-connected robots.

Paper ID: 387

A Collaborative UAV-UGV Heterogeneous System for Enhanced Search And Rescue In Complex Disaster Environments

Md. Nafis Mahmud*, Atia Shahnaz Ipa and Zibon Nesa Zim

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Abstract

In the immediate aftermath of structural collapses, rapid victim localization is severely bottlenecked by the physical limitations of single-platform robotics: Unmanned Aerial Vehicles (UAVs) lack indoor void penetration capabilities, while Unmanned Ground Vehicles (UGVs) suffer from slow transit speed and vertical obstacle barriers. This paper presents a heterogeneous, collaborative Master-Slave UAV-UGV system engineered to bridge this operational gap through targeted aerial transit and localized ground exploration. A high-payload Pixhawk controlled quadcopter navigates autonomously to designated entry waypoints and deploys a compact ESP32-driven 4WD rover using a custom 3D-printed electromagnetic docking mechanism triggered via auxiliary PWM channels. Upon release, the UGV executes localized reconnaissance while streaming real-time video telemetry over Wi-Fi and delivering vital emergency aid packages. Experimental evaluations demonstrate a 100% payload release success rate, a mean autonomous landing displacement error of 57.5 cm, an operational ground endurance of up to 48 minutes under a maximum 250 g cargo load, a communication latency under 120 ms, and an average total mission execution time of 4.2 minutes per target zone, providing a low-cost, high-reliability blueprint for urban search and rescue operations.

Paper ID: 449

HEXAGUARD: An IoT-Based Search and Rescue Rover for Hazard Monitoring and Emergency Response¹Md Zahid Hasan Zihad*, ²Imtiaz Ahmed, ²Md Toha Bin Asad Dip and ²Dr. Md. Mahfuzur Rahman¹Darsana Government College²Jashore University of Science and Technology

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Abstract

Abstract— Search and rescue missions in dangerous areas are often challenging and risky for emergency teams. Using a rescue rover can lower these risks by letting operators monitor the scene and carry out key tasks from a safe distance. This paper introduces HEXAGUARD, an IoT-based search and rescue rover designed for hazard monitoring and emergency response. The rover features a hybrid RF and Wi-Fi communication system, two ESP32 controllers, several gas sensors, two IP cameras, a 6-DOF robotic arm, and a water-based fire suppression system. Operators can use a web dashboard built with ESPHome and Home Assistant to see live sensor readings and camera feeds in one place. The prototype was tested outdoors to check its monitoring, movement, object handling, and fire suppression abilities. The results show that HEXAGUARD can support rescue teams by improving their awareness of the situation and reducing the need for people to enter dangerous zones. The system can also be improved in the future, such as by adding autonomous navigation or AI-based hazard detection. Keywords— IoT, Search and Rescue Rover, Hazard Monitoring, Emergency Response, ESP32, Home Assistant, Robotic Arm, Fire Suppression

TRACK 4: BIOSIGNAL PROCESSING, NEURAL INTERFACES AND WEARABLE HEALTH SYSTEMS (BNWH)

Paper ID: 263

DS-SleepNet: An Explainable Dual-Stream Meta Fusion Architecture for Sleep Disorder Diagnosis

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Abstract

Accurate sleep disorder diagnosis is critical for proactive healthcare to effectively mitigate severe systemic health consequences. However, traditional polysomnography remains highly resource-intensive, clinically restrictive, and largely inaccessible for widespread routine screening. While automated computational approaches offer scalable alternatives, existing diagnostic frameworks predominantly rely on single-stream models. These standalone architectures frequently encounter performance plateaus by struggling to balance latent feature extraction with structured rule optimization. Furthermore, they severely lack the explicit diagnostic transparency mandated for clinical medical adoption. To address this critical research gap, the proposed study introduces DS-SleepNet, a novel dual-stream diagnostic architecture. The methodology evaluates multidimensional clinical health records utilizing an optimized feature engineering pipeline. The framework parallelly deploys a Deep Neural Network and an eXtreme Gradient Boosting classifier. A regularized logistic regression meta learner dynamically synthesizes their predictive probabilities. The proposed architecture achieved a highly reliable classification accuracy of 97.33% alongside an Area Under the Curve of 0.991. For clinical implications, the framework integrates SHAP and LIME methodologies to ensure diagnostic transparency. This explicitly aligns automated decisions with established medical knowledge to foster practitioner trust. Finally, comprehensive ablation analysis confirms the systemic robustness of the framework, demonstrating that the meta fusion strategy consistently mitigates individual algorithmic bias.

Paper ID: 398

A Wearable Smart Diabetic Shoe with Hybrid Adaptive Multi-Index Classification for Early Detection and Prevention of Diabetic Foot Ulcers

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Abstract

Diabetic foot ulcer (DFU) is a common complication of diabetes that may lead to infection, hospitalization, and lower-limb amputation if not detected early. This study presents a wearable smart diabetic shoe for continuous monitoring and early risk assessment of DFU. The proposed system integrates force-sensitive resistor sensors, temperature sensors, and vibration modules to measure plantar pressure, foot temperature, and vibration sensation. A Hybrid Adaptive Multi-Index DFU Classifier (HAMDC) combines these measurements with age-related risk to generate a Composite Diabetic Foot Risk Index (DFRI). According to the estimated risk level, the system automatically adjusts vibration therapy to support preventive care. Evaluation using data from 345 diabetic patients achieved an overall classification accuracy of 91.01%, indicating that the proposed approach offers an effective and practical solution for continuous diabetic foot monitoring.

Paper ID: 500

Explainable Low Back Pain Classification Using Multifidus sEMG and Multi-Domain Feature Selection

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Abstract

Low back pain is the leading global cause of years lived with disability, yet objective neuromuscular assessment for LBP remains an open challenge. This paper proposes a multifidus-only binary LBP classification framework using only two bilateral sEMG channels extracted from 22 subjects. Total 49-features consisting time-domain (TD), frequency-domain (FD), and time-frequency domain (TFD) representations is evaluated across five epoch lengths (300ms–2.5s) under three metaheuristic feature selection strategies e.g., Grey Wolf Optimizer (GWO), Particle Swarm Optimization (PSO), and Boruta combined with nine classifier and stratified 5-fold validation. SVM with Boruta on 2.5s extension-phase epochs achieves 95% accuracy (precision: 0.90, recall: 1.00, F1: 0.95, AUC: 0.995) at 2.5s windows. Additionally, explainable technique e.g., SHAP is used to explain the model decisions. SHAP analysis consistently identifies LogEnergy, DASDV, SSC, Kurtosis, and Hjorth Complexity as physiologically interpretable, cross window stable biomarkers. The compact pipeline is suitable for wearable and edge deployment LBP screening and potential applications in biomedical research and healthcare.

Paper ID: 502

Adapting a Lightweight Time-Series Foundation Model for sEMG Hand Gesture Classification

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Abstract

Surface electromyography (sEMG)-based Hand Gesture Recognition (HGR) holds significant promise for human–computer interaction, but is limited by labeled-data requirements and substantial inter-subject variability. This study investigates the adaptation of Mantis, a lightweight, pretrained time-series classifier, as a data-efficient alternative for sEMG based gesture classification. Frozen-backbone (head-only) and full fine-tuning are compared across window lengths and overlap settings, followed by a six-fold subject-wise cross-validation. With a 50% window overlap, full fine-tuning achieved $92.12 \pm 0.65\%$ accuracy, exceeding head-only tuning by 5.94 percentage points. Without overlap, the two strategies produced nearly identical mean accuracies ($85.09\% \pm 1.69\%$ and $84.73\% \pm 0.58\%$, respectively). In subject-wise evaluation, head-only and full fine-tuning achieved 79.54% and 78.47%, respectively. These results show that full fine-tuning improves segment-level performance when overlapping windows are used, whereas freezing the backbone remains a competitive and computationally simpler strategy for unseen-subject evaluation.

Paper ID: 505

NeuroFocal-Net: A Multi-Scale CNN–Vision Transformer Hybrid with Focal Attention for Robust Five-Class EEG Seizure Detection

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Abstract

We present NeuroFocal-Net, a compact CNN– Vision Transformer hybrid for five-class brain-state classification on the Bonn University EEG corpus, and, more importantly, a corrected evaluation of it. An earlier version of this work reported 98.67 % accuracy on a partition that did not add up: Mixup was applied to all 500 segments before the split, and the validation set was drawn from the same 500 segments that fed training. Reproducing that protocol returns 99.47 % test accuracy, 16.27 percentage points above the same model trained and tested on a partition of the corpus into 350 training, 75 validation and 75 test segments with no overlap of any kind. On the corrected partition, over 10 independent stratified splits, NeuroFocal-Net reaches $83.87 \% \pm 5.74$, pooling to 83.87% over 750 test decisions (95 % Wilson interval 81.06–86.33 %). Stratified five-fold cross- validation gives $86.80 \% \pm 0.45$, and a contiguous-block partition, which keeps temporally adjacent segments out of different splits, gives $83.40 \% \pm 6.35$. Against four baselines re-trained under the same protocol and budget, the hold-out differences are significant against DeepConvNet, EEGNet, ShallowConvNet and not against BaselineViT; under the block partition, however, NeuroFocal-Net loses only -0.47 pp where the baselines lose -4.60 to -12.00 pp, and its +5.80 pp margin over BaselineViT there is significant ($p = 0.004$). Because the public Bonn release carries no subject identifiers, none of these protocols is subject-independent, and we make no claim of clinical generalisation. We publish the corrected numbers rather than the earlier ones, and argue that segment-level accuracy on Bonn has stopped discriminating usefully between methods.

Paper ID: 512

Adaptive Topology-Switched Wireless Power Transfer for Implantable Neural Stimulators: SAR-Constrained Optimisation and Misalignment Robustness Analysis

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Abstract

This work presents an adaptive, topology-reconfigurable inductive wireless power transfer (WPT) system for deep-brain neural stimulator implants at 13.56 MHz. Rather than fixing a single compensation network across the whole depth range, the architecture switches between Series-Series (SS) and LCC-S compensation according to real-time coupling-coefficient (k) estimates from back-channel telemetry so that each coupling regime is served by the topology appropriate to it: SS in the weakly coupled, bifurcation-free deep-implant regime ($k < k_{crit}$) and LCC-S in the over-coupled regime ($k > k_{crit}$) where frequency-splitting bifurcation would otherwise collapse SS efficiency. A MATLAB simulation suite – modified-Wheeler coil synthesis, elliptic integral mutual inductance computation, quasi-static SAR modelling and Pareto multi-objective optimisation – gives $k_{crit} = 1/\sqrt{Q_1 Q_2} = 0.0278$ against a nominal 30 mm DBS coupling of $k = 0.0125$. Link efficiency at any k is explicitly bounded by the fundamental two-coil limit $\eta_{max} = U^2 / (1 + \sqrt{1 + U^2})^2$, $U = k\sqrt{Q_1 Q_2}$; simulated PTE is reported consistent with this bound at $\approx 4.6\%$ at the nominal 30 mm weak-coupling point, rising to $\approx 50.6\%$ at $k = 0.08$ (shallow/over-coupled targets), where LCC-S's bifurcation immunity not raw efficiency – is the source of benefit. Peak brain SAR is 0.325 W/kg, screened against the IEEE C95.1-2019 local-exposure limit (a screening indicator only; formal implant certification requires ISO 14708-3 thermalise testing, not performed here). A Class-E driver delivers $P_{out} = 500\text{mW}$ with $C_{shunt} = 12.97\text{ pF}$ and a charge-balanced biphasic neural load draws 5.20 mW at $V_{dd} = 3.584 \pm 0.059\text{ V}$, consistent with a 124.8 mWh/24 h implant-side energy budget. A separate Pareto sweep identifies a shallower-coupling design point ($k \approx 0.28$, outside the 30 mm DBS envelope) attaining 82.1% PTE at 38.4 mW PDL. All results are simulation-only; hardware validation is identified as the primary limitation and immediate next step.

TRACK 5: POWER ELECTRONICS, RENEWABLE ENERGY AND GRID AUTOMATION (PREG)

Paper ID: 232

Explainable Hybrid ARIMA–ML Forecasting of Bangladesh’s GDP: Sectoral Energy Consumption and Carbon Emission Dynamics
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Abstract

Macroeconomic forecasting in developing economies such as Bangladesh is complicated by rapid structural transformation, rising sectoral energy demand, and sector-specific carbon dynamics that conventional linear stochastic models cannot represent. This paper develops an explainable Hybrid Residual Learning Ensemble (Hybrid-Stack) for gross domestic product (GDP) forecasting. ARIMA models the linear autoregressive dynamics; an ensemble of four machine-learning (ML) regressors, XGBoost, Random Forest, Gradient Boosting (GBM), and Support Vector Regression (SVR) is then trained on the ARIMA residuals to capture nonlinear structure. SHapley Additive exPlanations (SHAP) interprets the ML correction layer post-hoc, preserving econometric transparency while allowing flexible nonlinear modelling. The framework is evaluated under an expanding-window recursive walk-forward protocol and benchmarked against econometric and standalone ML baselines; pairwise Diebold–Mariano tests assess forecast-accuracy differences. Predictors include electric power consumption, urban population share, and sector-specific CO₂ emissions, proxying energy demand, structural transformation, and production intensity. The Hybrid-Stack achieves the best overall performance (MAPE=7.60%, PBIAS≈0.0024%), outperforming the ARIMA(3,1,2) benchmark (MAPE=8.01%) and every standalone ML baseline. The second-best standalone model, SVR (MAPE=7.86%), matches the Stack in accuracy but carries substantially larger systematic overestimation (PBIAS=−3.09%). The framework delivers a more balanced and reliable GDP forecast by combining linear economic dynamics with nonlinear environmental and structural information.

Paper ID: 259

Continuation-Accelerated Multi-Start Newton Solution of Selective Harmonic Elimination for a Seven-Level Inverter

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Abstract

Selective harmonic elimination pulse-width modulation (SHE-PWM) for a seven-level ($P=3$) staircase inverter requires the solution of a system of transcendental equations whose solution set is multivalued and, over portions of the modulation range, empty. In this work, the three-angle formulation that cancels the third and fifth harmonics subject to the fundamental constraint is solved exactly across the full modulation-index range $M \in [0.60, 1.00]$, sampled at a step of 0.00025 (1601 operating points), using a Continuation-Accelerated Multi-Start Newton (CAMS-Newton) procedure with an analytic Jacobian. Every accepted triplet satisfies the ordered constraint $0 < \alpha_1 < \alpha_2 < \alpha_3 < 90^\circ$ and a residual below 10^{-11} , and the largest residual observed over all retained operating points is 1.33×10^{-15} . The feasible region is shown to be genuinely discontinuous, existing only in $M \in [0.60000, 0.69050]$ and $M \in [0.80275, 0.81850]$, with no ordered solution elsewhere. The lowest total harmonic distortion attained by the eliminating solution, evaluated with an exact closed-form expression, is 11.90% at $M = 0.81350$ with switching angles

(8.62°, 28.77°, 54.89°). The optimal solution is validated in the MATLAB/Simulink environment on a reduced component-count multilevel inverter, where the third and fifth harmonics are suppressed and the simulated distortion agrees with the computed value. The complete angle-THD table and the verification routine are released as reproducible artifacts.

Paper ID: 260

A Constraint-Aware Neural Warm-Start with Newton Refinement for Selective Harmonic Elimination PWM: A Reproducible Study on the Three-Angle Fifth-Seventh Elimination Problem

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Abstract

Selective Harmonic Elimination Pulse-Width Modulation (SHE-PWM) requires solving a nonconvex system of transcendental equations for the converter switching angles at every modulation index; the difficulty lies in the initial guess, since a numerical solver converges quickly only when warm-started inside the correct basin. This paper presents a reproducible, no-fabrication study of a hybrid neural-numerical solver for the canonical three-angle problem that regulates the fundamental and eliminates the fifth and seventh harmonics. Ground truth is generated by a bounded trust-region least-squares solver under an explicit minimum-total-harmonic distortion branch policy, and every label is retained only after an ordering, bounds and equation-residual audit at a tolerance of 10^{-9} . A constraint-aware multilayer perceptron is proposed whose output parameterization guarantees the strict ordering $0 < \alpha_1 < \alpha_2 < \alpha_3 < 90^\circ$ for all weights, so feasibility is intrinsic rather than restored by sorting; the network is trained with a loss that couples angle error to the exact equation residuals. A three-step Newton refinement then polishes each prediction to solver precision, and a short local fallback resolves the known minimum-THD branch crossing. Across five seeds the network alone attains a mean absolute angle error of $0.307(216)^\circ$; after refinement the maximum residual over the feasible window is 2.0×10^{-7} with 100% convergence, the targeted harmonics reach machine precision, and inference is $9.4\times$ faster than a robust cold-start solver.

Paper ID: 305

A Venturi Wind Concentrator for Enhancing 16 kW PMSG Turbine Output and Capacity Factor at Low Wind Sites in Bangladesh

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Abstract

Along coastal Bangladesh the annual mean wind speed falls between 4 and 6 m/s, keeping conventional turbines below their rated point and the capacity factor low. This study addresses the problem with a venturi wind concentrator (VWC), a converging duct that seats the rotor at its throat, where continuity forces the local wind speed upward. Contracting the throat to 40 percent below the inlet area accelerates the flow by a factor of 1.67, and since turbine power scales with the cube of speed, the ideal power enhancement factor (PEF) reaches about 4.6. The concentrator is embedded in a MATLAB/Simulink model of a 16-kW permanent magnet synchronous generator (PMSG) feeding a three-phase load, with ambient speed swept from 1 to 6 m/s so that the reported quantity is delivered electrical output, not throat velocity. At the 6 m/s design point the VWC lifts the output from 3.6 kW to 15.6 kW, a PEF of about 4.3, and drives the capacity factor from 22 to 98 percent, with larger relative gains below the cut in threshold. A random forest surrogate trained on the area ratio reproduces the analytical PEF law with a holdout coefficient of determination of 0.99 and maps any target enhancement onto the required throat

contraction. Because the metric is a capacity factor and not an efficiency, the near rated output implies no breach of the Betz limit, and the VWC emerges as a low cost, retrofittable option for weak wind sites in Bangladesh.

Paper ID: 362

Fault-Ride-Through and Power Quality Enhancement of a Grid-Connected PV/PMSG-Wind Hybrid System Using DC-Link Chopper and LCL-Filtered Three-Level VSC

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Abstract

The paper describes a grid-connected hybrid renewable energy system consisting of 100 kW PV array and permanent magnet synchronous generator (PMSG) wind energy conversion system that can be used to improve fault ride-through performance and enhance power quality. The system consists of PV boost converter with MPPT control, wind side rectifier and boost converter, common 500V DC link, DC link chopper with braking resistor, three-level voltage source converter, LCL filter, step-up transformer, grid interface. In normal operation, the PV and wind sources are connected to the common DC-link and the grid-side converter sends the power generated to the utility grid with the power factor very close to unity. When there are grid faults, the reduced power transfer from the grid results in excess renewable energy battery charging the DC-link capacitor which causes overvoltage stress and poor grid recovery performance. This is done by using a DC-link chopper, which will be turned on if the voltage at the DC-link exceeds its maximum value, and will dissipate excess energy in the braking resistor. The results demonstrated by the simulation indicate that the proposed chopper control can successfully lower the peak voltage of the DC-link from 688 V to 524 V and the voltage overshoot from 37.6% to 4.8%. The settling time is also enhanced from 0.31 s to 0.06 s. Moreover, it can lower the THD of the grid current to 1.42% and the power factor to 0.998 for the three-level VSC which is filtered by LCL. The results show that the proposed configuration successfully improves the fault ride-through capability, DC-link stability, and power quality in the grid.

Paper ID: 447

A Multi-Paradigm Meta-Ensemble Framework for Short-Term Electricity Load Forecasting

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Abstract

Short-term electricity load forecasting (STLF) is one of the most important aspects of reliable operation of the power system, economic dispatch, and energy management. Nevertheless, the electricity demand is a complicated model due to its nonlinearity, time dependence, seasonal variations, and exogenous factors. In this paper, a multi-paradigm meta ensemble framework is proposed, which can combine heterogeneous forecasting methods, including the deep learning model CNN-LSTM, the statistical-machine learning hybrid model ARIMA-Random Forest, and gradient boosting models LightGBM and XGBoost. The framework integrates these complementary paradigms by stacking models together with a meta-learning approach and adds a periodic model refresh (PMR) mechanism to enable model adaptation based on new observations. True hourly electricity load data was utilized in the experiments, and true values were assessed using the MAE, RMSE, and MAPE. The forecasting accuracy was analyzed on a weekly basis, the daily peak demand was predicted, monthly performance trends were analyzed, and the performance was evaluated on a seasonal basis through comparative analyses. The experimental results show that the hybrid LightGBM XGBoost model has the lowest forecasting errors, and the proposed framework shows that the integration of different forecasting paradigms for short-term load forecasting and adaptive model management is effective. The proposed structure leads to a scalable base on which to create solid forecasting systems for the modern smart grid application.

TRACK 6: CYBERSECURITY, TRUSTWORTHY AI AND MEDIA FORENSICS (CTMF)

Paper ID: 224

SpatSeqFake: A Hybrid Spatial-Sequential Framework for Cross-Dataset Deepfake Image Detection¹Sadia Afroz Oishi, ¹Ariful Islam, ¹Md Sarwar Hosain*, ²Ohidujjaman, ¹Abul Fazal Mohammad Zainul Abadin and ¹Liton Chandra Paul¹Pabna University of Science and Technology²United International University

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Abstract

Deepfake detection is a major challenge in digital media forensics due to the rapid progress of generative adversarial networks and diffusion-based image synthesis. Existing deep learning methods often struggle to generalize across diverse manipulation settings and compressed images. This paper proposes a hybrid deepfake detection framework that combines EfficientNet-B4 for spatial feature extraction, a spatial attention module for artifact localization, and a Bidirectional Long Short-Term Memory (BiLSTM) network for modeling contextual dependencies among spatial feature patches. The proposed architecture captures both local manipulation artifacts and broader inter-patch relationships. Experiments on the CIFAKE and OpenForensics Wild datasets achieved an accuracy of 98.73% and an AUC-ROC of 99.89% on CIFAKE, and an accuracy of 98.85% and an AUC-ROC of 99.87% on OpenForensics Wild. Results under augmented and compressed settings indicate improved robustness and generalization across varying image conditions.

Paper ID: 240

Towards Intelligent SOC Automation: An Open-Source Analyst-in-the-Loop Framework with Dual-LLM Alert Triage and MITRE ATT&CK RAG¹Saif Ahmed Shuvo, ¹Sarfin Uddin Bhuiyan Anik, ²Md Saniul Basir Saz, ²Md Tuhin Reza, ²Dr. Md. Kamrul Islam, ²Monishanker Halder, ²Sk. Shalauddin Kabir and ²Dr. Mohammad Nowsin Sheikh*¹Faridpur Engineering College²Jashore University of Science and Technology

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Abstract

Enterprise Security Operations Centers (SOCs) often receive more than 10,000 security alerts each day, and many of them are benign. However, these alerts still require manual triage, which is time-consuming and increases the workload of security analysts. This paper proposes an open-source Analyst-in-the-Loop (AITL) SOC automation framework to support the triage of high- and critical-severity security alerts. The framework combines Wazuh for security monitoring, Suricata for network intrusion detection, n8n for workflow automation, TheHive for case management, and Cortex for threat enrichment in a six-stage alert triage process. It uses two large language models, with Groq Llama 3.3-70B as the primary model and Google Gemini 2.5 Flash-Lite as a backup if the primary model is unavailable. The framework also uses the MITRE ATT&CK knowledge base together with threat intelligence from AbuseIPDB, VirusTotal, AlienVault OTX and Hybrid Analysis to generate clear and structured triage reports for security analysts. The proposed framework provides a fully open-source approach to SOC automation that supports consistent and explainable alert analysis while keeping a human analyst involved in the final decision.

Paper ID: 385

Aligning the Unseen: MMD-Regularized Autoencoders for Cross-Device IoT Botnet Detection with Statistically Significant False Positive Reduction

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Abstract

The rapid proliferation of Internet of Things (IoT) devices has exposed critical infrastructure to sophisticated botnet attacks. Intrusion detection systems trained on one device type consistently fail when deployed on different hardware due to distribution shift. While pooling benign traffic across devices improves generalization, it still yields elevated false positive rates (FPR) because reconstruction error distributions remain misaligned across devices. We propose a lightweight domain-alignment method: a Maximum Mean Discrepancy (MMD)-regularized autoencoder trained on pooled benign data from four heterogeneous IoT devices, in which the alignment term explicitly minimizes the pairwise distributional discrepancy between every pair of devices' latent representations within each training batch. On the N-BalIoT dataset, our MMD-aligned autoencoder reduces the average cross-device FPR from 0.070 (pooled autoencoder) to 0.045---a 36% relative reduction---with a modest F1 change (0.982 to 0.955). A paired t-test across three random seeds confirms the FPR reduction is statistically significant ($t=90.98$, $p=0.0001$), while the F1 change is not ($t=0.73$, $p=0.5419$). We provide a full 4×4 pairwise transfer matrix, a per-attack-type breakdown, a lambda sensitivity analysis using the proposed alignment mechanism itself, and UMAP visualization of the latent space before and after alignment. The method adds no inference-time overhead and stores only a single global threshold, demonstrating that simple latent alignment can materially improve the practicality of IoT botnet detection without adversarial training or complex ensembles.

Paper ID: 437

Dialect-Aware Real vs. Telephone-Processed Voice Classification Using AASIST-Lite for Bangladeshi Regional Speech

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Abstract

Voice-based biometric systems face growing risks from advanced spoofing and voice manipulation, especially when differentiating genuine human speech from telephone-processed or synthetic audio. Existing anti-spoofing models often ignore regional linguistic and dialectal variations, causing reduced performance in diverse language settings. In Bangladesh, distinct phonetic differences across regional dialects make reliable voice identification even more challenging. To address this, we propose a dialect-aware voice anti-spoofing framework designed to distinguish real and telephone-processed speech while accounting for regional linguistic nuances. The system is built on the AASIST-Lite architecture, combining 2D CNNs for spectro-temporal feature extraction, Transformer Encoders for long-range temporal modeling, and an Attentive Pooling mechanism for highlighting critical acoustic cues. The model was trained on a 30-class dataset covering 15 regions of Bangladesh (e.g., Barishal/Barishal Telephone, Chittagong/Chittagong Telephone, Pabna/Pabna Telephone), including both real and telephone-processed versions of each dialect. Experimental evaluation shows a 95.87% validation accuracy, with most classes scoring above 0.95 in precision, recall, and F1, and some achieving 1.00, demonstrating strong robustness. Overall, this work's main contribution is the effective application of a dialect-aware strategy for voice anti-spoofing and the large-scale validation of AASIST-Lite on a regional Bangladeshi voice dataset, paving the way for more secure and inclusive

authentication and forensic systems. Index Terms—Voice Anti-spoofing, Regional Dialects, AASIST- Lite, Deep Learning, Speech Classification, Real vs. Telephone- Processed Voice.

Paper ID: 452

Compact and Calibrated: Multi-Component Knowledge Distillation for Lightweight LLM Jailbreak Prompt Detection

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Abstract

Guardrail classifiers filter unsafe or jailbreak-style prompts before LLM response generation, but face a persistent accuracy–efficiency trade-off: accurate open-weight encoders are too large for low-latency moderation, while compact cross entropy-trained encoders sacrifice accuracy and calibration for speed. Prior distillation methods supervise a single signal in isolation and rarely target sub-40M-parameter models with an explicit calibration objective, leaving open whether small models can match larger teachers on both discrimination and calibration. We propose a six-component distillation objective jointly supervising output, representation, and calibration-level signals: Hinton soft-label distillation, training-time Brier calibration, decoupled supervised contrastive learning, FitNets feature matching, relational distillation, and multi-label harm-category distillation. By distilling a bge-large-en-v1.5 teacher (336.6M parameters) into a 33.801M-parameter student (9.96× more compact, 1.87× faster at batch size one), we obtain a model that attains ROC-AUC 0.9050 and PR-AUC 0.9226 on held out BeaverTails (n = 7,766), surpassing the teacher (ROC AUC 0.8994, PR-AUC 0.9184) and exceeding a same-size non distilled baseline by 1.91 ROC-AUC and 1.39 PR-AUC points, with under 1.1% extra parameters. We further report two model-agnostic findings: all trained models show a consistent V-shaped reliability diagram and an ECE floor of approximately 0.29~0.31 resistant to training-time and post-hoc calibration, and the teacher’s embedding topology does not visibly transfer to the student despite explicit alignment losses, even as classification performance does.

Paper ID: 515

Privacy Preserving Federated Learning for Early Student Dropout Prediction: Combining FedProx, Differential Privacy and Institutional Non-IID Partitioning

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Abstract

The early identification of at-risk students faces barriers related to the protection of privacy and non-IID data in institutions. In this paper, we introduce a DP-FL pipeline aimed at predicting early dropout using three techniques: (i) FedProx proximal term technique for client drift; (ii) Opacus-based DP SGD for formal privacy and (iii) realistic non-IID partitioning in which clients are clustered based on real-life partitions (course modules) as opposed to synthetic splits. In this regard, using the Open University Learning Analytics Dataset (OULAD), we partitioned our dataset based on four biggest modules per student and learned a tabular MLP model using centralized, vanilla and DP-federated setups ($\epsilon \in \{1.0, 3.0, 8.0, \infty\}$). Our results show that formal privacy guarantees are attainable at a strict budget ($\epsilon = 1.0$) with only a marginal F1-score reduction relative to the non-private federated ceiling, while the advantage of FedProx over FedAvg under differential privacy remains small and requires multi-seed validation before it can be claimed as a robust effect. Herein, we discuss the detailed protocol of our experiment, which involved identifying and resolving two implementation issues (a problem with the Opacus noise calibration in federated clients’ ephemeral initialization and the bias in fixed threshold evaluation in the DP-SGD gradient clipping scheme) that influence the reported utility and are applicable to other DP-FL researchers using simulators like Flower.

TRACK 7: EXPLAINABLE CLINICAL PREDICTION AND ASSISTIVE HEALTH MONITORING (ECAH)

Paper ID: 1

NightCare-VideoMAE: A Framework for Low-Light Patient Activity Recognition

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Abstract

Patient Activity Recognition (PAR) plays a crucial role in healthcare environments, particularly in low-light conditions, to enhance patient safety and care. Patient activities such as drinking, eating, falling, sitting, standing, and walking provide useful information about daily behavior, mobility, posture changes, and emergency risks. However, most existing video-based human activity recognition methods are developed using RGB videos captured under normal lighting conditions, while low-light patient monitoring is not well studied. Moreover, real low-light patient activity datasets are limited, which makes it difficult to train robust models. To overcome these challenges, this research introduces a GAN-assisted Video Masked Autoencoder (VideoMAE) framework for low-light patient activity recognition. Six patient-related activities (drink, eat, fall floor, sit, stand, and walk) are selected from the HMDB51 dataset. The LLVIP and MSRS visible-infrared datasets are used to train CycleGAN and IRCycleGAN for visual domain translation, and the trained generators produce low-light and infrared-like activity domains. A Kinetics-pretrained VideoMAE model is fine-tuned with test-time augmentation and weighted dual-seed ensemble fusion. This framework performs 93.23%, 92.24%, and 91.84% accuracy on Original HMDB51, CycleGAN-HMDB51 and IRCycleGAN-HMDB51 respectively. On a self-captured real low-light medical-room dataset the Original HMDB51 model achieves only 50.00% accuracy, while the translated-domain models achieve 97.44% and 94.23% accuracy. The results demonstrate the ability of GAN-based domain translation to enhance patient activity recognition results in real low-light scenarios.

Paper ID: 326

A Comparative Study of Federated XGBoost for Stroke Prediction under IID and Non-IID Healthcare Data Distributions

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Abstract

Across the world, stroke continues to rank among major causes of persistent physical impairment and death, highlighting the significance of early risk prediction in healthcare. Despite machine learning being considerably promising for stroke prediction, healthcare institutions store patient data individually, and privacy policies constrain the direct sharing of information. Federated Learning (FL) addresses this challenge through training models collaboratively without exchanging raw data. This study utilized a public dataset to build a centralized Extreme Gradient Boosting (XGBoost) baseline model after preprocessing. Then, the scaled training dataset was partitioned into three virtual hospitals under both Independent and Identically Distributed (IID) and Non-IID settings. The latter was generated using a Dirichlet distribution. A total of five experimental setups were evaluated by applying two federated approaches: Parallel bagging and sequential cyclic training. To prioritize the identification of stroke cases, a recall-constrained threshold optimization was employed during model selection. An Area Under the Curve (AUC) of 0.783 was attained by the centralized XGBoost baseline, whereas two Non-IID federated models recorded AUCs of 0.763 (bagging) and 0.765 (cyclic). Across five additional runs, Non IID configurations showed higher mean AUCs and recalls, although their advantage over IID was not consistent. Since the federated simulations were obtained by deriving virtual clients from a centralized dataset, the assessment of privacy advantages was conceptual rather than

validated in real multi-institutional deployment. These findings support the potential of federated XGBoost for privacy-aware healthcare applications.

Paper ID: 443

An Explainable Ensemble Learning Framework for IoT-Based Elderly Healthcare Monitoring and Fall Detection Using Multi-Dataset Validation

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Abstract

Elderly individuals face serious risks when physiological decline or falls go unnoticed, making continuous remote monitoring essential. However, deploying machine learning on edge hardware remains challenging due to sensor noise and poor transparency. This paper presents an IoT healthcare framework using a custom ESP32 wearable and a Raspberry Pi 4 edge node, enabling real-time physiological risk assessment and fall detection without cloud dependency. A two-level stacked ensemble (LightGBM, XGBoost, ExtraTrees, and a Logistic Regression meta-learner) performs health risk classification. Fall detection uses a standalone XGBoost classifier, selected for low inference latency on constrained hardware. Both pipelines integrate SHAP-based explainability, providing feature-level justifications for individual predictions. Evaluation used four datasets: a public sensor benchmark (D1, 18,090 samples), a large-scale clinical dataset with significant domain shift (D2, 200,020 samples), raw ESP32 prototype telemetry (D3, 200 samples), and an independent fall detection dataset (1,750 samples). The stacked ensemble achieved 99.28% accuracy and 99.94% ROC-AUC on D1. On D2, accuracy dropped to 69.97% while sensitivity remained 99.36%. External validation on ESP32 hardware showed the D2-trained model generalized better (90.50% accuracy) than the D1-trained model (76.50%). For fall detection, the model achieved 91.14% accuracy and 90.0% ROC-AUC. Emergency alerts with GPS coordinates are sent redundantly via GSM SMS and Telegram. A FastAPI-based web dashboard enables caregivers to track real-time vitals and historical records. Index Terms—Internet of Things (IoT), elderly healthcare monitoring, wearable sensors, edge computing, stacked ensemble learning, XGBoost, explainable artificial intelligence (XAI), SHAP, fall detection, Raspberry Pi.

Paper ID: 459

Lightweight Two-Stage CNN-to-LLM Pipeline for Automated Chest X-Ray Report Generation

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Abstract

Automated radiology report generation from chest X-rays (CXRs) is a clinically valuable but computationally demanding task. Existing vision-language approaches often require large co-trained architectures, substantial compute, and yield limited interpretability. We propose a lightweight, modular two-stage pipeline that decouples visual understanding from language generation: a convolutional or transformer-based backbone, pre-trained on the NIH-CXR14 dataset and subsequently fine-tuned on a 10,000-image MIMIC-CXR subset, predicts per-disease probabilities for 14 standard chest pathologies; these probabilities are converted into structured clinical statements via per-disease optimized thresholds and passed as a natural-language prompt to Gemma-2-2b-it, a 2.3-billion-parameter instruction-tuned language model fine-tuned with Low-Rank Adaptation (LoRA). We systematically compare four visual backbones — DenseNet201, EfficientNetV2-S, ViT-B/16, and ViT-Large — across both classification and report generation quality. Explainability is provided via Grad-CAM heatmap anchored to the highest-probability pathology. The best-performing model achieves a BERTScore F1 of 0.8094, BLEU-4 of 0.1381, and ROUGE-L of 0.2870, demonstrating competitive performance at a fraction of the computational cost of end-to-end vision-language models.

Paper ID: 521

From Perception to Reconstruction: Type-Aware Semantic Color Priors for Color Vision Deficiency

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Abstract

Color Vision Deficiency (CVD) affects the perception of chromatic information and reduces the accessibility of color-dependent visual content. Existing correction methods are predominantly type-agnostic, overlook severity variations and lack mechanisms for estimating restoration reliability. This paper presents ColorReasoner, a type-aware semantic color restoration framework that integrates CVD classification, semantic prior learning, adaptive restoration and uncertainty estimation within a unified architecture. The framework first employs a multi-task EfficientNet-B0 classifier to jointly predict CVD type and severity, achieving 99.61% classification accuracy and a severity estimation MAE of 0.0577. Semantic priors extracted from a frozen DINOv2 encoder are fused with predicted type and severity information to condition a FiLM-modulated transformer-based restoration network. Experimental results demonstrate that ColorReasoner achieves state-of-the-art restoration performance, obtaining 33.3133 dB PSNR, 0.9621 SSIM and a ΔE of 3.8271, outperforming Restormer, SwinIR and UNet-based baselines. Furthermore, the uncertainty estimation module achieves an AUSE of 0.0008, indicating strong alignment between predicted uncertainty and restoration error. The results demonstrate that semantic conditioning and type-aware restoration significantly improve both reconstruction fidelity and perceptual color accuracy for personalized CVD assistance.

Paper ID: 539

Enhancing Explainable Diabetes Prediction with CTGAN Based Data Augmentation: A Comparative Study with SMOTE

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Abstract

Type 2 diabetes is a growing global health concern, making early risk prediction essential for timely intervention and improved patient outcomes. Although machine learning (ML) has shown promising performance in diabetes prediction, many existing studies overlook class imbalance and potential data leakage during model development. This study proposes a leakage-free ML framework that integrates data augmentation, feature engineering, and Explainable Artificial Intelligence (XAI). Three data balancing settings are evaluated: no augmentation, Synthetic Minority Oversampling Technique (SMOTE), and Conditional Tabular Generative Adversarial Network (CTGAN). To prevent data leakage, all preprocessing and augmentation steps are performed only on the training data within each cross validation fold. Five ML models, including Logistic Regression, Random Forest, Support Vector Machine, XGBoost, and a soft-voting ensemble, are compared using five-fold stratified cross-validation and an independent test set. Among the evaluated models, XGBoost trained with CTGAN-generated data achieved the best weighted F1-score. SHAP and LIME were used to explain model predictions, consistently identifying polyuria, polydipsia, and sudden weight loss as the most influential features, in agreement with established clinical knowledge. The study also discusses current limitations and highlights directions for future validation using larger and more diverse datasets.

TRACK 8: INTELLIGENT TRANSPORTATION AND URBAN INFRASTRUCTURE AUTOMATION (ITUI)

Paper ID: 101

Design and Implementation of a Hybrid IoT-Based Emergency SOS Alert System Using GSM and MQTT

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Abstract

Reliable emergency communication remains a critical challenge, particularly in environments where internet connectivity is unavailable or intermittent. Existing safety systems are often constrained by dependence on either GSM-based messaging or internet-based protocols, which reduces their reliability under real-world conditions. This paper presents the design and implementation of a hybrid IoT-based emergency SOS alert system that integrates GSM and MQTT communication to provide both offline and real-time emergency notification capabilities. Our work employs a dual-controller architecture: an Arduino microcontroller handles GPS data acquisition and GSM communication, while an ESP32 module manages MQTT-based cloud connectivity. The multi-layered button interface allows the user to initiate specific types of emergency responses, such as sending an SMS alert, initiating a voice call, and resetting the alert. The combined mode communication approach guarantees efficiency, thanks to the use of GSM for the offline alert system and MQTT for monitoring purposes. In terms of experiment-based performance results, the average time required to send an SMS message is 3--8 seconds, and the delay between MQTT updates does not exceed 2 seconds, while GPS accuracy is no more than 5--10 meters. Unlike current solutions, the experimental model is more efficient, faster, and does not require either a smartphone or Internet connectivity.

Paper ID: 315

PDRL-TSC: Predictive Deep Reinforcement Learning for Hardware-Constrained Adaptive Traffic Signal Control

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Abstract

As mixed-autonomy traffic becomes a reality, traditional point-based queue estimation fails to capture the complex longitudinal dynamics of approaching vehicles, as they rely heavily on instantaneous spatial snapshots rather than continuous movement trends. This paper proposes a deep learning framework Predictive Deep Reinforcement Learning (PDRL-TSC) with two models for adaptive signal control at an isolated 2-phase intersection. We utilize a Long Short-Term Memory (LSTM) network, trained on the empirical NGSIM Lankershim dataset, to predict high-fidelity vehicular trajectory offsets over a 6-second horizon. These predictions are fed into a Proximal Policy Optimization (PPO) reinforcement learning agent. PPO dynamically maps these predictive states to optimal phase actuations by balancing the exploration of new timing strategies with the exploitation of known safe transitions via its clipped surrogate objective function. To ensure real-world deployability, hardware-in-the-loop constraints including a 5-second National Electrical Manufacturers Association (NEMA) polling cycle and an actionmasking Malfunction Management Unit (MMU) safety veto are enforced within the simulation environment. Furthermore, the system broadcasts Signal Phase and Timing (SPaT) data to optimize Connected and Automated Vehicle (CAV) eco-driving. Our results demonstrate that this neural-inference approach reduces cumulative stop-delay by 84.31% and enhances fuel efficiency to 11.03 Km/L at a 20 percent CAV penetration rate, while mitigating the computational latency bottlenecks of traditional Receding Horizon Control (RHC) optimization algorithms. Index Terms—Deep

Reinforcement Learning, LSTM, Proximal Policy Optimization, Mixed-Autonomy Traffic, Intelligent Transportation Systems (ITS), Road Speed Profile (RSP), Signal Phase and Timing (SPaT)

Paper ID: 357

Multi-Objective Adaptive Signal Control of Urban Roundabouts Using Genetic Algorithms

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Abstract

Urban roundabouts can provide safe and efficient operation under moderate and balanced traffic, but their performance often deteriorates when peak-hour demand is high and directionally asymmetric. This paper develops and evaluates a multi-objective adaptive signal control framework for a congested three-leg urban roundabout in Icheon-si, South Korea. The framework integrates field-derived turning movements, microscopic traffic simulation in SUMO, real-time controller interaction through TraCI, and a real-valued Genetic Algorithm (GA). Three strategies are compared under identical calibrated demand and geometry: unsignalized priority operation, Webster-based fixed-time signal control, and GA-based adaptive signal control. The GA updates green times every 60 s and minimizes a weighted objective combining average delay, maximum queue length, and CO₂ emissions, while fuel consumption is evaluated as an additional energy-related indicator. Results show that the GA controller reduces average delay by 62%, maximum queue length by 61%, travel time by 39.7%, CO₂ emissions by 39.3%, and fuel consumption from 287.4 to 181.7 mL relative to the unsignalized baseline. Demand sensitivity tests and statistical validation confirm robust and significant improvement at the 95% confidence level. The study demonstrates that adaptive, sustainability-oriented signal control can substantially improve roundabout operation under congested and imbalanced demand.

Paper ID: 359

A Multi-Paradigm Forecasting Framework for Urban Traffic Volume Prediction Using Data-Driven Time Series Modeling

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Abstract

Accurate traffic volume forecasting is essential for intelligent transportation systems, freeway management, signal coordination, and congestion mitigation. This paper presents a multi-paradigm forecasting framework for hourly urban traffic volume prediction by comparing statistical, machine learning, and deep learning models under a unified experimental setting. The framework is evaluated using the Metro Interstate Traffic Volume dataset, which contains 48,204 hourly westbound traffic observations collected on Interstate 94 between Minneapolis and St. Paul, Minnesota, USA, from 2012 to 2018. Four models are examined: ARIMA, SARIMAX, XGBoost, and LSTM. The workflow includes chronological preprocessing, temporal and calendar feature extraction, cyclical encoding, lag and rolling statistics, weather variables, rolling one-step-ahead evaluation, and residual diagnostics. Experimental results show that XGBoost achieves the best performance with MAE of 149.64, RMSE of 233.07, and MAPE of 6.76%,

outperforming LSTM, SARIMAX, and ARIMA. The findings demonstrate that feature-engineered tree-based learning can provide highly competitive short-term traffic forecasts for structured hourly traffic data.

Paper ID: 361

Agentic Vision-Language Framework for Resilient Railway Track Defect Detection and Maintenance Orchestration

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Abstract

Automated railway track inspection increasingly relies on computer vision models trained to detect predefined defect categories. Although these models perform well for commonly represented defects, their effectiveness decreases when rare or long-tail anomalies fall outside the distributions observed during training. In addition, conventional detection pipelines typically terminate at object localization and classification, providing limited support for condition interpretation and maintenance decision making. To address these limitations, this study proposes M²-Rail, a Multi-Model, Multi-Modal framework that integrates YOLOv9-based visual detection, Multimodal Large Language Model (MLLM) reasoning, and structured maintenance orchestration. YOLOv9 serves as the Vision Layer and processes track imagery at high frame rates to localize candidate defects and identify Regions of Interest (ROIs). ROIs associated with uncertain detections are selectively forwarded to the MLLM Reasoning Layer for zero-shot anomaly interpretation and severity assessment. The resulting semantic outputs are subsequently processed by the Orchestration Layer using retrieval-augmented generation to produce structured maintenance information. The framework was evaluated using the Rail-5k dataset together with synthesized long-tail variants. The proposed approach increased Mean Average Precision (mAP) from 81.0% for the YOLOv9 baseline to 93.4% on the evaluated long-tail defect classes, while the Vision Layer maintained a throughput of approximately 32 frames per second. Ablation experiments with alternative vision-language backbones further examined the relationship between reasoning performance and computational requirements. These results indicate that selective integration of visual detection and multimodal reasoning can extend railway inspection beyond predefined defect recognition toward contextual anomaly interpretation and structured maintenance support.

TRACK 9: ROBOT DESIGN, KINEMATICS AND MACHINE VISION (RDKV)

Paper ID: 238

Design of a Low-Cost Humanoid Robot for STEM Education in Resource-Constrained Classrooms¹Tanbir Hossain, ¹A A M Muzahid*, ²Md Awsaf Iqbal Chowdhury, ¹Rasheduzzaman Md, ¹Osama Ibn Mahfuz, ¹Jiadian Wang and ³Taffhimul Islam¹Shanghai University of Engineering Science²Daffodil International University³Noakhali Science and Technology University

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Abstract

Access to robotics education remains limited in resource constrained regions, where commercial humanoid robots often exceed USD 800. This paper presents EDRO-16 (Education Driven Robot with 16 Degrees of Freedom), a low cost humanoid platform for Science, Technology, Engineering, and Mathematics (STEM) education. The system features 16 degrees of freedom across the head, arms, legs, and waist, actuated by 16 MG90S servo motors via a PCA9685 PWM driver. An ESP32-WROOM microcontroller programmed in C++ using Visual Studio Code and PlatformIO provides wireless Wi Fi control. The mechanical structure is fully fabricated using consumer grade Fused Deposition Modeling (FDM) 3D printing with Polylactic Acid (PLA) filament, enabling a total cost of USD 58.09. The robot measures 24.23 cm in height and weighs 450–500 g, supporting portable classroom deployment. Experimental results demonstrate stable bipedal walking at 0.87 ± 0.24 cm/s with a 100% success rate, and battery runtime ranging from 303 minutes in idle mode to 75 minutes in dancing mode. Educational evaluation with 20 students and 9 experts yielded mean scores of 4.40 ± 0.66 and 4.26 ± 0.80 , confirming strong effectiveness for STEM learning. Source code and SolidWorks design files are available at <https://github.com/OsamalM/EDRO-16>.

Paper ID: 325

Latent Geometry-Guided Gaussian Splatting for Depth-Free Novel View Synthesis¹Mst. Jeba Shazida, ¹Md. Mahafuj Anam Murad*, ¹Dr. Md. Mahfuzur Rahman and ²Shariful Haque¹Jashore University of Science and Technology²International American University

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Abstract

Gaussian Splatting has emerged as a powerful method for rendering images and for synthesizing views. Nevertheless, current techniques rely on explicit depth supervision or depth-based initialization, which might not be available or reliable in challenging visual scenes. In this study, a depth-free latent geometry prior-based view synthesis model based on Gaussian Splatting is suggested. The technique learns visual features pertaining to geometry based on the RGB images without ground-truth depth maps. Image features are extracted with a multi-scale feature encoder, and the learned latent geometry prior controls the Gaussian representation and differentiable rendering process. The proposed model was trained and tested on the BlendedMVS dataset that includes various and difficult scenes, such as cities, architectural buildings, sculptures, and tiny objects. The experiment has demonstrated that the method obtained a PSNR of 19.03, SSIM of 0.829, and LPIPS of 0.131. These findings suggest that the given method can maintain structural information and the perception of similarity in synthesized images. The findings also suggest that dependency on depth supervision can be reduced when using latent geometry priors and enhances view synthesis using Gaussian Splatting. This model has potential applications in AR/VR, digital heritage, virtual education, simulation, and low-cost visual documentation.

Paper ID: 328

Comparative Analysis of 6-DOF and 7-DOF Robotic Manipulators Through Structural, Kinematic, and Motion Planning Evaluation.

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Abstract

The aim of this paper is to analyze both 6 DOF and 7 DOF robotic arms through mechanical design, structural analysis and robotic control therefore comparing their difference, accuracy level also evaluate development and optimization of robotic manipulators in industrial and research applications. Both the models are designed using SolidWorks and finite elements analysis was done in ANSYS also the kinematic modeling and Motion planning simulation were done in MATLAB and ROS 2 with MoveIt 2. The manipulator contains comparable link geometry and material properties to establish a control comparison. Key evaluations like von Mises stress distribution along with deformation reachable, inverse kinematics with workspace analysis and motion planning performance under obstacle constraints highlight the impact of both arms during comparison. This evaluation describes that the 7 DOF shows kinematic redundancy along with obstacle avoidance capability whereas the 6 DOF manipulator proves greater structural stiffness. This result eventually clarifies that both the manipulators must be guided by the specification of work and also its accuracy level on certain tasks.

Paper ID: 369

A Comparative Study of Skeleton and Silhouette Features for Drone-Based Gait Person Identification Using Deep Learning

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Abstract

Human gait has emerged as a promising biometric trait for person identification due to its non-intrusive nature and ability to operate at a distance without subject cooperation. Among the various gait representations, skeleton-based and silhouette-based features have gained significant attention because they capture complementary structural and appearance-related information. This study presents a comprehensive comparative analysis of these two feature modalities for gait-based person identification using a newly developed drone-captured gait dataset consisting of 50 individuals. The collected video sequences were processed to extract synchronized skeleton and silhouette representations, resulting in 9,342 samples for each modality. Four state-of-the-art deep learning architectures, namely EfficientNet-B0, ResNet50, MobileNetV3-Large, and DenseNet-121, were evaluated to determine the most effective model for person identification. Experimental results demonstrated that DenseNet-121 achieved the best overall performance for both modalities. The skeleton-based model obtained 99.00% test accuracy, a mean F1-score of 0.9899, and an AUC of 1.0000, while the silhouette-based model achieved 99.22% test accuracy, a mean F1-score of 0.9912, and an AUC of 1.0000. Comparative analysis indicates that silhouette representations provide slightly superior identification performance, whereas skeleton representations offer compact structural information and robustness against appearance variations. The findings provide valuable insights into the relative effectiveness of gait feature modalities and establish a foundation for future multimodal gait recognition systems and intelligent drone-based surveillance applications.

Paper ID: 492

A Low-Cost IMU-Based Wearable Device for Real-Time Elbow Range-of-Motion Monitoring with Clinically Acceptable Accuracy

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Abstract

Range of motion (ROM) assessment is an important clinical aspect of monitoring rehabilitation. There are a number of solutions, but they are hampered by cost, accessibility and suitability for unsupervised home use. This paper introduces the design, implementation and validation of a wearable device, with the cost of approximately 30 USD, for real-time measurement and monitoring of elbow ROM. The Madgwick quaternion-based sensor fusion algorithm is used to track movement with an inertial measurement unit (IMU). The system architecture is composed of an ESP32 microcontroller, a 6-axis IMU (MPU6050), wireless data transmission using UDP protocol, and an ergonomic 3D-printed housing design. The validation was performed in single trial keeping in comparison to three reference instruments: manual goniometry, commercial dynamometer ActivForce 2 and Kinovea video analysis software. The results of measurements from 22 healthy adult volunteers with BMI ranging from 15.82 to 27.49 were analyzed and good agreement with the reference measurements was obtained. The device had a Pearson correlation coefficient of 0.887, mean absolute error of 4.59°, and intraclass correlation coefficient of 0.858 compared to Kinovea reference measurements. Bland-Altman analysis revealed minimal systematic bias (−1.91°) with 95% limits of agreement of −13.38° to +9.56°. Results show that the device can achieve clinically acceptable measurement accuracy similar to manual goniometry and has real-time feedback function at a lower cost than the commercial device, which is suitable for home-based rehabilitation monitoring in resource-limited environments.

Paper ID: 543

VISTA RT: Heterogeneous Local Global Residual Fusion for State-Invariant Object Recognition and Retrieval

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Abstract

Recognizing the same physical object across changes in state, viewpoint, pose, illumination, and background is difficult because these variations enlarge intra-object differences while visually similar instances retain small inter-object differences. This paper presents VISTA-RT, a heterogeneous three-encoder framework for state-invariant object and category recognition and retrieval. Its object representation combines a GeM-pooled EfficientNetV2-S branch with a ConvNeXt-Tiny branch through a bounded learned scalar residual coefficient. A separate ConvNeXt-Tiny encoder produces a category embedding, reducing direct parameter sharing between object-identity and category supervision. Training uses identity-balanced sampling with ArcFace, supervised contrastive, batch-hard triplet, and global preservation objectives. Under the official OWSC-SI protocol, the model obtains 93.58% single-view object recognition accuracy and 94.26% object retrieval mAP, exceeding the strongest published values by 16.01 and 15.04 percentage points, respectively. Multi-view object recognition and retrieval reach 95.47% and 97.33%. Under OWSC-GN, the same checkpoint obtains 85.95% single-view and 98.67% multi-view object recognition, with retrieval mAP of 82.17% and 99.33%. These results establish strong benchmark performance for the complete framework; component-level attribution requires matched official-protocol ablations.

TRACK 10: DEEP LEARNING FOR MEDICAL IMAGE ANALYSIS (DMIA)

Paper ID: 207

A Cascaded Pipeline for Melanoma Detection, AJCC Staging, and Skin-Lesion Classification with Grad-CAM Interpretability

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Abstract

Most automated dermoscopy systems label a lesion once. Clinical triage is not a single decision, though: a clinician asks whether a lesion is melanoma, how advanced it is if so, and which disease it represents if not. We built a cascade that follows that order. A binary detector (Stage-1A) screens each lesion; suspected melanomas continue to an AJCC staging model (Stage-2), and the rest are sent to a separate non-melanoma classifier. Stage-1A combines three backbones by soft voting: ConvNeXt-Small, ConvNeXt-V2-Base, and EfficientNetV2-M. We assembled the data from HAM10000, ISIC 2019, and ISIC 2024, removing cross-source duplicates by image identifier to give 23,814 images. Every number we report comes from patient-grouped five-fold out-of-fold predictions, and we fix decision thresholds on held-out folds so that no test image helps choose its own operating point. Since melanoma prevalence is only about 18%, accuracy alone is misleading, so we foreground AUROC, AUPRC, sensitivity, and macro-F1. Stage-1A reaches 0.919 AUROC (95% CI 0.915–0.924) and 0.788 AUPRC; staging yields a 0.783 macro-F1 at 96.1% accuracy; and the three-class branch yields a 0.878 macro-F1 at 91.7% accuracy. We also give a compositional, prevalence-weighted estimate of overall behaviour and examine the pipeline's failure modes, notably a large domain-shift gap on the ISIC 2024 total-body-photography images. Training ran on a single 8 GB consumer GPU supplemented by free cloud accelerators.

Paper ID: 209

CaneScan-XAI: An Explainable Ensemble Deep Learning Framework for Automated Sugarcane Leaf Disease Classification

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Abstract

Sugarcane is an important agricultural crop that supports both the sugar and bioenergy industries, supporting the livelihoods of millions of farmers. However, early detection remains a major challenge, as manual disease diagnosis methods are time-consuming, costly, and highly dependent on expert knowledge. This study introduces an explainable deep learning framework for effective sugarcane leaf disease classification using 9,269 images representing 14 disease classes collected from two publicly available Mendeley datasets. The dataset was preprocessed and divided into training, validation, and testing sets with a 70:20:10 ratio. Three pre-trained CNN architectures—Xception, InceptionV3, and DenseNet201 were initially evaluated, followed by the enhancement of the best-performing models using SE and CBAM attention mechanisms. A weighted soft-voting ensemble approach was applied to improve classification performance further, achieving an accuracy of 97.20%. To address the need for model transparency, Grad-CAM was integrated to provide visual explanations of model predictions. Furthermore, a lightweight Streamlit-based application was developed to deliver real-time disease detection with explainable outputs, demonstrating the superior performance of explainable AI in supporting smarter and more sustainable sugarcane farming.

Paper ID: 358

Explainable AI-Enhanced Deep Learning Models for Automated White Blood Cell Subtype Classification in Blood Smear Images

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Abstract

Accurate classification of white blood cell (WBC) subtypes is important for supporting the early detection and monitoring of hematological abnormalities. This study evaluates four pretrained convolutional neural network (CNN) architectures, namely ResNet50, Xception, VGG16, and MobileNetV2, for multiclass classification of five WBC subtypes: Basophil, Erythroblast, Monocyte, Myeloblast, and Segmented Neutrophil. A publicly available dataset containing 5,000 peripheral blood smear images was used in this study. The dataset was divided into training, validation, and test sets using a stratified 70:15:15 split, and data augmentation was applied only to the training set. All models were trained using transfer learning with custom classification heads. Among the evaluated models, MobileNetV2 achieved the highest test accuracy of 99.87%, outperforming Xception, VGG16, and ResNet50. Model performance was assessed using accuracy, loss curves, confusion matrix, classification report, and ROC analysis. To improve interpretability, Explainable AI (XAI) techniques, including Grad-CAM and LIME, were applied to visualize the image regions that influenced model predictions. The results indicate that MobileNetV2 provides a strong balance between classification performance, computational efficiency, and interpretability, making it a promising model for automated WBC classification support in resource-constrained medical environments.

Paper ID: 433

Benchmarking Attention-Enhanced Encoder–Decoder Models for Polyp Segmentation in Colonoscopy Images

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Abstract

Even though colorectal cancer (CRC) is one of the most frequent cancers worldwide, it persists to be a considerable cause for cancer mortality in society. Colonoscopy has a very important role to play in the early identification of colorectal polyps to prevent malignant progression. On the other hand, automated polyp segmentation is a challenging problem owing to varying morphologies of polyps, ambiguous boundaries and complicated mucosal backgrounds. In this study, we present an attention-augmented encoder–decoder framework for automated polyp segmentation in colonoscopic frames. The irregular attention gates proposed run on top of the skip connections to automate dark the unwanted background noise and light up the clinically important spatial features. The proposed model was extensively experimented on the two benchmark datasets, Kvasir-SEG and CVC-ClinicDB. The proposed model achieves mDice of 0.94, IoU of 0.89, Precision of 0.93, Recall of 0.92 and F1- Score of 0.93 on the removal segmentation task better than some state-of-the-art baseline architectures like U-Net, Attention U-Net and Double U-Net. Ablation studies also confirm that components contribute to the overall segmentation performance. The findings show that the proposed framework represents a robust and clinically applicable method for computer-aided polyp detection.

Paper ID: 508

Hierarchical Attention and Dynamic Gating Network for Medical Image Classification

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Abstract

The combination of CNNs with Vision Transformers (ViTs) is a modern diagnostic system. Local patterns are captured by CNNs, while global relationships are modeled by Vision Transformers. These features are usually combined by fixed concatenation, which can create semantic gaps and restrict model flexibility. A hierarchical attention model is proposed to address this issue. Dynamic gating is employed to integrate the outputs from a multi-scale CNN with global representations of ViT. Local features are extracted using an EfficientNet backbone in a hierarchical fashion with a dual attention block. Fine tuned feature maps are then used as input for a small transformer encoder to produce global context information. Sample-specific fusion weights are computed by a dynamic gating network. The dataset for validation is the Alzheimer MRI dataset having four stages of the disease. Evaluation is done using 5-fold cross validation. The gating mechanism learns the relative contribution of both pathways for each input. HAT-Net has an average cross validation accuracy of 96.5%. The micro-average area under the ROC curve equals 0.998. The results surpass the performance of ResNet-50, EfficientNet, and static feature fusion models. Classifier confidence for the reported Grad-CAM examples ranges from 0.95 to 0.98, while the Grad-CAM maps highlight relevant brain regions associated with neurodegeneration.

Paper ID: 518

HCSNet: Hybrid CNN–Swin Transformers with Diffusion Augmentation for Explainable Breast Ultrasound Cancer Classification

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Abstract

One of the leading causes of cancer-related death for women globally is still breast cancer, and early, accurate diagnosis plays a critical role in patient outcomes. This paper proposes two hybrid CNN Swin Transformer architectures for breast ultrasound classification and systematically compares three training regimes, no augmentation, geometric augmentation, and diffusion-based synthetic augmentation, on the Breast Ultrasound Images (BUSI) dataset. A class-conditional denoising diffusion probabilistic model (DDPM) with classifier-free guidance is trained from scratch to generate synthetic ultrasound images, which are then quality-filtered in feature space before being merged with real training data. The proposed hybrid models, which fuse multi-scale EfficientNet-B0 features with a Swin Transformer backbone through a learned gate (HCSNet-G) and a multi-head cross-attention mechanism (HCSNet-X), are benchmarked against four widely used CNN baselines (ResNet50, InceptionV3, MobileNetV3, VGG16) under all three augmentation regimes. Diffusion-based augmentation consistently yields the largest improvement for the proposed hybrid models, with HCSNet-X (cross-attention fusion) achieving the best overall test accuracy of 90.60%, an improvement of over 3 percentage points over its no-augmentation counterpart. Grad-CAM visualizations on selected test cases suggest that both hybrid models attend to clinically relevant lesion regions, addressing the interpretability gap identified in prior work. These results demonstrate that a lightweight, dataset-specific diffusion model can outperform conventional geometric augmentation for hybrid CNN-Transformer architectures on limited medical ultrasound data.

Paper ID: 552

Whose Cuisine? Cultural Bias in Digital Eating Support Systems¹Sharmin Sultana Lincoln , ^{2,3}Soham Biren Katlariwala , ⁴Fahima Hossain Muna¹Military Institute of Science and Technology, Dhaka, 1216, Dhaka Division, Bangladesh²Gujarat Technological University, Ahmedabad, 382424, Gujarat, India³University of the People, Pasadena, CA, 91101, USA⁴University of Dhaka, Dhaka, 1000, Dhaka Division, Bangladesh

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Abstract

Cultural bias in language models is a growing concern as these systems increasingly mediate everyday food decisions, and unreliable cultural representation can compound risks such systems already carry in eating-disorder-adjacent contexts. This paper examines whether five language models, differing in scale, provider, and hosting, generate culturally reliable meal suggestions for diaspora personas across repeated conversational use. We benchmark Llama 3.2 1B/3B, Llama 3.1 8B, Gemma 2 2B, and Gemini 3.5 Flash-Lite over 900 livequeried, multi-turn conversations spanning five heritage personas and two framings, scoring outputs with a twenty-class cuisine classifier validated against independent human judgment. The central finding is a structural, model-independent failure: no model reliably generates a West African dish the scoring corpus can recognize, because the corpus contains no West African category, and retraining on a broader resource does not fix it. Cultural stability otherwise tracks parameter scale within a model family more than provider identity or framing, with no erosion over conversational turns and no convergence on a “safe” Italian default. Closing representational gaps in training data thus contributes more to cultural reliability than model scale or provider choice. Further work shall extend the persona set to additional underrepresented cuisines and test provider-identity effects at production scale

TRACK 11: POWER SYSTEMS, SOLAR ENERGY AND ELECTRICAL MACHINE AUTOMATION (PSEM)

Paper ID: 69

Leakage-Resistant Physics-Informed Learning for Multiclass Transmission-Line Fault Diagnosis

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Abstract

Data-driven fault diagnosis can appear nearly perfect when temporally adjacent electrical samples are randomly divided between training and test sets. This paper develops a leakage-resistant and physics-informed framework for six-class diagnosis from three-phase instantaneous currents and voltages. The supplied data contain 7,861 samples arranged in six long class-contiguous segments, making ordinary row-wise validation scientifically unsafe. Each segment is divided into non-overlapping 128-sample macro-blocks; 32-sample windows are confined to one block, and the blocks adjacent to every outer test group are purged from training. The representation contains 111 descriptors combining robust channel statistics with Clarke-domain residuals, interphase magnitudes, unbalance indices, and cross-phase coupling. Under repeated purged stratified-group five-fold validation, PI-LightGBM achieves accuracy 0.9538 ± 0.0378 and macro-F1 0.9419 ± 0.0512 , significantly outperforming logistic regression, RBF-SVM, and random forest after Holm correction. A matched ablation raises macro-F1 from 0.7398 to 0.9562 when physics descriptors are added. Grouped permutation analysis identifies Clarke/zero-sequence and interphase quantities as the dominant feature families. Calibration and additive-noise audits show that temperature scaling degrades probability quality and that clean-data performance does not imply measurement-noise robustness.

Paper ID: 187

ESP32-Based Outage-Resilient Solar Tracking and IoT Energy Analytics

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Abstract

Low-cost solar IoT prototypes typically assume continuous cloud connectivity (e.g., Blynk, ThingSpeak) an assumption that fails in grid-unstable, low-connectivity regions such as rural Bangladesh. We present a single microcontroller architecture that unifies differential-LDR solar tracking, hysteresis-based battery protection, autonomous fault detection, and bidirectional Telegram-Bot control on one ESP32, with flash-persisted logging that keeps the energy ledger intact through Wi-Fi or server outages. A seven-day field trial against an identical fixed-panel baseline yields a 30.7% mean energy improvement (25--35% daily range), sub-2~s alert latency, and 98% uptime, at an estimated \$35 bill of materials. Benchmarked against recent IoT solar-tracking systems, the design trades the higher yield of dual-axis Kalman/PID and ML-adaptive trackers (43--55%) for outage resilience, bidirectional low-bandwidth control, and integrated battery/fault management a combination absent from every reviewed system making it suited to distributed, off-grid energy access deployments.

Paper ID: 210

High-Efficiency Lead-Free $\text{Cs}_2\text{PdBBr}_6$ Perovskite Solar Cells: A SCAPS-1D Numerical Study of Transport Layer Optimization

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Abstract

In this study, a lead-free $\text{Cs}_2\text{PdBBr}_6$ -based perovskite solar cell (PSC) has been numerically simulated systematically with help of SCAPS-1D software with an FTO/ETL/ $\text{Cs}_2\text{PdBBr}_6$ /HTL/Au architecture. Three electron transport materials (WO_3 , TiO_2 , and SnS_2) and three hole transport materials (MoO_3 , CuI, and GO) were comparatively evaluated across varying layer thicknesses. The optimum thicknesses of the FTO and absorber layers are determined to be 0.01 μm and 5 μm , respectively. Among the investigated transport materials, WO_3 and MoO_3 are used as the electron transport layer (ETL) and the hole transport layer (HTL), respectively, due to their favorable energy band alignment and efficient charge carrier extraction capability. The combined effect of absorber defect density (1010 to 1020 cm^{-3}) and operating temperature (280 K to 360 K) was further analyzed, confirming that crystal quality is the dominant performance-limiting factor. The fully optimized The FTO/ WO_3 / $\text{Cs}_2\text{PdBBr}_6$ / MoO_3 /Au device achieves a PCE of 30.62%, $V_{oc} = 1.30$ V, $J_{sc} = 27.29$ mA/cm^2 , and $FF = 86.31\%$ at 300 K, surpassing all previously reported $\text{Cs}_2\text{PdBBr}_6$ -based configurations.

Paper ID: 266

Field-Oriented Control of Enhancing Torque Ripple and Flux Estimation Accuracy of Induction Motor based on Multi-Level Inverter

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Abstract

This research presents improved control strategy to reduce torque ripple and enhanced rotor flux estimation accuracy for three phase induction motor drives (IMDs), that combines Field-oriented control (FOC) with nine level cascaded H-bridge inverter. Conventional two-level inverters (2LI) introduce significant harmonic distortion, raising torque ripple and degrading drive performance. The proposed nine-level inverter (9LI) uses level-shifted SPWM (LS-SPWM) to substantially improve output voltage quality. Two rotor flux estimators are compared: Current Model Estimator (CME) is a sensor-based and Voltage Model Estimator (VME) is a sensorless. The system is validated in MATLAB/Simulink under step speed, speed reversal, and load disturbance tests. Results show that the 9LI reduces load voltage THD from 97.90% to 22.55% under the CME, and from 100% to 22% under the VME, compared to the 2LI. More significantly, it reduces steady-state torque ripple by approximately 85% under the CME (1.35 to 0.20 Nm) and 74% under the VME (1.15 to 0.30 Nm), with lower overshoot and faster rise time. These results confirm that the 9LI delivers consistent improvements in torque quality and waveform fidelity across both estimation methods, with ripple suppression as its principal advantage. All results reported here are obtained by simulation in MATLAB/Simulink; experimental hardware validation has not been performed and is stated explicitly as a limitation of this study.

Paper ID: 334

PowerMamba-BD: A Selective State-Space Forecaster with Inference-Time Conservation Enforcement for Demand and Load Shedding on the Bangladesh National Grid

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Abstract

Short-term forecasting of electricity demand and regional load shedding is critical for the Bangladesh Power Development Board (BPDB), whose grid runs under chronic supply constraints. Existing deep forecasters achieve competitive demand accuracy but produce internally inconsistent triples of demand, supply, and load shedding, with as many as 18 percent of test predictions reporting physically impossible negative load shedding. We propose PowerMamba-BD, which separates predictive accuracy from physical consistency. A time-axis selective state-space backbone trained on six years of BPDB daily records predicts demand and load shedding, and a post-hoc inference step clamps load shedding to non-negative values and derives regional supply from the conservation identity $D=S+L$, verified within plus or minus 5 MW on 99.99 percent of records. Because enforcement touches only load shedding and supply, demand accuracy is preserved exactly. Across five seeds on the 2024 test fold, PowerMamba-BD reaches a demand MAPE statistically indistinguishable from the strongest deep baselines (12.08 plus or minus 0.13 percent), cuts the load shedding under-prediction rate from 31.7 to 22.8 percent, and produces zero conservation residual and zero negative load shedding across every seed, region, and horizon. A four-origin backtest reproduces this behavior. Inference-time enforcement outperforms training-time alternatives at zero accuracy cost.

Paper ID: 413

Data-Driven Techno-Economic and Environmental Assessment of Rooftop Solar PV

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Abstract

Rooftop solar photovoltaic (r-PV) systems are increasingly central to decarbonizing electricity supply in dense South Asian cities, yet techno-economic evidence for academic buildings remains largely simulation-based. This paper presents a data-driven assessment of an 18 kW r-PV system at Uttara University, Dhaka, Bangladesh, using six months (June–November 2024) of metered monthly energy totals supported by archived hourly profile records. The system achieves a mean load-coverage ratio of $27.30\% \pm 5.21\%$, a specific yield of 1,501 kWh/kWp/yr, and avoids 18.09 tons of CO₂ and 453,952 L of water annually. Baseline economic analysis yields a 3.70-year payback period and an LCOE of \$0.0355/kWh. A 100,000-trial Monte Carlo simulation propagating realistic parameter uncertainty confirms these outcomes are robust, yielding a probabilistic payback of 3.60 years (P10–P90: 2.91–4.43 yr) and a 98% probability of payback within five years. Benchmarking against three independent studies confirms representative performance. The results show that r-PV effectively offsets daytime demand but requires storage integration to close the evening deficit.

TRACK 12: ROBOT MANIPULATION, NAVIGATION AND HUMAN-ROBOT INTERACTION (RMHI)

Paper ID: 49

A Voice-Based Semantic Task Interface for Sequential Navigation in ROS 2-Based Systems

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Abstract

Autonomous indoor navigation systems usually operate using direct single-goal execution, which constrains their ability to perform structured multi-stage tasks within a single interaction cycle. However, Navigation2 (Nav2) provides reliable path planning and obstacle avoidance; it does not inherently support semantic, voice-driven sequential navigation with intermediate stops while maintaining stable navigation behavior. This work introduced a voice-based task interface with semantic command interpretation, where spoken instructions were mapped to predefined goal poses, enabling multi-goal execution through a state-driven framework operating external to the Nav2 stack. The proposed approach decoupled task-level sequencing from navigation control, allowing multiple goal poses to be generated from a single command and executed sequentially in a controlled, non-overlapping manner without modifying Nav2 internals. Experimental evaluation was conducted across direct navigation, intermediate-stop navigation, and dynamic obstacle scenarios. Execution time increased from approximately 53 s for direct navigation to 296 s for dual intermediate-stop tasks, reflecting expected scaling with increased task and path complexity, without introducing additional navigation overhead. Voice-processing latency remained below 1 ms, ensuring real time and non-blocking operation. These results indicated that the proposed interface enabled structured sequential navigation while preserving the Nav2 performance characteristics in indoor environments.

Paper ID: 277

Hand-Gesture-Guided One-Demo-to-Many Robot Skill Generation with Teacher-Weighted Residual Correction in MuJoCo¹Fahim Shahriar*, ²M A Hafiz, ³Tasfia Noor and ³Wasima Noor Iqra¹Southwest Petroleum University²Research Buddy AI Tech³Chittagong University of Engineering & Technology

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Abstract

Robot manipulation policies often require many demonstrations, making data collection burdensome for non-expert users. We present a hand-gesture-guided one-demo-to-many framework in which one dual-Franka manipulation trajectory is cleaned, rigidly translated in the horizontal plane, and validated through MuJoCo replay. To test generalization without trajectory-point leakage, the 50 generated episodes in each of two manipulation conditions are split by complete spatial offset into 30 training, 10 validation, and 10 held-out test episodes. Two-layer multilayer perceptron policies are trained with five independent seeds. On held-out offsets, the proposed rigid-translation augmentation achieves 98.0%±4.5% success in Condition A and 96.0%±5.5% in Condition B, compared with 48.0%±13.0% and 26.0%±8.9%, respectively, for a matched-size naive-noise baseline. The directional paired Wilcoxon test across seeds gives $p = 0.03125$ in each condition. The $w = 0.60$ teacher-blending setting was selected post hoc from preliminary experiments and fixed before the held-out evaluation. Exact same offset teacher blending at $w = 0.60$ and direct teacher replay at $w = 1$ reach 100%, but are explicitly treated as oracle-assisted reference ceilings rather than autonomous generalization. These results show that geometry-preserving augmentation from one gesture demonstration can support strong interpolation to unseen spatial offsets in the tested simulation conditions.

Paper ID: 281

IRIS: Intelligent Robotic Interactive System with Multimodal Sensing for Autonomous Object Manipulation

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Abstract

We present IRIS (Intelligent Robotic Interactive System), a five DOF robotic arm that uses vision, voice, and touch sensors to carry out autonomous pick and place tasks. YOLO based object detection, ArUco marker homography, Radial basis function (RBF) interpolation for inverse kinematics, and Gemini driven natural language processing runs on a host computer. Joint angle commands are sent over WiFi to an ESP8266 microcontroller, which drives five servos through a PCA9685 PWM driver. We integrated custom fabricated capacitive touch sensors (1 cm × 1 cm) to the gripper fingertips and tested the system on different types of medicinal pills. The integration of tactile sensing increased the object pick and place success rate from 60% to 85%. IRIS is particularly suited for health care task, such as handling medicines, delivering the supplies and allows a hands free operation in a clean or a sterile environment. Index Terms—Robotic manipulation, Intelligent Interactive system, Multimodal sensing, YOLO object detection, human robot interaction

Paper ID: 288

Human-Demonstration-Guided Warehouse Navigation with Safety-Aware Gap Reasoning for Mobile Robots

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Abstract

This paper studies human-demonstration-guided local navigation for a TurtleBot3 Burger in a dynamic warehouse. A multi-layer perceptron (MLP) maps a 17 dimensional LiDAR, odometry, and dynamic-risk state to velocity commands, while an interpretable rule layer arbitrates GO, SLOW GO, WAIT, side-wall avoidance, and corner recovery. To isolate the contribution of each component, we evaluate MLP only, rules-only, and hybrid controllers over 30 independent 600-s runs per mode. All modes remained active for the full duration, but their safety and behavior differed substantially. MLP-only spent $64.98 \pm 22.35\%$ of its time below 0.15 m clearance, versus $2.34 \pm 1.61\%$ for rules-only and $2.43 \pm 1.57\%$ for hybrid. Relative to rules-only, the hybrid reduced recovery initiations by 40.8% ($p = .0022$) and reduced normalized distance from the human command distribution by 16.7% ($p < 10^{-10}$). A separate Nav2 diagnostic on the same warehouse map reached its long-route goal in 9 of 30 trials; its different goal-reaching objective precludes a direct success-rate ranking. The results show that explicit safety arbitration suppresses prolonged close-obstacle exposure, while the learned component reduces recovery demand and better preserves demonstrated motion behavior.

Paper ID: 456

An Automated Scan-to-Simulation Pipeline for Indoor Digital Twin Construction Using BLK360 and NVIDIA Isaac Sim

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Abstract

Constructing a simulation environment that faithfully reproduces a real indoor workspace is a major bottleneck when deploying digital twins for mobile robots: scenes are typically modeled by hand, which is slow, subjective, and quickly outdated. This paper presents an automated scan-to-simulation pipeline that converts a registered terrestrial laser scan of an industrial indoor space into a physics-enabled, photorealistic NVIDIA Isaac Sim environment, and couples it with a real mobile manipulator to form a live bidirectional digital twin. A Leica BLK360 point cloud of 68.5 million points is automatically processed into (i) a 2D occupancy grid via floor-plane estimation and height-band rasterization, (ii) a collidable wall geometry layer for physics simulation, and (iii) a dense colored point-cloud layer for visual fidelity. A swerve-drive mobile manipulator modeled from 2D CAD drawings is embedded in the scene, and a ROS 2 DDS bridge mirrors the real robot pose at 30 Hz while navigation goals selected inside the simulator are dispatched to the on-board navigation stack. A one-shot anchor-based calibration aligns the robot localization frame with the scan frame. The full conversion completes in under 80 seconds on a desktop workstation, and the resulting 3.7-million-point scene simulates at 5.6 times real time in physics-only mode and above 100 rendered frames per second in the interactive viewport. On-site experiments with the physical robot show that the reconstructed scene preserves metric scale to within 2% of tape-measured room dimensions, that end-to-end twin positioning agrees with tape-measured checkpoint distances to 5.5 cm mean absolute error, and that navigation goals issued inside the simulator drive the real robot to the commanded locations.

Paper ID: 479

Low-Cost 3D Environment Reconstruction from 2D LiDAR Data Using the OMRON TM5-900 Robotic Manipulator

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Abstract

Abstract-This work evaluates a local three-dimensional reconstruction procedure in which an RPLIDAR A2M8 is carried by the end-effector of an OMRON TM5-900. The sensor is translated along the programmed P48 to P49 path, so successive 2D profiles can be organized along a mechanically generated third coordinate. Two independent acquisitions were obtained from a laboratory scene formed by a workbench and an approximately orthogonal wooden board. Within the 150 to 1000 mm region of interest, acquisitions A and B contained 3,782 and 3,820 valid measurements; preprocessing retained 2,752 and 2,684 points. The same preprocessed pair was evaluated with point-to-point, point-to-plane, and hybrid ICP, producing RMSE values of 4.901, 4.933, and 4.944 mm, respectively, with fitness = 1.000. Rigid registration preserves the input cardinalities; integration of the aligned pair therefore yields 5,436 points. Sequential RANSAC classified 3,644 points in Plane 1, 1,539 in Plane 2, and 253 as residual, and estimated an inter-plane angle of 89.113 degrees. Because the experiment does not include an independent metrological reference or point-wise LiDAR-TCP synchronization, these results quantify inter-acquisition consistency and repeatability rather than absolute reconstruction accuracy. Keywords-3D reconstruction, 2D LiDAR, OMRON TM5-900, point cloud registration, ICP, RANSAC, robotic sensing.

TRACK 13: EXPLAINABLE MACHINE LEARNING FOR CLINICAL RISK PREDICTION (ECRP)

Paper ID: 254

Machine Learning Based Survival Prediction in Lung Cancer: A Comparative Study of TNM and UICC Staging Systems

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Abstract

Accurate survival prediction is essential for risk stratification in lung cancer. This study compared the prognostic performance of overall Union for International Cancer Control (UICC) stage grouping and individual tumor- node metastasis (TNM) components using machine learning survival models. A lung cancer survival SEER dataset was analyzed using two predictor frameworks: UICC stage grouping and individual TNM variables. CoxPH, RSF, DeepSurv, and TabNet models were developed using label and one- hot encoding. Five fold cross- validation was used for internal validation. Performance was evaluated using the concordance index, integrated Brier score, and mean time dependent AUC. SHAP analysis, calibration, and Kaplan- Meier risk stratification were performed for model interpretation and validation. All models achieved good predictive performance, with C- index values of 0.805- 0.820 and mean AUC values of 0.844- 0.859. DeepSurv with one- hot encoded TNM variables achieved the best performance (C- index: 0.820 ± 0.005 , IBS: 0.143 ± 0.003 , mean AUC: 0.859 ± 0.005). DeepSurv using one hot encoded UICC stage showed comparable performance. M stage and N- stage were the most influential TNM predictors. Both UICC stage grouping and TNM components provided strong prognostic value. TNM based DeepSurv showed marginally better discrimination, whereas UICC stage achieved comparable performance with a simpler staging representation.

Paper ID: 268

HepaPredict: A Web-Driven Hybrid LightGBM-CatBoost Framework for Early Liver Disease Prediction

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Abstract

Liver disease has become a major global health concern, with the number of cases and its impact escalating annually. In recent years, liver-related deaths have accounted for nearly 3%-5% of all deaths worldwide. The HepaPredict platform effectively processes large-scale patient data to offer accurate disorder detection and treatment advice. Liver disease is one of the most common and serious health issues affecting people around the world. Finding this condition early allows individuals to live longer and healthier lives. Within HepaPredict, various machine learning models were trained on relevant datasets to identify and assist in diagnosing this disease at a primary stage. The primary objective of this research is to analyze medical data to determine an effective machine learning classifier capable of accurately detecting liver disease in comparison with healthy individual data. These integrated features render diagnosis and healthcare more accessible, streamlined, and efficient. To enhance diagnostic precision, HepaPredict leverages advanced machine learning methodologies, including LightGBM, CatBoost, AdaBoost, Gaussian Naïve Bayes (Gaussian NB), and a unique LCBBoost hybrid architecture (LightGBM + CatBoost). Furthermore, by applying robust preprocessing techniques such as categorical encoding and feature scaling, we significantly

elevated the overall performance and predictive accuracy of the models. Our proposed models were compared with several recent studies, and the results demonstrate that the proposed approaches achieve a higher accuracy improvement ranging from 0.48% to 30.67%, depending on the dataset and the selected machine learning algorithm. For further validation, we chose the algorithm that achieved the highest accuracy with the dataset. Finally, the best-performing algorithm was integrated into a web application using the Python Flask framework to enable real-time predictions of liver disease.

Paper ID: 276

How Little Is Enough for CPU-Only Surface-Defect Classification? A Controlled Comparison of Classical and Compact Convolutional Models

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Abstract

Automated visual inspection supports zero-defect smart manufacturing, but deep learning inspectors are usually presented as tools that need GPUs, pre-trained backbones, and large labelled datasets. Small and medium manufacturers rarely have all three. We ask how much of this machinery is really needed on a standard surface-defect benchmark, and we answer with a controlled study on a single machine. On NEU-CLS (1,800 images, six classes) we test two sub-megabyte, GPU-free classifiers under the same preprocessing and the same single-core CPU timing: a 142-d texture fusion (multi-scale LBP, Gabor bank, GLCM) with an RBF-SVM, and a 24k-parameter CNN trained from scratch. Both reach about 99% accuracy. A pretrained ResNet-18, fine-tuned under the same CPU protocol, gives a direct measured anchor for the claim that pre-training is not necessary here. We then go beyond clean accuracy. A shared- pipeline corruption study over five corruption types and five severities shows that the compact CNN degrades more slowly than the texture+SVM (mean corruption accuracy 54.0% against 39.2% on NEU-CLS), mainly under JPEG compression and Gaussian blur. A per-corruption breakdown and a replication on the Magnetic-Tile-Defect benchmark confirm that equal clean accuracy does not mean equal reliability after deployment. The replication also shows that these orderings depend on the domain. On the imbalanced MT dataset the from-scratch CNN collapses to near-majority-class prediction (macro-F1 0.177), the clean-accuracy parity breaks, and the apparent reversal of the robustness ordering is an artefact of this collapse, not a real property of the models.

Paper ID: 395

EDRP-XAI: Explainable Diabetes Risk Prediction Using Optimized Ensemble Learning for Smart Healthcare

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Abstract

Diabetes Mellitus (DM) is a major chronic disease which leads to complications like cardiovascular disease, renal failure and loss of vision. While machine learning has great potential to predict diabetes, interpretability and class imbalance are important obstacles for clinical use. This paper introduces EDRP-XAI, a new explainable ensemble method that is tested on the extensive BRFSS dataset (original shape: 253,680 records), compared to other studies that use similarly sized data. It has feature engineering, Robust Scaler normalization, mutual-information feature

selection, training-set-only SMOTE balancing, and Optuna Bayesian hyperparameter optimization, all of which are combined with cross-validation. Five base classifiers were compared and the top two (CatBoost, LightGBM) classifiers were combined using Soft Voting and Stacking. All individual learners and Stacking were outperformed by the tuned and optimized Soft Voting ensemble with its accuracy of 83.87%, precision of 81.03%, recall of 83.87% and F1 score of 80.89%. The reasoning of the dominant predictors (BMI, general health, health-risk score and age) was coherent across the different risk levels, as consistently identified by SHAP, LIME and permutation importance. These findings make EDRP-XAI a scalable, interpretable and practical diabetes screening candidate for the early stages of the disease.

Paper ID: 523

Explainable and Uncertainty-Aware Machine Learning for Predicting Cryotherapy Treatment Success in Cutaneous Warts

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Abstract

Cryotherapy stands as a first-line treatment for cutaneous warts, but clinical success rates are heterogeneous among patients and small retrospective cohorts, restraining the generalisability of purely accuracy-based predictive models. This paper introduced a hierarchical, interpretable and uncertain-aware machine learning pipeline for forecasting cryotherapy success on a public cryotherapy database. The pipeline went stepwise from transparent baselines (logistic regression, a spline-based generalised additive model, shallow decision tree) to high performing ensembles (random forest, XGBoost, under three class-imbalance regimes) and finally a Bayesian logistic regression fitted via Hamiltonian Monte Carlo, providing per patient posterior credible intervals. Repeated stratified 5-fold cross-validation (50 outer folds) with bootstrap confidence intervals, and Wilcoxon signed-rank and McNemar tests are used for model comparison. A spline-based additive model achieved the best transparent model accuracy of 91.3%, with a ROC-AUC of 0.954. SHAP explanations fell similarly among model families, with treatment time, patient age and wart burden as leading predictors. An ensemble of 91%-fidelity decision-tree surrogates stratified the patients into six strata of predicted success probabilities, which ranged from 6.1% to 97.4%. Net clinical benefit over default treat-all/treat-none strategies was positive in decision-curve analysis, and a subset of near-boundary patients warranting clinician review were flagged with Bayesian credible intervals. This study examines the interpretation of these data, as well as the limitations relevant to the small, single-site sample.

Paper ID: 538

A Machine Learning-Based Clinical Decision Support System for Early Detection of Chronic Kidney Disease

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Abstract

Chronic Kidney Disease (CKD) develops silently over months or years, and by the time symptoms become noticeable, a large share of kidney function may already be lost. This makes early, low-cost screening genuinely valuable, especially in settings where routine nephrology access is limited. This paper presents an end-to-end CKD risk-prediction system, CKD Predict, that combines a thoroughly benchmarked machine learning pipeline with a deployed full-stack web application. The system was built on the UCI/Kaggle Chronic Kidney Disease dataset (400

patient records, 24 raw clinical attributes). The data was cleaned, missing values were imputed using a distance-weighted K-Nearest Neighbours imputer, and the feature space was reduced to fifteen clinically meaningful attributes through a combined Chi-squared, mutual-information, and Random Forest importance ranking. Class imbalance was corrected with SMOTE applied strictly to the training partition, and features were scaled with a RobustScaler. Fourteen classifiers and ensemble strategies were trained and compared under 10-fold stratified cross-validation, with Random Forest, Extra Trees, LightGBM, and three ensemble variants all reaching 100% test accuracy and AUC-ROC, while Logistic Regression reached 96.25% accuracy with a stable, low variance cross-validated F1-score of $98.46\% \pm 1.72\%$. Logistic Regression was ultimately selected as the production model for its interpretability and resistance to overfitting on a relatively small dataset.

Paper ID: 566

HealthCal-XAI: An Explainable Multi-Endpoint Machine Learning Framework for Preventive Disease Risk Screening

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Abstract

Machine learning (ML) tools for preventive health screening are often presented as isolated disease classifiers with limited transparency, weak discussion of class imbalance, and little connection between offline evaluation and deployable user interfaces. This paper presents HealthCal-XAI, an explainable, deployment-aware, multi-endpoint framework for preliminary risk screening across four tabular health tasks: diabetes, heart attack risk, stroke risk, and maternal health risk. The proposed framework integrates heterogeneous public datasets, endpoint-specific preprocessing, lightweight classifier selection, risk-aware evaluation, explainability-by-design, and a Flask-based web deployment layer. Unlike conventional single-disease prototypes, HealthCal-XAI treats each endpoint as a separate prediction problem and selects models using accuracy, macro-F1, weighted-F1, and balanced accuracy. It further introduces a practical explainable AI (XAI) layer based on global feature attribution, local prediction explanation, model cards, and non-diagnostic risk communication. Standardized evaluation shows competitive performance for diabetes (75.3% accuracy), heart attack risk (82.2%), maternal health risk (83.0%), and stroke risk (91.5% accuracy with low macro-F1 due to severe imbalance). The central contribution is a reproducible and deployment-ready health-AI pipeline that connects model development, explainability, and responsible deployment for low-resource preventive screening.

TRACK 14: CYBERSECURITY, FRAUD DETECTION AND DIGITAL FORENSICS (CFDF)

Paper ID: 219

Real-Time Browser Threat Detection Security Agent via Local VLM–LLM¹Dhiman Kumar Sarker, ²Fyaz Nafin Rahman, ¹Md Rezaul Karim Khan, ³Sushmoy Paul Rohit, ⁴Tanvir Ahmed and ⁵Md Mizanur Rahman*¹Maharishi International University²Prairie View A&M University³James cook University⁴Independent University⁵Green University of Bangladesh

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Abstract

Web-borne threats, including phishing, adult content, illegal gambling, and clickunder redirects, pose significant risks to Internet users. Existing defenses frequently depend on static URL blocklists, limiting their effectiveness against zero-day, obfuscated, and rapidly changing threats. This paper presents, an autonomous browser-security framework that runs entirely on consumer hardware without cloud-based inference. The framework integrates 4-bit NormalFloat (NF4)-quantized Qwen2.5-VL-3B-Instruct and Qwen2.5-3B-Instruct models within a 6 GB VRAM environment. Its two-pass vision-language pipeline first extracts visible webpage text and subsequently classifies potential threats using the extracted evidence. In parallel, the language model analyzes URL structures for phishing, gambling, and affiliate-scam patterns. These outputs are fused with a curated knowledge base containing more than 300 trusted domains, 120 adult and gambling domain signatures, and 50 clickunder URL patterns. A Manifest V3 Chrome extension processes browser tabs through a serial scanning queue and automatically closes confirmed threats. PhishGuard was evaluated on 3,688 URLs collected from PhishTank, PhiUSIIL, URLhaus, and manually curated sources. It achieved a macro precision of 91.4%, recall of 88.7%, F1-score of 0.901, and false-positive rate of 4.8%, while using 4.51 GB of GPU memory. The results demonstrate that multimodal analysis, URL reasoning, and knowledge-guided score fusion can provide effective, privacy-preserving browser protection on resource-constrained consumer hardware. The source code is available at <https://github.com/rafi79/phishguard>.

Paper ID: 236

A Comparative Deep Learning Framework for Phishing Detection Using Transformer and Reconstruction-Based Models

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Abstract

Phishing attacks continue to pose a significant threat in modern cybersecurity, as attackers increasingly rely on deceptive URLs to evade rule-based and blacklist-based detection systems. This paper presents a hybrid deep learning framework combining a fine-tuned RoBERTa transformer for contextual URL understanding with a reconstruction-based autoencoder for anomaly scoring. Each URL is processed through two parallel branches: the raw URL string is analyzed by the transformer branch, while 84 engineered numerical features spanning lexical, domain, content, and reputation characteristics are processed by the autoencoder branch. The branch outputs are combined via a weighted fusion mechanism, with the fusion weight tuned on the inner validation fold, to produce a final phishing risk score. All experiments used a deduplicated dataset of 10,474 unique URLs, 51.8% synthetically generated rather than collected from live traffic (Section III), under a 5-fold nested stratified cross-validation protocol, with all preprocessing fitted only on training folds to prevent leakage. The framework achieves AUC 0.9883 ± 0.0010 , F1 0.9490 ± 0.0032 , and accuracy 0.9471 ± 0.0036 , with inference latency of 10.231 ms per URL. Against an XGBoost baseline on the same features under the identical protocol (AUC 0.9953 ± 0.0009), the hybrid model's

gain over its own RoBERTa-only branch (AUC 0.9881 ± 0.0010) is not statistically significant on AUC ($p=0.374$), F1 ($p=0.182$), or accuracy ($p=0.277$; paired t-test, 5 outer folds), and XGBoost significantly outperforms the hybrid model on all three metrics ($p<0.001$). This indicates limited practical benefit from the fusion mechanism relative to both a strong transformer-only baseline and a cheaper gradient-boosted one. We report this finding transparently and discuss its implications for the framework's contribution.

Paper ID: 278

Beyond Learned Fusion: Modality-Aware Max-OR for Asymmetric Deepfake Detection in Short Audio-Visual Clips.

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Abstract

Audio–visual deepfake detectors often reduce the two streams to a single learned fusion score. In an asymmetric forgery, one stream is manipulated while the other remains authentic, so a strong cue can be weakened by the apparently genuine modality. This paper introduces Max-OR, a parameter-free decision rule that retains the learned Dual-Branch Late Fusion Network (DBLF-Net) score together with separate audio and visual specialist outputs. The specialist scores are combined using a soft-OR operation, and the larger value between this score and the learned fusion score is used for the final decision. Adding the rule does not require retraining either encoder, and the operating threshold is selected only from validation data. On 2,098 FakeAVCeleb v1.2 test clips from 44 identities, Max-OR obtained 96.62% accuracy, 98.25% F1-score, 97.22% area under the receiver operating characteristic curve, and 96.65% macro sensitivity across the audio-only and video-only fake categories. Accuracy improved by 3.57 percentage points over learned fusion, with a paired identity-cluster 95% confidence interval of 1.99 to 5.42 points. Under alternative one-second audio crops, aggregate accuracy decreased by less than one percentage point, although audio-only forgeries showed substantially larger changes. The results indicate that preserving modality-specific evidence improves one-sided forgery detection, while audio-segment selection remains an important limitation.

Paper ID: 283

VulPatchNet: A Dual-Encoder Cross-Attention Framework for Multi-Language Vulnerability Patch Generation

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Abstract

Automated vulnerability repair has advanced rapidly with pretrained code models, yet most existing approaches are restricted to one programming language and depend on just a single encoder, limiting their ability to capture both structural and generative code semantics. In this paper, we propose VulPatchNet, a dual-encoder cross-attention framework for multi-language vulnerability patch generation. VulPatchNet fuses GraphCodeBERT, which captures data-flow and structural dependencies, with CodeT5, which provides sequence-to-sequence generation priors, through a cross-attention bridge that conditions the generation encoder on structural representations. A fix-mask generator further produces token-level vulnerability salience scores, guiding the decoder toward regions requiring repair. The model is trained, tested, and evaluated on a curated dataset of 78,715 vulnerability-patch pairs spanning eight programming languages. The experimental results indicate that VulPatchNet obtains a BLEU score of 23.89, CodeBLEU of 25.10, and exact match of 2.64%, which outperforms the CodeT5 and VulRepair baselines by a large

margin, with statistical significance confirmed via Wilcoxon signed-rank test at $p < 0.05$ after Bonferroni correction. An ablation analysis examines the impact of each architectural element on performance. These findings demonstrate that combining structural and generative pretrained representations offers a promising direction for scalable, language-agnostic automated vulnerability repair.

Paper ID: 312

An In-Depth Comparative Analysis of Credit Card Fraud Detection Using A Hybrid Machine Learning Framework and Explainable AI

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Abstract

With the rapid growth of online transactions, credit card fraud has become a critical issue. Numerous machine learning techniques have been implemented to achieve strong fraud detection performance. Over time, fraud detection techniques have evolved from rule-based approaches to supervised and unsupervised machine learning algorithms, each having its own advantages. Subsequently, hybrid architectures combining both supervised and unsupervised approaches have been incorporated to harness the strengths of these two different types of techniques. In this research, we propose a hybrid fraud detection approach using the unsupervised Autoencoder technique and the supervised CatBoost technique on three credit card fraud detection datasets: PaySim, BankSim, and Sparkov. The proposed hybrid architecture achieved the highest accuracy of 99.97% and an F1-score of 0.90, demonstrating high predictive performance. However, due to the performance–interpretability tradeoff, understanding the decision-making process of hybrid techniques becomes more complex. Therefore, SHapley Additive exPlanations (SHAP) is employed to enhance interpretability and provide an in-depth analysis of the fraud detection datasets.

Paper ID: 384

Intrusion Detection System Using Optimized Machine Learning with Explainable AI

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Abstract

Modern cyber threats are becoming increasingly sophisticated, and intrusions detection systems (IDS) need to be capable of delivering high detection accuracy and be explainable in their decision-making process. Despite their success, Machine Learning (ML) models makes them less trusted and trustworthy for real-world cybersecurity environments. In order to fill this gap, this study proposes an optimized and explainable IDS framework based on stacking ensemble method, where Random Forest (RF), XGBoost, Gradient Boosting (GB) and K-Nearest Neighbors (KNN) are used as base classifiers and Logistic Regression is used as meta-learner trained using the out-of-fold predictions. The framework is built and tested on NSL-KDD dataset where classes Normal, DoS and Probe are the majority classes. In order to balance the class distribution, the Synthetic Minority Over-Sampling Technique (SMOTE) is applied to the training set only, while the test set is left imbalanced so as to assess the generalization ability in a realistic way. The results of experimental study reveal that the proposed stacking ensemble outperforms between baseline models. The proposed model performed accuracy 89.92%, precision 90.36%, recall 89.92%, f1-score 89.81%. To keep transparency, SHAP and LIME are combined to deliver global or instance-level explanations, which led to the most influential features being identified for DoS and Probe attacks. Overall, the findings suggest that a balanced performing stacking ensemble can be effectively provide a good tradeoff between detection accuracy and transparency, which is a step towards more trustworthy and analyst-centric IDS systems.

TRACK 15: ROBOT DESIGN, KINEMATICS AND MECHATRONIC SYSTEMS (RDKM)

Paper ID: 225

Design and Experimental Evaluation of a Low-Cost Autonomous Solar Panel Cleaning Robot Using Line-Following Navigation

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Abstract

Dust accumulation on solar photovoltaic panels reduces power output by up to 30% in high-soiling environments. This paper presents a low-cost, Arduino-based autonomous solar panel cleaning robot designed for small-scale photovoltaic installations. The prototype integrates line-following navigation using a TCRT5000 infrared sensor array, an LDR-assisted stopping mechanism for panel boundary detection, an event-driven finite-state machine controller, and a triangular rotating cleaning head driven by an MG995 servo motor. Experimental evaluation demonstrates a cleaning efficiency of approximately 75% under standardized dust loading conditions. A power consumption profile across five operational states is documented, showing a maximum current draw of 2.5 A during cleaning operations. Cost analysis reveals a prototype unit cost of approximately 32,460 BDT (~\$295 USD), representing a 78% reduction relative to commercial robotic cleaning systems. Furthermore, testing demonstrates a 15% improvement in actual PV power output after cleaning, and the robot successfully navigates and cleans panels inclined up to 25° with minimal efficiency loss. These results confirm the feasibility of semi-autonomous solar panel cleaning at accessible cost points for developing-country contexts, with clear pathways to closed-loop tilt optimization and vacuum-assisted cleaning in future work.

Paper ID: 294

Kinematic Analysis, Trajectory Generation, and Low-Cost Hardware Evaluation for a Quadraped Robot Leg

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Abstract

Inverse kinematics is a fundamental component of quadraped robot locomotion that converts desired foot positions into actuator commands. This paper presents a coordinate-keyframe-driven analytical inverse kinematics framework for a 3-degree-of-freedom quadraped robot leg. The implementation combines closed-form geometric inverse kinematics and Cartesian keyframe trajectories with a low-cost, embedded servo-controlled prototype. To simplify the inverse kinematics problem, the 3-D foot position is decomposed into an x-z projection for lateral shoulder rotation and an r-y two-link planar model for shoulder and knee motion. Forward kinematics is used to reconstruct and evaluate the resulting trajectory. A front-right leg prototype was built using high-torque servo motors, carbon-fiber leg sections, an ESP32-C6 controller, and a PCA9685 servo driver. The prototype was evaluated using a scale-calibrated side-view camera-tracking setup. Experimental results showed that the measured foot path followed the predicted IK-to-FK trajectory, with a symmetric nearest-curve mean deviation of 0.37 cm and an approximate symmetric Hausdorff maximum deviation of 1.74 cm. The higher phase-matched RMSE of 4.11 cm was mainly attributed to servo delay and vertex rounding. The results indicate that the proposed framework is simple and interpretable for planar single-leg control and provides a practical basis for future low-cost educational quadraped development.

Paper ID: 360

A Low-Complexity 3-DOF Collaborative Robot with Vision-Based Expiry Date Recognition for Retail Automation

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Abstract

Effective management of perishable goods becomes a considerable challenge in grocery retail, as manual inspection of expiry dates leads to food waste, increased labour costs, and operational inefficiencies. This paper introduces a cost-effective, task-orientated collaborative robotic framework designed for automated expiry-date inspection and shelf-level intervention in retail settings. The proposed system integrates a three-degree-of-freedom (3-DOF) collaborative manipulator with a ROS2-based control architecture, YOLOv10n-based object detection, optical character recognition (OCR), and a rule-based decision engine to establish an end-to-end detect–decide–act workflow. The robotic manipulator was designed using computer-aided design (CAD) software, converted into a Unified Robot Description Format (URDF) model, and analytically modelled through forward and inverse kinematics. The analytical model was validated through MATLAB simulations, showing positional discrepancies of less than 0.01 mm between the analytical and numerical solutions. An economic assessment utilising representative UK grocery retail benchmarks estimated a capital cost of around £4,000, annual operating costs of about £178, a payback period of roughly 2.7 months, and a return-on-investment surpassing 400%. Perception evaluation produced a 90.8% image-level detection rate for YOLOv10n and a 64.31% exact-date recognition accuracy using EasyOCR, indicating the feasibility of low-complexity collaborative robots alongside current limitations of the proposed vision pipeline.

Paper ID: 445

Design, Development, and Experimental Demonstration of a Low-Cost Additively Manufactured Morphing Quadcopter

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Abstract

Conventional drones or quadcopters have a fixed geometry, which reduces the adaptability and maneuverability in confined or narrow spaces. To solve this problem, the current study shows the design and development of a morphing quadcopter, which is capable of transforming into three distinct configurations (X, H, and O). The quadcopter frame is designed using CAD software. The frame is fabricated using additive manufacturing (FDM) for lightweight and to obtain a structurally efficient frame for the morphing mechanism. The quadcopter incorporates locally sourced hardware components for its electrical and power systems, highlighting the feasibility of developing advanced UAV platforms using readily available resources in Bangladesh. The study also shows the methods of servo motor calibration for morphing and integrating the morphing mechanism with the flight controller. The proposed morphing quadcopter is experimentally validated through successful flight demonstrations in X, H, and O configurations.

Paper ID: 495

Design, Development, and Workspace Evaluation of a Ceiling-Mounted Four-Axis SCARA Robotic Arm for Light-Duty Automation

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Abstract

This paper presents the design, development, and workspace evaluation of a Ceiling-Mounted Four-Axis SCARA Robotic Arm developed for workspace optimized light-duty automation applications. This architecture is integrated with two rotational arm mechanisms and horizontal rail translation and vertical positioning mechanisms which are not found in the conventional fixed-base SCARA system to achieve the advantages of good workspace utilization and maintaining a safe distance from the ground. The proposed configuration had a planar workspace area gain of 110.4% over an equivalent fixed base SCARA configuration. The system is composed of an Arduino based control architecture, dedicated motor drivers, belt pulley transmission, and a modular end-effector mechanism, all of which are optimized to provide an effective solution. A Denavit-Hartenberg (DH) model is developed according to the customized kinematic configuration, to formulate the forward kinematics and inverse kinematics of the manipulator. Reachability of the proposed configuration was investigated using a Python-based computational framework and compared to an equivalent fixed-base SCARA configuration. The analysis demonstrates the effect of the additional translational degrees of freedom on the reachable workspace and provides a quantitative assessment of workspace enhancement. The proposed design is applicable in the robotic systems where the space constraints are present with respect to the traditional floor mounted systems.

Paper ID: 542

Inverse Kinematics of 6-DOF IRB 2400 Robots with Customized MLPs and Explainable Joint Feature Analysis

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Abstract

The inverse kinematics (IK) problem of six-degree of-freedom (6-DOF) robotic manipulators is a fundamental real time robotics problem, requiring high accuracy and low latency. While classical analytical and numerical methods frequently suffer from high computational cost, the presence of singularities, and limited robustness, many learning-based methods focus on accuracy rather than computational cost and interpretability. Rather than seeking to maximize raw accuracy, this paper bench marks whether a compact, interpretable multi-layer perceptron can meet real-time latency requirements for IK on the ABB IRB 2400 robot, evaluating it against tree-based ensembles and a deeper neural network using regression accuracy, statistical tests, inference latency, and explainability analysis. The results demonstrate that tree-based ensembles achieve higher accuracy but incur higher, less stable latency. At the same time, the proposed compact MLP provides fast, stable inference with competitive accuracy, making it more suitable for real-time control. The learned representations are also consistent with the robot's kinematic structure, as demonstrated by SHAP and LIME analyses. These results point to a clear trade-off between accuracy and efficiency, and show that lightweight models can achieve practical IK while maintaining interpretability.

TRACK 16: BANGLA NATURAL LANGUAGE PROCESSING AND LANGUAGE TECHNOLOGIES (BNLP)

Paper ID: 203

Bangla Sentence Function Classification: Corpus Development, Model Benchmarking, and Interpretability

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Abstract

Automatic sentence function identification is important for many downstream natural language processing (NLP) applications such as dialogue systems, text-to-speech synthesis, and machine translation. However, benchmark resources for Bangla sentence function classification remain limited. To mitigate this gap, this paper introduces a corpus of 10,000 Bangla sentences, manually annotated into four functional categories, namely declarative, interrogative, imperative, and exclamatory. The corpus is nearly balanced across the four classes, with high annotation reliability reflected by a Fleiss' Kappa of 0.82. Furthermore, we evaluate multiple feature representations, including Bag-of-Words (BoW), TF-IDF, and Word2Vec, with several classical machine learning classifiers. In addition, two heterogeneous ensemble models, namely Single-Level Ensemble (SLE) and Double-Level Ensemble (DLE), are utilized to improve classification performance. Experimental results show that TF-IDF consistently outperforms Word2Vec, likely due to its ability to emphasize discriminative lexical cues associated with sentence functions, particularly given the relatively small corpus used to train Word2Vec. The DLE model with TF-IDF features achieves the best performance with accuracy and macro-F1 of 0.95, demonstrating the effectiveness of sparse lexical representations and heterogeneous ensemble learning for this task. Further cross-validation confirms the robustness of the approach, while LIME-based interpretability provides insights into model predictions. The developed corpus and model benchmarking establish strong baselines for Bangla sentence function classification.

Paper ID: 213

Leakage-Aware Sentiment Analysis of Bangla-English E-Commerce Reviews Using RoBERTa with Rigorous Cross-Validation

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Abstract

Automated sentiment classification of e-commerce reviews helps platforms understand customer opinions and improve services. However, reported performance is often inflated by label leakage, single train-test split evaluation, or reliance on accuracy for imbalanced datasets. This paper presents a leakage-aware framework for sentiment analysis on a Bangla-English e-commerce review corpus of 78,130 reviews from two South Asian platforms. A leakage audit shows the Emotion field is a perfect proxy and Rating a strong proxy for the sentiment label, with rule-based predictors reaching macro-F1 up to 1.0000. A further audit shows 96.8% of reviews share a product with another review, creating a second, product-level leakage surface that a naive split would not close. Emotion and Rating are therefore excluded from all model inputs, and cross-validation uses product-grouped stratified folds so no product appears in both training and validation. The framework uses a pretrained RoBERTa encoder with an attention refinement module and a regularized MLP classifier, trained with a class-weighted, label-smoothed loss for the 86/14 class imbalance. The model is evaluated with 5-fold product-grouped stratified cross-validation with early stopping, reporting mean and standard deviation across folds. Results achieve a mean macro-F1 of 0.8559 +/-

0.0078, weighted-F1 of 0.9298 $\pm$ 0.0037, and ROC-AUC of 0.9586 $\pm$ 0.0040, with a paired t-test confirming the model significantly beats the strongest naive baseline ($p = 0.0023$). An ablation study, run under the same full protocol as the main results, shows the added self-attention layer modestly outperforms a bare linear head, indicating the architectural choice earns its place given comparable training. The dataset is publicly available for future research and reproducibility.

Paper ID: 264

Fine-Grained Sentiment Classification of Bangla Book Reviews: A Large-Scale Benchmark Dataset and Ensemble Learning Approach

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Abstract

Sentiment analysis for fine-grained sentiment at the level of Bangla reviews has been largely unexplored, especially sentiment intensity classification of reviews due to the lack of large corpora with manually annotated data. This paper aims to fill this research gap and proposes a large-scale manually annotated dataset with 20,107 Bangla book reviews gathered from Rokomari.com and divided into five ordered sentiment categories: Strongly Negative, Negative, Neutral, Positive, and Strongly Positive. The annotation process was validated by calculating the inter-annotator agreement using Cohen's Kappa and securing an impressive agreement coefficient of kappa = 0.889. In order to create a complete benchmark for fine-grained sentiment classification in the Bangla language, two traditional machine learning algorithms, namely Logistic Regression and Support Vector Machine, and four pre-trained transformers including mBERT, MuRIL, BanglaBERT and XLM-RoBERTa, were rigorously tested within a single experimental paradigm. Besides, a validation-based weighted ensemble learning framework was proposed for taking advantage of the predictions of the evaluated models. This proposed ensemble yielded the highest performance score of all, achieving an accuracy rate of 90.59%, a Macro F1 score of 80.15%, and a Weighted F1 score of 90.49%. The statistical test of significance by McNemar confirmed that the differences in the performance of the ensemble compared to individual baselines were indeed statistically significant. The resultant benchmark and dataset, as well as the weighted ensemble model, provide a common experimental ground for fine-grained, intensity-based sentiment analysis for the Bangla language.

Paper ID: 352

Multi-Modal Stacking Ensemble with SHAP Explainability for Bangla Clickbait Detection

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Abstract

Clickbait remains a persistent challenge on video-sharing platforms, where sensationalized titles mislead viewers and erode trust, particularly in low-resource settings like Bangla. The BaitBuster-Bangla dataset offers 253,070 labeled YouTube videos across metadata, text, and engagement dimensions, but prior work has relied solely on text-based features with classical machine learning, reaching at most 85% accuracy, without incorporating engagement signals or explainability. This study proposes a multi-modal stacking ensemble that combines multilingual sentence embeddings of video titles with engagement-derived features (view, like, dislike, and comment ratios) to detect Bangla clickbait. The ensemble XGBoost, Random Forest, and LightGBM base learners with a logistic regression meta-learner achieves 96.35% accuracy and an F1-score of 0.968 on the human-annotated subset, substantially outperforming existing benchmarks. An ablation study confirms that combining textual and engagement features

yields a statistically significant improvement (McNemar's test, $p < 0.001$) over text-only and engagement-only variants. SHAP analysis shows the like-to-dislike ratio is the most influential engagement signal, with case studies illustrating correct and misclassified examples. A separate label-reliability analysis across the dataset's human, automatic, and AI-generated labeling strategies finds near-perfect agreement (Cohen's Kappa > 0.96) on the curated human-labeled subset but much weaker agreement (Kappa = 0.517) on a broader, randomly sampled portion. These results establish a new state-of-the-art benchmark on BaitBuster-Bangla while highlighting critical limitations in the dataset's labeling consistency at scale.

Paper ID: 516

Real-Time Isolated Bangla Sign Language Recognition Using MediaPipe Hand Landmarks and LSTM

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Abstract

Bangla Sign Language (BdSL) recognition is essential for bridging communication between the deaf community and the general public, yet real-time recognition remains challenging due to the complexity of hand motion and the computational cost of video-based models. This paper proposes MediLSTM-BSL, a lightweight real-time recognition system for Bengali Sign Language that uses MediaPipe hand landmark extraction combined with a multi-layer Long Short-Term Memory (LSTM) network. Instead of raw video frames, the system processes 63-dimensional skeletal landmark sequences, reducing computational cost while preserving the temporal structure of gestures. The model is evaluated on the publicly available BdSL video dataset containing 40 everyday signs. Under signer-independent 5-fold cross-validation, MediLSTM-BSL achieves a mean accuracy of 99.06% (± 0.42) and a macro F1-score of 99.06%, outperforming evaluated MLP, 1D-CNN, GRU, LSTM, and BiLSTM baselines under the same signer-independent protocol. End-to-end streaming latency on a CPU is 41.3 ± 3.8 ms (median 40.7 ms, P95 48.9 ms), corresponding to ≈ 24 FPS, confirming suitability for real-time interactive applications.

Paper ID: 553

Bengali Dialect Classification via Feature-Enriched RAG and Multi-Strategy Prompting Ensembles

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Abstract

Bengali dialect classification is challenging due to the linguistic diversity across regional varieties in Bangladesh and West Bengal. In everyday communication, the same meaning may appear in different dialectal forms, such as “আমরা” (Standard Bangla) and “আই যাইতাছি” (Chittagong dialect), highlighting the difficulty of automatic identification. This paper proposes an advanced framework that combines retrieval-augmented generation (RAG), linguistic feature engineering, and ensemble prompting to improve dialect classification. The system integrates FAISS based semantic retrieval using LaBSE embeddings with linguistic features including phonetic markers, verb conjugation patterns, and character n-grams. Experiments on the BanglaDial dataset containing 59,232 sentences across twelve dialects demonstrate that the proposed approach achieves 97.08% accuracy using Llama-3.1-8B-Instruct. The framework also maintains strong performance with smaller models, achieving 86.25% accuracy using a compact 3B parameter model, indicating its suitability for resource-constrained environments. Source code is available at <https://github.com/rafi79/bengali-dialect-classification>.

TRACK 17: ARTIFICIAL INTELLIGENCE FOR CROP DISEASE AND PRECISION AGRICULTURE (ACPA)

Paper ID: 63

Performance Analysis of Sequential CNN Architectures for Regional Fruit Leaf Disease Detection

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Abstract

The early and precise identification of fruit plant diseases is paramount for enhancing crop yield and ensuring economic sustainability in regional agriculture. Traditional manual inspection is labor-intensive and prone to error, particularly for tropical species like Sapodilla (*Manilkara zapota*), Mango (*Mangifera indica*) and Lychee (*Litchi chinensis*). This paper presents a comprehensive performance analysis of a Sequential Convolutional Neural Network (CNN) architecture optimized for localized disease detection. We utilize a newly curated dataset of 6,000 high-resolution images capturing both healthy and diseased specimens—specifically Powdery Mildew, Gall Midge and Litchi Mite—sourced directly from real-world agricultural environments in the Faridpur region. The methodology employs a robust technical pipeline including pixel-level normalization, standardized resizing to 180x180 pixels and extensive data augmentation techniques such as rotation, zooming and flipping to mitigate overfitting and enhance model generalization. Experimental results demonstrate that the proposed model achieves a classification accuracy of 96% across six distinct classes within 50 training epochs. By evaluating the model's efficacy in extracting critical morphological and vein-based features from regional leaf samples, this study provides a resource-efficient framework suitable for real-time diagnostic applications in precision agriculture.

Paper ID: 145

Comparative Analysis of CNN Architectures for Hog Plum Leaf Disease and Maturity Stage Classification

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Abstract

This paper presents the first deep learning framework for identify hog plum leaf diseases and growth stages automatically. A total of 5,425 field pictures were gathered and manually tagged into eleven classes, comprising of healthy leaves, varied developmental phases, fungal attacks, and insect attacks. A preprocessing pipeline was created in a structured way to improve the quality of images by eliminating noise, correcting illumination, and increasing contrast. VGG19, DenseNet 201, and CoAtNet three deep learning models were trained and tested using the same conditions to investigate the most effective architecture on this task. The experimental findings indicate that CoAtNet was the most accurate with 98.63 %, followed by DenseNet201 with 97.26%, and lastly, VGG19 with 92.66 %. CoAtNet had a high performance in the differentiation of visually similar disease patterns and high generalization in a wide range of field conditions. The proposed framework offers a stable basis on which effective diagnostic tools can be developed to facilitate an early detection of the disease and the ability to control the disease properly in the cultivation of hog plums.

Paper ID: 157

Deep Learning-Based Morphological Segment Classification of Piper Chaba: A Benchmark Study with Class-Imbalance Mitigation

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Abstract

Piper chaba (Bengali: Chui Jhal) is a medicinally important climbing pepper native to the Indian subcontinent. Accurate identification of its morphological parts: aerial stem, primary rhizome, and fibrous root mass is critical for herbal medicine quality control and botanical research, yet manual identification remains error-prone due to inter-class visual similarity and severe class imbalance in field-collected data. This paper presents the first benchmark study applying transfer learning to Piper chaba morphological part recognition. We introduce a dataset of 697 field-collected images across three classes (Aerial Stem: 422, Primary Rhizome: 196, Fibrous Root Mass: 79) from Khulna, Bangladesh, and evaluate four architectures: MobileNetV3-Large, ResNet-50, EfficientNet-B3, and ViT-B/16. Class imbalance is addressed via weighted random sampling and inverse-frequency weighted cross-entropy loss. EfficientNetB3 achieves the best macro-F1 of 0.8738 (90.91% accuracy), with Grad-CAM analysis confirming attention to botanically meaningful features including vascular bundle cross-sections and surface textures. Our results establish a strong baseline for automated Piper chaba part identification and demonstrate the effectiveness of lightweight architectures with imbalance mitigation on small scale botanical datasets.

Paper ID: 293

Bridging Local and Global Feature Learning: An Explainable Ensemble Framework for Intelligent Crop Disease Diagnosis

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Abstract

Effective and accurate crop disease diagnosis is crucial to improve the efficiency of agriculture and reduce crop losses. Traditional diagnostic methods are subject to labour intensity and are often impractical in rural farm settings. This paper proposes an interpretable DL solution for automated multi-crop disease classification and localization. Under various environmental conditions, a large-scale, region-specific dataset consisting of over 160,000 images of 13 major crop species with 52 healthy and diseased classes has been created. The proposed model is a hybrid of five CNN architectures: ResNet152, DenseNet201, MobileNetV3, EfficientNetB7, and InceptionResNetV2, and four Transformer architectures: Vision Transformer (ViT), Swin Transformer, DeiT and Convolutional Vision Transformer (CvT), all fine-tuned with transfer learning. A performance-weighted ensemble approach with validation F1-scores is used to increase robustness and reduce model-specific bias. In addition, YOLOv12 is used for the localization of the disease, and Grad-CAM is used to improve the interpretability of the model by visualizing disease-relevant regions. Results of the experiments show that the classification accuracy rate is between 92.00% and 99.36%, and the macro F1-scores are between 0.92 and 0.97. The YOLOv12 model attained an mAP@50 of 88.9% for localization tasks. The results validate the effectiveness and scalability of the suggested framework for intelligent crop disease diagnosis and precision agriculture systems.

Paper ID: 544

An Explainable Machine Learning Framework for Crop Recommendation Using Soil and Environmental Features

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Abstract

Sustainable crop recommendation is a critical component of modern agriculture, enabling optimized resource utilization, improved yield, and enhanced food security, particularly under challenges such as soil heterogeneity, complexity of nutrient management, and climatic variability. Farmers routinely have to decide which crop suits a given plot of land without much guidance on how soil chemistry, nutrient levels, and weather interact to affect yield. This paper builds a machine learning pipeline that recommends crops from environmental and soil measurements, and tests it on two datasets that differ in size by roughly an order of magnitude (10,000 and 100,000 records) to see whether the approach holds up as data volume grows. Ten classifiers, ranging from logistic regression to several tree-based ensembles, were trained and compared on both datasets. None of the individual models matched the Adaptive Weighted Stacking Ensemble (AWSE) introduced here, which combines Random Forest, XGBoost, CatBoost, LightGBM, and Extra Trees through a meta-classifier that learns how much weight to give each base learner rather than averaging them equally. AWSE reached 84% accuracy on the smaller dataset and 94% on the larger one, ahead of every baseline tested, while also keeping per-class performance more balanced across individual crop types. Because a model farmers cannot question is unlikely to be trusted, SHAP was applied throughout, at the level of individual features, individual predictions, and overall model behavior, so the reasoning behind a recommendation can be inspected rather than taken on faith. Together, the accuracy gained from AWSE and the transparency added by SHAP point toward a crop recommendation tool that is accurate enough to be useful and explainable enough for farmers and policymakers to act on.

Paper ID: 550

TriAttenNet: A Lightweight Multi-Branch Separable-Attention Network for Explainable Ash Gourd Leaf Disease Classification

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Abstract

Due to the visual variability and overlapping symptoms of plant leaf diseases and the background clutter in the field, accurate diagnosis of field diseases is challenging. Heavy convolutional neural networks (CNNs) have been proven to be accurate, but they are too expensive to compute on resource-constrained agricultural devices. In this paper, we proposed a lightweight CNN model with a good balance of accuracy and model size for ash gourd leaf disease classification, called TriAttenNet. We proposed a lightweight CNN model with good accuracy and model size for ash gourd leaf disease classification, TriAttenNet in this paper. The network extracts multi-scale lesion features via three parallel depthwise separable convolution (DSC) branches of different kernel sizes. It then merges with the 1x1 pointwise convolution and sharpens them using a dual attention module, which is a combination of channel and spatial attention. TriAttenNet is compared with five backbones (DenseNet121, Efficient-NetB0, ResNet50, MobileNetV2 and VGG16) trained under the same conditions on the public Cucurbit Leaf Disease Dataset (4121 field images, four classes), with a stratified 70:15:15 split and stratified 3-fold cross-validation. The model has a mean validation accuracy of 98.18% and 0.997M parameters. Visual evidence of the relevant regions for prediction is

interpretable and provided by qualitative SHAP and LIME analyses. The results show that TriAttenNet is a compact option for resource-constrained agricultural applications, and further external validation and embedded device benchmarking is required to assure deployment readiness.

Paper ID: 562

A Robust Weighted Ensemble Framework with Explainable AI for High-Precision Cotton Pest Identification

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Abstract

Fast and accurate insect pest recognition is vital in securing yield stability and quality of cotton growing. Although automated pest identification has improved greatly with deep learning, the lack of transparency of these models and statistical reliability remains a challenge for deployments in the field. In this paper, we present a high performance eighted ensemble approach of synergizing three state-of-the-art architectures namely NASNetMobile, ConvNeXtTiny and InceptionV3. In order to obtain a specialized dataset, 1625 images were carefully selected and geometrically and photometrically augmented in strategic way to 7000 images, which was also enhanced by applying CLAHE and amma correction to produce enhanced features. Using 5-fold cross-validation and 95% confidence intervals to ensure a strong generalization, the proposed ensemble model had the best classification rate of 98.95%. To explain the "black-box" nature of neural networks, the Explainable AI (XAI) techniques such as LIME and UMAP were used to obtain "local feature interpretability" and "global manifold visualization. In addition, calibration analysis took place to verify the predictive validity and confidence scores' trustworthiness. The outcomes reveal that the combination of ensemble learning with XAI offers a clear, effective, and scalable solution to enhance precision cotton pest management, enabling high-performing models to link with ease of agricultural application.

TRACK 18: INDUSTRIAL AUTOMATION, DEFECT INSPECTION AND PREDICTIVE MAINTENANCE (IADP)

Paper ID: 94

Fatigue-Aware Break Schedule Optimization for Garment Assembly Lines Using Discrete Event Simulation and NSGA-II

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Abstract

Garment assembly workers face high rates of musculoskeletal disorders from repetitive manual tasks over long shifts, yet break scheduling in the industry follows fixed patterns that ignore how fatigue accumulates and recovers. This paper presents a simulation-optimization framework that couples a dual-component fatigue model with a discrete event simulation of a 20-station polo shirt assembly line and optimizes break schedules using the NSGA-II multi-objective genetic algorithm. The fatigue model combines peripheral muscle fatigue, using a three-compartment controller with rest recovery, with a cumulative strain accumulator for progressive central fatigue. Three objectives are optimized simultaneously: daily throughput, peak worker fatigue, and a modeled injury risk index. Model parameters are grounded in published biomechanical data and in production records from a Bangladeshi garment factory of 1,197 observations. Across 30 independent replications, the optimized schedule reduces peak fatigue by 10.4% and injury risk by 17.1% relative to the industry-standard pattern, at a cost of 5.1%

in throughput, with all differences statistically significant ($p < 0.001$) and large effect sizes. The framework gives garment factory managers a data-driven tool for designing break schedules that balance productivity against worker health.

Paper ID: 308

Adjacency-Aware Graph Transformers for Learning Mesh Dynamics

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Abstract

Learning-based physics simulators have emerged as a promising alternative to computationally expensive numerical solvers for modeling physical systems on complex and irregular meshes, such as those arising in Computational Fluid Dynamics (CFD) and engineering simulations. Existing graph neural network (GNN)-based approaches, such as MeshGraphNets, effectively capture local interactions through message passing but often struggle to model long-range dependencies and global physical correlations. To address this limitation, we propose an adjacency-aware graph transformer architecture for mesh-based physical simulation. A compact stack of GNN layers encodes the mesh topology and derives an adjacency-aware bias, which is incorporated directly into the Transformer's multi-head attention, thereby grounding global attention in the underlying structural connectivity. Together with geometric positional encoding, this unifies local structural information and global contextual interactions within a single framework. By facilitating efficient information propagation across the mesh, the proposed approach aims to improve predictive accuracy and long-horizon rollout stability in complex physical systems. The experimental study on benchmark mesh simulation datasets is encouraging, and we outline the remaining steps toward establishing the proposed architecture as an effective and stable alternative to existing graph-based simulators.

Paper ID: 416

TRUST-RUL: Reliability Auditing of Turbofan Remaining Useful Life Models Under Critical-Region, Uncertainty, and Domain-Shift Stress Tests

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Abstract

Remaining useful life (RUL) studies are often summarized using a single in-domain error value, although reliability-sensitive prognostics also require late-life accuracy, uncertainty calibration, shortcut resistance, and evidence of transportability. This paper presents TRUST-RUL, a reproducible audit protocol and reviewer-facing evidence card for turbofan RUL models. The workflow combines leakage-safe engine-level splits, lead-time-weighted XGBoost, cycle-only and sensor-only ablations, engine-grouped cross-validation-plus intervals, matched temporal baselines, balanced transfer experiments, domain-shift diagnostics, and an external N-CMAPSS DS02-006 stress test. Across the four C-MAPSS subsets, lead-time weighting reduced critical-region root mean squared error by 14.7–20.3%, while global error changed by –2.1% to +5.1%. Cycle-only models performed 44.6–51.4% worse than full-feature models. However, marginal 90% interval coverage of 0.910–0.923 concealed wide intervals and worst-subgroup coverage of 0.774. Direct-transfer error inflation ranged from 16.6% to 130.5%, while the effectiveness of simple adaptation methods remained dependent on the source–target pair. On N-CMAPSS DS02-006, the full model achieved an RMSE of 9.12 and a critical-region RMSE of 8.45; nevertheless, cycle-only performance remained close, and empirical interval coverage decreased to 0.832. These findings demonstrate that benchmark accuracy alone is insufficient for establishing model reliability and identify the additional validation required before making deployment-related claims.

Paper ID: 525

A Multi-Task U-Net Model with ASPP and Attention for Joint Segmentation and Classification of Industrial Defects

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Abstract

In the manufacturing industry, assuring product quality is crucial for sustaining operational efficiency and meeting customer satisfaction standards. This research introduces a novel fine tuning deep learning model capable of performing both segmentation and classification simultaneously. The proposed model utilizes a U-Net architecture for segmentation and a dedicated classification head for object categorization. The Atrous Spatial Pyramid Pooling (ASPP) module records multi-scale features, which makes the model more accurate. Attention blocks help the model focus on important parts of the picture. To enhance model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is implemented, allowing visualization of the regions the model prioritizes during predictions for both segmentation and classification tasks. This research introduces a newly merged dataset that combines two distinct types of industrial products: hexagonal nuts and casting products (pump impellers), including both defective and non-defective samples. The dataset is annotation-based and has been labeled using the online VGG Image Annotator tool. The U-Net model demonstrates high performance in segmentation, achieving a mean pixel accuracy of 99%, mean IoU of 97%, and mean Dice coefficient of 98%. The model achieves exceptional results in the classification domain, with an F1-score of 99.36%, a precision of 99.37%, a recall of 99.35%, and an accuracy of 99.35%. The findings validate the effectiveness of deep learning in automating defect identification in industrial products, thus diminishing human error, eliminating material waste, and enhancing overall production quality and efficiency. A comparative study demonstrates how much better the suggested strategy is than the current techniques. To further optimize defect classification, future studies should concentrate on improving the segmentation of problematic regions and investigating other performance measures.

Paper ID: 551

RoadDG-Lite: Source-Only Cross-Country Generalization and Confidence Calibration for Lightweight Road Damage Detection

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Abstract

Road damage detectors trained on pooled multi- country data degrade when deployed to unseen geographic or acquisition-style environments, yet standard benchmarks conceal this through pooled evaluation. We present RoadDG-Lite, a lightweight source-only framework for cross-country road damage detection evaluated under a strict six-country leave- one-country-out (LOCO) protocol on RDD2022 [1] with an additional zero-shot test on a Bangladesh road-damage dataset. Three modules address the generalization gap without any target data: Domain-Balanced Class-Aware Sampling (DBCS) corrects country-size and class-frequency bias; Damage-Preserving Cross-Country Fourier Style Mixing (DP-CCFSM) transfers road texture statistics across source countries while protecting thin- crack bounding boxes; and source-validation isotonic calibration produces reliable confidence estimates. RoadDG-Lite improves macro-average mAP@0.50:0.95 from 23.3 to 27.6 (+4.3 pp) and worst-country mAP from 18.9 to 23.2 (+4.3 pp) across all six held-out countries, while reducing Detection-ECE from 0.148 to 0.071, with only 3.0 M parameters and 8.1 GFLOPs for field deployment.

TRACK 19: TRUSTWORTHY, FEDERATED AND PRIVACY-PRESERVING MEDICAL AI (TFMA)

Paper ID: 108

Voice-Guided Traffic Sign Classification Using Progressive Deep CNN with Batch Normalization for Driver Assistance

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Abstract

Traffic sign recognition is a safety-critical component of modern intelligent transportation systems and autonomous driving platforms. This paper proposes a deep Sequential Convolutional Neural Network (CNN) that integrates progressive filter expansion, Batch Normalization, and Dropout regularization for robust multi-class traffic sign classification on the German Traffic Sign Recognition Benchmark (GTSRB) dataset. The key novelty lies in a unified end-to-end framework combining high-accuracy CNN classification with a real-time Google Text-to-Speech (gTTS) voice alert module, enabling immediate auditory driver notification upon sign detection without requiring visual attention to a display. The proposed system achieves 99.16% test accuracy on the GTSRB dataset, surpassing the established community baseline of 98.06% by 1.10%, and results confirm strong generalization with minimal overfitting, validating the effectiveness of the combined regularization strategy.

Paper ID: 257

Fed-Physio-Tunnel: Secure Physiological Gating for Reversible Memory Editing in Federated Cognitive Digital Twins

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Abstract

Unlike biological systems, Cognitive Digital Twins are not biologically adaptable and are unable to change memory in response to physiological state, or to process information in a uniform manner across different physiological conditions. This contrasts with a human “cognitive tunnel,” in which acute stress overrides any non-essential memories to focus on tasks critical for survival. We present Fed-Physio-Tunnel, a framework that allows Digital Twins to replicate this phenomenon in a privacy-preserving and cryptographically verified manner, using memory editing. The Stress Evaluator is a light-weight model, trained using VeriFed’s verifiable federated learning, which tracks ECG and EDA signals on edge devices without revealing raw biometrics. When it senses critical stress thresholds, the system produces zk-SNARKs that cryptographically verify true stress conditions, so they cannot be falsified by an adversary. Under certain high-stress conditions, CoV-ME is dynamically triggered, limiting the LLM’s context window to information that is critical for survival. This is reversible, and is done without loss of memory, unlike static pruning. Fed-Physio-Tunnel performs better both in terms of accuracy of high-priority tasks (4.2% accuracy improvement during stress) and inference latency (22% latency reduction while performing memory tasks). The framework reaches high stress detection accuracy of 94.2% while maintaining a low level of privacy leakage (LiRA = 0.09) and robustness against poisoning attacks (ASR = 2.1%), setting the benchmark for secure, biological and adaptive AI that performs under stress.

Paper ID: 487

RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI¹Md Abu Sayeb Rayhan*, ¹Mosammat Fahimah, ¹Md Tarek Mahmud, ²Hustagna Hiya and ¹Ajoy Shutra Dhar¹Comilla University²IIT, Jahangirnagar University

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Abstract

Deep skin-lesion classifiers rival dermatologists on curated benchmarks, yet they fail silently once deployed, where many inputs fall outside the training distribution. No single notion of uncertainty explains this failure: ensembles stay jointly overconfident under covariate shift, evidential deep learning degrades once features shift and conformal prediction certifies only marginal coverage. We address this gap with Reliability-Gated Evidential Fusion (RGEF), which integrates these paradigms through a learned, input-conditional gate that allocates trust on a per-sample basis. Rather than conditioning the gate on the task backbone, whose representation collapses under shift, we condition it on frozen self-supervised DINOv2 embeddings, which are weak detectors in isolation but domain-invariant routers. When an input drifts outside the gate's calibrated support, a density-based fallback smoothly reverts fusion to its static mode. Benchmarked against eleven published post-hoc detectors on an internal near-OOD split (HAM10000, with df/vasc held out), RGEF attains an AUROC of 0.9594 (10-seed mean 0.9596, 95% CI [0.9585, 0.9608]) versus 0.7805 for MSP, 0.8179 for the best single pillar and 0.8239 for a scalar meta-learner, each significant at $p < 10^{-4}$ and cuts FPR@95 from 52.6% to 23.2%. The advantage persists under external shift: on ISIC 2019 (unseen SCC) AUROC rises from 0.6705 to 0.8144 and on PAD-UFES20 (smartphone acquisition) from 0.5567 to 0.6148 above the static and supervised-gate variants, though not above the strongest scalar baselines, a residual modality gap we analyse rather than conceal. A per-class error audit and a three-tier triage protocol connect the score to clinical use. Reliability-aware fusion, rather than any individual estimator, is the key ingredient for safe dermatology AI.

Paper ID: 489

Failure-Aware Brain Tumor Segmentation with Tumor Type Conditioned Reliability Estimation and Explainability Guided Refinement¹Md. Masud Rana Redoy*, ²Tanvir Ahmed Tamim, ¹Sk. Md. Tashfin Sami, ¹Md Abdul Masud and ¹Md. Samsuzzaman¹Patuakhali Science and Technology University²BRAC University

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Abstract

Accurate brain tumor delineation from MRI is essential for surgical planning, radiotherapy targeting, and longitudinal monitoring in neuro-oncology. However, mean Dice alone can conceal catastrophic per-case failures, creating a trustworthiness gap in clinical deployment. We present a unified framework that shares an EfficientNet-B4 encoder across a U-Net decoder, a four-class tumor-type classifier, and a tumor-type conditioned failure prediction head that regresses per-case Dice from spatial mask features and classifier probabilities, exploiting the association between tumor type and segmentation difficulty. High-risk cases are routed to a failure-aware refinement network, where Grad-CAM saliency derived from the failure head provides spatial guidance for targeted correction while also explaining the predicted failure, without requiring ground truth at inference. On BRISC 2025 (6,000 annotated T1-weighted MRI scans), the framework achieves a segmentation Dice of 0.8736 (95% CI: 0.861 0.885; 5-fold CV: 0.8644 ± 0.0041), failure detection AUROC of 0.9908, Average Precision of 0.8187 (22.7× the random-baseline prior), Pearson $r = 0.8104$, and classification accuracy of 99.3% (macro F1 = 0.993), with no significant segmentation degradation relative to a single-task baseline (Wilcoxon $p = 0.981$, $d = 0.025$).

Paper ID: 496

A Triple-Layer Privacy-Preserving Federated Learning Framework for Secure Medical Image Classification

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Abstract

Medical imaging data is highly sensitive, yet training accurate diagnostic models typically requires aggregating patient records across institutions. Federated Learning (FL) avoids raw data sharing, but exchanged updates can still leak training information through gradient, model inversion, and membership inference attacks. Differential Privacy (DP) bounds this leakage but injects noise that degrades accuracy on high-dimensional models, while Homomorphic Encryption (HE) protects aggregation but is computationally prohibitive for large networks. We propose a triple-layer framework integrating FL, DP, and HE through a detached-head architecture, where a frozen pretrained ConvNeXtV2-Nano backbone extracts shared features while only a lightweight classifier head (~2,564 parameters) is trained collaboratively; Opacus DP-SGD protects client gradients and CKKS encryption (TenSEAL) secures aggregation. On COVID-19 Radiography (4 classes, 21,165 images) at $\epsilon = 0.75$, we reach 83.79% accuracy and 83.58% F1, with HE adding only 1.22% overhead and 0.15 points of accuracy cost over DP alone. A model inversion attack shows plain FL+DP allows class-prototype reconstruction at ~100% confidence, which HE-encrypted aggregation reduces to uninformed guessing, while five empirical membership inference attacks remain well below the DP-implied bound. The framework generalizes across non-IID partitions, client counts, and a second imaging modality.

Paper ID: 514

Beyond Data Mirage: Hallucination Detection and Mitigation in Small Language Models

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Abstract

Hallucination is a false perception experienced without any stimulus. In language models, hallucination refers to the response when a language model generates a flawless response which is logically and factually incorrect. Language Models (LMs) predicts answer based on statistical pattern instead of ensuring the accuracy of the verdict. With rise in generative AI adoption, enterprises are inclining more to adopt small language models (SLMs). The goal of this work is to compare the hallucination rate of three small language models (SLMs), Phi, Qwen, Gemma under resource constrained condition. Four mitigation techniques, Self Consistency, Few-shot Prompting, Structured Reasoning and one NLI (Natural language Inference) based contradiction filter were applied to check the performance of the mentioned models under similar constraint. Three hallucination benchmark datasets have been used to test model performance and form a comparative analysis. Qwen model showed the highest hallucination rate of 70.83% across all three datasets under the given constraint whereas Phi model showed the lowest hallucination rate of 68.65% before application of mitigation techniques. Finally, a comparative analysis was performed between mitigation techniques to offer an effective resilience against hallucination confirming the potential of this approach in reducing inconsistency of model response. All the proposed techniques showed a great reduction in hallucination rate, while all the proposed techniques provided the most mitigation rate in Gemma 2 model, the mitigation technique, Structured Reasoning reduced the most hallucination rate with a percentage of 84.21% of correct response generation across all three models followed by Self-consistency, Few-shot Prompting and NLI (Natural Language Inference) Filter respectively.

Paper ID: 558

IB-Net : An Explainable Hybrid CNN-Transformer Model for Brain Tumor Classification¹Sad Ibne Musa Nabil, ²Saifuddin Sagor*, ¹Fuad Hossain, ²Shahariar Hossain, ¹Md. Shafiul Alam Chowdhury and ³Md Masum Billah¹Uttara University²Daffodil International University³Independent University

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Abstract

The automatic classification of brain tumors based on Magnetic Resonance Imaging (MRI) data is an essential part of computer aided clinical diagnosis. Modern deep learning systems, however, tend to have a dilemma of either being local multi-scale feature extractors, global contextual models, or being clinically interpretable. This paper presents the new hybrid CNN-Transformer architecture IB-Net (InceptionV3-BEiT), which addresses the above deficiencies. The IB-Net combines the powerful long-range contextual understanding of BEiT Transformer with the powerful multi-scale feature extraction ability of InceptionV3, using factorized convolutions, in a synergistic way. The proposed model was trained and validated on a large and diverse number of T1-weighted (T1w) contrast-enhanced MRI data (12064 scans), with a thorough multi-step preprocessing protocol (CLAHE and non-local means denoising, and gamma correction) that enhanced the visibility of pathological features. The experimental results confirm that the maximum accuracy of IB-Net is 99.03%, which is far higher than the accuracy of the CNN alone and the other hybrid methods. In addition to predictive performance, we will tackle the "black-box" aspect of deep learning using Explainable AI (XAI) techniques. We show that IB-Net always focuses on the clinically relevant tumor areas while simultaneously showing diagnostic transparency and trustworthiness, by combining the use of Grad-CAM visualizations with a quantitative Intersection over Union analysis. The results indicate that IB-Net is an extremely accurate and interpretable tool for assessing neuro-oncology patients, and has significant potential to be integrated into clinical practice.

TRACK 20: NEUROLOGICAL SIGNAL ANALYSIS AND WEARABLE HEALTH MONITORING (NSWH)

Paper ID: 252

An Ordinal Deep Learning Framework with FT-Transformer and CORAL for Assessing Parental Perception of Smartphone-Related Harm in Bangladeshi Children

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Abstract

Smartphone use among children in Bangladesh has increased rapidly, raising concerns about its effects on academic performance, sleep, and behaviour. Most existing studies treat parental perception of harm as a categorical outcome, overlooking its ordinal nature: neutral, somewhat harmful, and very harmful. To address this gap, we propose an ordinal deep learning framework that combines a Feature Tokenizer Transformer (FT-Transformer) with a CORAL head to predict parental perceptions of smartphone-related harm from a survey of 351 Bangladeshi respondents. The survey includes information on demographics, phone access, content and app use, child outcomes, and parental characteristics. We compare five models under identical 5-fold stratified cross-validation: Random Forest, XGBoost, TabNet, FT-Transformer with cross-entropy, and the proposed FT-Transformer + CORAL. Performance is evaluated using Accuracy, Macro-F1, Quadratic Weighted Kappa (QWK), Adjacent Accuracy, and Kendall's τ , with statistical support from Wilcoxon signed-rank tests, McNemar's tests, and bootstrap confidence intervals. FT-Transformer + CORAL achieves the best ordinal agreement, with a QWK of 0.849, Kendall's τ of 0.833, and Adjacent Accuracy of 0.957, while XGBoost attains the highest raw accuracy of 0.903. An ablation study shows that replacing the cross-entropy head with CORAL on the same backbone improves QWK from 0.675 to 0.849 and Macro-F1 from 0.743 to 0.856. SHAP and Integrated Gradients identify pre-bedtime phone use, homework focus, and educational app use as the most influential predictors. These findings demonstrate that ordinal learning better captures the ordered structure of parental perceptions than conventional multiclass classification, providing more reliable predictions and greater interpretability for educational and public health decision-making.

Paper ID: 258

BrainGATv2-Mamba: Spatial-Temporal Learning for Schizophrenia

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Abstract

Schizophrenia is a neuropsychiatric disorder that affects a person's perception, cognition, emotions, and behavior. Even today, its neurological substrates remain poorly understood. Resting-state functional Magnetic Resonance Imaging (fMRI) provides a non-invasive window into brain connectivity, yet existing automated classification methods largely discard the temporal evolution of functional coupling. To resolve this challenge, this research proposes BrainGATv2-Mamba, a hybrid deep learning framework that encodes dynamic functional connectivity (dFC) through two complementary mechanisms, Graph Attention Network v2 (GATv2) for spatial modeling of inter-regional dependencies within each time window, and Mamba State Space Model (SSM) for capturing long range temporal dynamics across the sequence of sliding window dFC matrices. The proposed method is evaluated on COBRE dataset. BrainGATv2-Mamba achieves an accuracy of 0.7931 ± 0.079 , outperforming SVM, Flat-Mamba, and GATv2-only baselines. A comprehensive ablation study spanning nine architectural variants demonstrates that both the dFC pipeline and each individual component are essential for optimal classification performance. Gradient based

saliency analysis further identifies mid-scan windows as the most discriminative for schizophrenia, consistent with previously reported fronto-parietal dysconnectivity.

Paper ID: 465

Prediction Epileptic seizures using deep learning

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Abstract

Abstract The timely prediction of epileptic seizures is of immense importance to increase safety and quality of life for patients with epilepsy. The electroencephalogram (EEG) signal is widely used for seizure prediction; however, nonstationarity and class imbalance remain open problems for EEG signal analysis. In this paper, a novel method is proposed for early seizure prediction up to 15 minutes prior to onset using a hybrid model of Transformers and BiLSTM for scalp EEG signal prediction. The performance of the proposed method is evaluated using the CHB-MIT dataset with a within-patient split to ensure the evaluation of the model on individual subjects. A unified framework is used to maintain consistency while evaluating the proposed method. To address the problem of class imbalance, data augmentation is used only on the training data to avoid data leakage. The experimental results demonstrate the effectiveness of the proposed method, and the hybrid model achieved an accuracy of 94%, precision of 96%, recall of 95%, F1-score of 96%, and an area under the ROC curve (AUC) of 95%

Paper ID: 468

Explainable Tinnitus Detection Using Audiograms: An Image-Based Deep Learning Approach

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Abstract

Tinnitus is a common auditory disorder that is usually diagnosed through manual inspection of audiograms. Previous works have primarily relied on pure tone audiometry (PTA) or numerical/tabular features for detection. This study focuses on the integration of Deep Learning to automate tinnitus detection by treating audiograms as images for a vision architecture. To this end, a custom CNN architecture has been designed, trained, and evaluated against several pretrained vision models. The proposed model achieved an accuracy of 92%, with a micro-average AUC of 0.971 and an F1-score of 0.92, performing competitively with pretrained baselines an order of magnitude larger in parameter count, while using only 455,682 trainable parameters. Grad-CAM visualizations suggest the model attends to distinct, class-specific regions of the audiogram, with activation patterns for Tinnitus cases concentrating around the point of steepest threshold decline. Findings from the study suggest that CNN-based methods can offer a viable, lightweight, and interpretable approach to audiogram-based tinnitus screening, with potential application to clinical audiology.

Paper ID: 274

Bidirectional LSTM with Temporal Pooling for Cross-Day Robust Intrusion Detection on CIC-IDS2017

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Abstract

The standard approach to evaluating IDSs involves random train-test splits that can overestimate the detection performance in the real world because of the common statistical characteristics of training and test samples.

Network traffic changes over time, and models are subjected to attack patterns that are different from those in the training set. This study investigates the cross-day robustness of the intrusion detection system (IDS) on the CIC-IDS2017 dataset. A time-based split is used — Tuesday through Friday morning traffic is used to train the model, and Friday afternoon PortScan/DDoS traffic is used for testing. A bidirectional LSTM with global average pooling alleviates the short-term dependence constraint across the flows while maintaining the stability of the model with temporal distribution shift. The 1.7M-flow training set is processed by a streaming pipeline without memory overflow or data leakage. The model performance under cross-day evaluation is comparable to 0.929 AUC-ROC and 0.917 balanced accuracy, which is comparable to the near-0.99 performance under a conventional random split. This gap indicates that standard evaluation overestimates the actual detection performance, and temporal partitioning is required to accurately assess it.

Paper ID: 534

SMOTE-Augmented Gradient Boosting for Epileptic Seizure Detection from Imbalanced Scalp EEG

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Abstract

The difficulty in detecting epileptic seizures from scalp EEG is primarily because ictal events are rare. In a continuous recording, seizure segments are a small proportion of the data, and many published detection systems have been able to achieve a high overall accuracy whilst detecting almost no of the real seizures, a failure mode the literature refers to as the Accuracy Paradox. This paper proposes a lightweight, interpretable pipeline that combines temporal, spectral and non-linear features of each scalp EEG epoch, represented in a 19-dimensional hybrid feature vector, and corrects for class imbalance using the Synthetic Minority Over-sampling Technique (SMOTE), which is applied strictly to the training partition after a stratified train-test split to prevent leakage of synthetic samples into evaluation. The same feature representation is used to train four classifiers: XGBoost, Random Forest, Support Vector Machine (SVM), and one-dimensional Convolutional Neural Network (1D-CNN), where the only variable is the architecture of the classifiers. The clinically imbalanced CHB-MIT Scalp EEG Database (ictal-to-inter-ictal ratio close to 1:344) and the pre-balanced UCI/Bonn benchmark (1:4) are two datasets at the extremes of the imbalance spectrum. CHB-MIT achieves the highest sensitivity of 73.50% and an AUC of 0.945, outperforming the feature-vector 1D-CNN with 47.39%, and Random Forest has the highest specificity of 99.59%. In this controlled, feature-vector setting, gradient-boosted ensembles are much more reliable clinically and cost less to operate than convolutional architectures for imbalanced, resource-constrained seizure detection.

TRACK 21: EMBEDDED IOT, HARDWARE SECURITY AND NETWORKED SENSING SYSTEMS (EIHs)

Paper ID: 333

Lightweight CNNs Under Resource Constraints: A Controlled Multigenerational Evaluation of Architecture, Initialization, Training Budget, and Efficiency Tradeoffs

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Abstract

Lightweight convolutional neural networks are widely used when image classification must operate under limits on storage, computation, memory, latency, or training budget. Newer lightweight CNNs are often presented as improving this balance, but such claims are difficult to compare across model families, checkpoints, input resolutions, training recipes, and execution environments. This paper evaluates nine public lightweight CNN model packages across CIFAR-10, CIFAR-100, and Tiny ImageNet under a shared downstream protocol. We report top-1 accuracy, macro F1, top-5 accuracy, parameter count, FP32 storage, GMACs, batch-size-1 latency on an NVIDIA L4 and AMD Ryzen 5 5500U CPU, peak PyTorch CUDA allocated tensor memory, point-estimate Pareto frontiers, EfficientNet-B0 scratch training, and paired model comparisons. EfficientNetV2-S gives the highest observed top-1 accuracy on CIFAR-10 and CIFAR-100 at 97.57% and 86.98%, while RepViT-M1.0 leads Tiny ImageNet at 79.87%. EfficientNet-B0 remains within 0.22, 0.85, and 1.79 points of the best result across the three datasets. Relative to EfficientNetV2-S, it uses about 79% fewer parameters and 86% fewer GMACs. A separate EfficientNet-B0 initialization study shows that even a scratch schedule of 100 epochs does not close the transfer learning gap: scratch training remains 3.29, 10.10, and 17.54 percentage points below pretrained EfficientNet-B0 on CIFAR-10, CIFAR-100, and Tiny ImageNet, while requiring approximately five times the recorded training time. MobileNetV3-Small has the lowest GMAC count, the fastest CPU latency, and higher point-estimate accuracy than MobileNetV4-Conv-S under the tested pretrained and scratch settings. The results show selective rather than universal gains from newer designs and support treating lightweight CNN selection as a controlled multiobjective deployment problem rather than a single accuracy or GMAC leaderboard.

Paper ID: 349

A Decentralized P2P Offline Mesh Networking System Using Drones

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Abstract

This project, developed by our team, presents a low-cost, decentralized drone-based search and communication system engineered to locate humans and establish critical connectivity in environments lacking internet infrastructure, such as disaster zones. The prototype utilizes a fleet of autonomous drones, each integrated with a Pixhawk flight controller, a Raspberry Pi edge computer, and a Raspberry Pi Camera V2. By employing OpenCV and the YOLO image detection library, the system performs real-time human detection while traversing pre-assigned flight paths at altitudes of 10–15 meters. Upon successful identification of an individual, the drone initiates a hover sequence and activates a localized Wi-Fi hotspot. Detected individuals are redirected via a captive portal to an offline-based chatting site, enabling communication without traditional cellular infrastructure. For robust, decentralized networking, the drones are interconnected through LoRa radio modules, providing an effective communication range of up to 3 km in line-of-sight conditions and approximately 350 m in obstacle-rich ground environments. Practical testing with a two-drone prototype confirmed successful inter-node communication,

achieving a localization precision within 3 meters and an average flight endurance of 20–25 minutes using 5500 mAh LiPo batteries. Although challenges regarding waterproofing and GPS accuracy persist due to budget constraints, the system demonstrates a resilient, scalable, and cost-effective solution for search, rescue, and emergency communication. By integrating autonomous vision-based detection with decentralized mesh networking, this research offers a viable framework for enhancing operational capabilities in critical, high-stakes environments.

Paper ID: 353

Design and Evaluation of Lightweight Adiabatic CMOS PUF for IoT Device Authentication

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Abstract

Physically Unclonable Functions (PUFs) have emerged as an attractive hardware security primitive for resource-constrained Internet of Things (IoT) devices by exploiting intrinsic manufacturing variations to generate unique device identities. This paper presents a simplified adiabatic CMOS PUF architecture based on a six-transistor SR-latch structure that reduces circuit complexity by eliminating additional power management requirements while maintaining desirable PUF performance characteristics. Unlike previous adiabatic PUF designs that require additional circuitry and multilevel power supplies, the proposed design eliminates the need for a tri-level power rail and reduces the transistor count through a simplified query mechanism using complementary power clocks. The proposed architecture was designed and evaluated using Cadence Spectre simulations in a 180 nm CMOS process. Performance evaluation using Monte Carlo simulations across 100 process variations and temperature analysis performed from 0°C to 100°C demonstrates an energy consumption of only 15.09 fJ/Cb-cycle, a uniqueness value of 50.02%, a reliability of 99.3%, a uniformity of 49.88%, and an average bit-aliasing of 49.88%. These results show that the proposed PUF achieves near-ideal randomness and robustness while providing competitive energy efficiency with reduced circuit complexity, making it well-suited for secure authentication in low-power IoT applications.

Paper ID: 462

High-Isolation Orthogonally Oriented 2-Port MIMO Vivaldi Antenna for Ku/Ka-Band Satellite Applications

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Abstract

This paper presented an orthogonally oriented 2-port antipodal Vivaldi antenna (AVA) MIMO configuration for Ku/Ka-band satellite communication applications. The proposed antenna was designed and analyzed on a Teflon substrate ($\epsilon_r = 2.1$, $\tan\delta = 0.0002$) and excited through a 50- Ω microstrip feed line, covering an operating frequency range from 14 GHz to 40 GHz. Initially, a single AVA element was designed and analyzed, where a reflection coefficient as low as -40 dB was achieved at 16.03 GHz, while a peak gain of 10.84 dB was obtained at 39 GHz. To achieve wide-angle coverage, two antenna elements were positioned orthogonally, providing nearly 180° directional coverage. A half-circular ground structure was incorporated into the ground plane beneath the flare region to suppress mutual coupling and improve isolation performance. The proposed AVA MIMO-2 configuration achieved S-parameter values below -10 dB and isolation better than -35.5 dB across the entire operating band, while maintaining a peak gain of 8.35 dB. Furthermore, the antenna demonstrated excellent diversity performance with an ECC below 0.00018, DG of 9.9991 dB, CCL less than 0.4 bits/s/Hz, and TARC below -10 dB. The obtained results confirmed that the proposed antenna was a suitable candidate for high-isolation wideband Ku/Ka-band satellite communication systems.

Paper ID: 499

Adaptive Replanning for Multi-UAV Swarm Search-and-Rescue Missions Under Failure and Priority Constraints

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Abstract

Multiple-UAV swarms can improve search-and rescue missions by covering disaster areas quickly and maintaining operation after UAV failures. However, fixed or distance-based task-allocation methods may perform poorly when drones fail, batteries become limited, or high-priority victim regions must be searched first. This paper presents ARS-Swarm, an adaptive replanning strategy for multi-UAV search-and-rescue missions under failure and priority constraints. The method assigns unfinished search sectors using a mission-aware score that combines UAV-to-sector distance, estimated energy demand, workload, battery risk, local hazard, and victim-priority reward. A Gazebo disaster-field environment is developed with ten UAVs, twenty search sectors, twelve victim locations, fire zones, damaged roads, collapsed structures, reduced-battery UAVs, and injected failure events. ARS-Swarm is evaluated across three Gazebo scenarios and compared with Static Sector Allocation, Greedy Nearest Search, Frontier Search, and Auction Replanning. All metrics are computed from Gazebo entity-state logs. The results show that ARS-Swarm achieves $90.00 \pm 8.66\%$ coverage, detects 11.33 ± 0.58 victims, and detects all 7.00 ± 0.00 high-priority victims at the 60 s mission deadline. Although Static Sector Allocation achieves higher coverage, it detects fewer high-priority victims. These results show that ARS-Swarm provides an explainable priority aware and failure-aware replanning framework for degraded multi-UAV SAR missions while maintaining competitive overall mission performance.

Paper ID: 503

A Dual-Controller Internet of Things Programmable Electronic Load for Autonomous Multimode Battery Characterization

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Abstract

Low-cost electronic loads typically provide limited automation, remote supervision, and battery-health assessment, while many Internet of Things (IoT) monitoring systems support only passive observation. This paper presents an edge-assisted IoT programmable electronic load for autonomous battery characterization and real-time telemetry. A dual-controller architecture uses an Arduino Nano for deterministic sensing and load regulation and an ESP32 for wireless monitoring, dashboard operation, and comma-separated values (CSV) logging, thereby separating communication tasks from the load-control loop. A shared adaptive error-band controller supports constant-current (CC), constant-power (CP), and constant-resistance (CR) modes without separate gain tuning. A ramp-hold-pulse-recovery (RHPR) sequence enables pulse-based internal-resistance estimation, Coulomb-counting state-of-charge estimation, capacity-based health indication, and a rolling-baseline anomaly detection mechanism. Bench and field tests using three DC sources demonstrated voltage tracking deviations below 3% and current regulation deviations below 0.6% across the tested operating conditions. CR operation over the tested 30–50 Ω range resulted in a mean absolute deviation of 0.303 Ω and RMSE of 0.375 Ω , with the values reported as descriptive metrics over the evaluated setpoints, while the 300 mA hold stage settled at 300.3 ± 0.3 mA.

Paper ID: 524

ULTRAFLOAT: An ESP32-S3-Based Non-Contact Water Level Sensing and Pump Control Platform with Dynamic Speed-of-Sound Compensation

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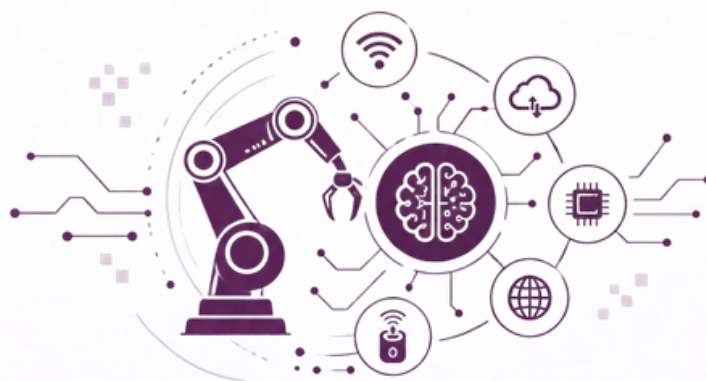
Abstract

The ULTRAFLOAT intelligent, contactless, temperature compensated water level monitoring and automated pump control system is presented to overcome the basic drawbacks of mechanical float switches, which are binary-only state reporting, mechanical wear and corrosion, and lack of a remote monitoring function. In an ultrasonic distance measurement, one of the most important requirements is temperature compensation, since the speed of ultrasonic waves depends on the ambient temperature ranging from 340 m/s at 15 °C to 353 m/s at 36 °C, causing up to 2% of systematic measurement errors. The ULTRA- FLOAT algorithm rejects this drift completely with a real-time dynamic compensation system that is based on the relation $V = 331.3 + 0.606 \times T$ m/s, where T is temperature measured by an AHT20 sensor located at the tank site. It is based on the ESP32-S3 microcontroller with a waterproof ultrasonic sensor JSN-SR04M that works on the TOF (time of flight) principle, which is powered by a specially designed Switched-Mode Power Supply (SMPS) delivering 12 V DC power with a conversion efficiency of 96%. The Distributed Hardware Design consists of a single RJ45/CAT5 cable that carries power and communications to the remote pump, which is controlled by a responsive single- page web dashboard communicating via a WebSocket link to the Control Unit to provide real-time telemetry. This makes it an affordable, efficient and almost attainable alternative to a conventional float switch solution.

**IEEE 5TH INTERNATIONAL CONFERENCE ON
ROBOTICS, AUTOMATION, ARTIFICIAL-INTELLIGENCE
AND INTERNET-OF-THINGS (RAAICON) 2026**



CO-LOCATED EVENTS



Jointly Organized By:



Hosted By:



Jashore University of
Science and Technology (JUST)

2026 IEEE REGION 10 · HUMANITARIAN TECHNOLOGY ACTIVITIES (HTA)

Humanitarian Technology Exhibition & Volunteer Training Program

Held alongside IEEE RAAICON 2026 | 19 September 2026 | Jashore University of Science and Technology

The proposed IEEE R10 Humanitarian Technology Activities (HTA) project is intended to address critical needs within our local community by leveraging appropriate and sustainable technologies to improve quality of life and community resilience. Many underserved communities in our region continue to face challenges related to limited access to technical knowledge, digital resources, and sustainable practices. This project aims to bridge these gaps through practical, low-cost, and community-oriented technological activities supported by engineering expertise and volunteer engagement.

The initiative will also emphasize sustainable development by promoting technology-enabled solutions, local capacity building, and knowledge transfer to ensure long-term benefits. Through training, awareness programs, and hands-on activities, the project will empower students, educators, youth, and community members to adopt innovative approaches for solving local challenges.

Alongside IEEE RAAICON 2026, this humanitarian project exhibition will be conducted following the volunteer training program, where a panel of judges will assess each project. A certificate will be issued to every participant, and awards will be granted to the top three project teams.

AT A GLANCE	
Date	September 19, 2026
Venue	Jashore University of Science and Technology, Jashore 7408, Bangladesh
Organized by	IEEE Region 10 · Bangladesh Section
Co-Organized by	IEEE JUST Student Branch
Events	Held alongside IEEE RAAICON 2026
Team size	1 to 5 members
Focus	Humanitarian technology for underserved communities, spanning health, education, agriculture, water & sanitation, disaster resilience, energy, and livelihoods

Why This Event Exists

Across Bangladesh and the wider Region 10, many communities still face limited access to essential services, such as healthcare, clean water, reliable energy, education, and economic opportunity. This IEEE Region 10 Humanitarian Technology Activities (HTA) project asks students to bridge these gaps with practical, low-cost, community-oriented engineering.

The work is explicitly framed around the UN Sustainable Development Goals; not a single goal, but the broader set that touches underserved and marginalized communities: good health and well-being (SDG 3), quality education (SDG

4), clean water and sanitation (SDG 6), affordable and clean energy (SDG 7), decent work and economic growth (SDG 8), reduced inequalities (SDG 10), sustainable communities (SDG 11), and climate action (SDG 13).

SDG 3 Good Health & Well-being	SDG 4 Quality Education	SDG 6 Clean Water & Sanitation	SDG 7 Affordable & Clean Energy	SDG 8 Decent Work & Economic Growth	SDG 10 Reduced Inequalities	SDG 11 Sustainable Communities	SDG 13 Climate Action
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Projects are judged on whether they could genuinely improve lives, widen access to essential services, or address a socio-economic or environmental challenge in an underserved setting.

Why Take Part

Open to engineering, science, and social-development-minded students with an interest in humanitarian work, whether the idea is a health device, an education tool, a livelihood platform, or a disaster-response system.

Show your work

Present to academics, industry experts, and humanitarian professionals, and get judged feedback you can act on.

Learn from experts

Technical workshops and industry-focused sessions delivered by distinguished IEEE and academic speakers.

Get certified

Every participant receives an official certificate recognizing their participation.

Share ideas

Trade approaches with students and researchers from universities across Bangladesh and Region 10 working on similar problems.

Win prizes

Prize money and certificates for the champion and runners-up, awarded at the closing ceremony.

Build your network

Meet IEEE leaders, humanitarian professionals, researchers and students from across Region 10.

Program — Three Segments

SEGMENT 1 Volunteer Training Program

Session I Humanitarian Project Design

The key aspects of designing a humanitarian project that is both innovative and sustainable, from problem framing within a community's real socio-economic context to a solution the community can keep running on its own.

Speaker: Prof. Shaikh A Fattah

Professor, Dept. of EEE, BUET, Dhaka · Past Chair, Education, IEEE Humanitarian Activities Committee (HAC)

Session II Technology-Driven Solutions for Underserved Communities

Using appropriate technology, low-cost digital tools and community training to improve education, agriculture, healthcare awareness, disaster resilience, clean energy access and digital/financial literacy across SDG themes rather than one sector alone.

Speaker: Prof. Celia Shahnaz

Professor, Dept. of EEE, BUET, Dhaka · Chair, IEEE SIGHT Group FLASH, Bangladesh Section

SEGMENT 2 Humanitarian Project Exhibition

For teams exhibiting a working prototype, model, or project that tackles a socio-economic, environmental, or public-service challenge in an underserved community. A panel of judges evaluates every project; certificates go to all participants and prizes go to the top three teams.

- **Presentation mode:** Oral presentation, plus a live demonstration where the project allows it.
- **Team size:** Minimum 1 member, maximum 5 members per team.
- **On the day:** Bring a fully functional prototype. Winners are announced at the closing ceremony.

FOCUS AREAS

01	Low-cost healthcare & diagnostic devices for underserved communities	06	Education & digital-literacy tools for underserved learners
02	Assistive & accessibility technologies for persons with disabilities	07	Livelihood & economic-empowerment technology
03	Agricultural technology & food security solutions	08	Community informatics, data-for-good & digital inclusion platforms
04	Clean water, sanitation & affordable/renewable energy solutions	09	Rehabilitation, physiotherapy & mental health support tools
05	Disaster response, early-warning & resilience systems	10	Wearable & remote health-monitoring systems

SEGMENT 3 Keynote Talk

Keynote Empowering IEEE Volunteers

How the IEEE Humanitarian Technology Board inspires and equips volunteers worldwide to run humanitarian technology activities at a local level, across the full range of SDGs.

Speaker: Amira Ouerfelli

2026 IEEE SIGHT Chair · IEEE MGA Representative, Humanitarian Technology Board (HTB)

HTA PROJECT EXHIBITION

Participating Projects

31 projects | 9 universities

Team Name: ResiliX

Project Title: FloodGuard Insight: Explainable AI for Spatiotemporal Flood Impact Mapping and Humanitarian Resource Optimization

Saiyedul Abrar, K.M. Shafayat Shapnil, Shahriar Rashid
Computer Science and Engineering, Bangladesh Army University of Engineering & Technology

Abstract

FloodGuard Insight is an explainable artificial intelligence and geospatial decision-support platform designed to improve flood preparedness and humanitarian response in underserved communities. Unlike conventional flood-monitoring systems, it evaluates not only flood risk but also population vulnerability, shelter accessibility, medical access, road conditions, resource shortages, and prediction uncertainty. The software prototype generates realistic synthetic rainfall, water-level, and community data. AI models analyze these time-series data to predict flood-risk levels and estimate future impacts. The platform divides the study region into geographic grid cells and generates interactive heatmaps showing flood probability, predicted water depth, vulnerability, shelter inaccessibility, resource shortages, uncertainty, and overall humanitarian priority. An explainability module identifies why a specific area is considered high-risk and recommends suitable actions. A resource-optimization module then suggests how limited rescue teams, food, water, medicine, and shelter capacity should be allocated to the most vulnerable locations. The current system is fully software-based and uses simulated data for demonstration. Its architecture is prepared for future integration with low-cost IoT devices, such as ESP32-based water-level and rainfall sensors connected through LoRa, GSM, or Wi-Fi. FloodGuard Insight aims to transform flood data into clear, equitable, and actionable humanitarian decisions.

Team Name: AXIUM

Project Title: ResQNet - AI-Powered Community Disaster Intelligence & Emergency Response Network

Md. Abu Sayem, Misbahul Huq Sazid, Mst. Sabrina Yasmin, Wasiur Rahman
Computer Science and Engineering, Bangladesh Army University of Engineering & Technology

Abstract

ResQNet is an advanced, localized disaster intelligence network designed to enhance community resilience through real-time data fusion and response coordination. ResQNet is built around distributed intelligent edge nodes (the hardware prototype shown above) that track critical environmental parameters.

This demo integrates key sensors—including gas detection (MQ-series type) for hazardous material tracking and ultrasonic distance sensing (HC-SR04) for flood level or structural impact monitoring—with an Arduino microcontroller core. These live sensor streams are processed by embedded artificial intelligence algorithms on-board for real-time localized risk assessment. The system, as envisioned, computes a localized threat level and presents it immediately (e.g., “STATUS: SAFE RISK: 19%”), providing proactive visual and audible community alerts. The edge nodes then push their prioritized intelligence to a central network hub, creating a common situational map, which may optimize emergency response efforts and resource allocation prior to and during disaster events.

Team Name: Team Andromeda

Project Title: IoT based health monitoring system

Md. Abu Obaida Maaz
Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology

Abstract

Designed and built an IoT-based patient health monitoring system on ESP32, integrating 8 sensors (MAX30102, DS18B20, DHT11, BMP280, MQ2, Flame IR) to track BPM, SpO2, body/room temperature, humidity, pressure, and gas/fire hazards in real time.

Developed firmware in C++/Arduino to process multi-sensor data, implement threshold-based critical-condition alarms with buzzer alerts, and display live readings on an I2C LCD and Arduino Cloud web dashboard.

Team Name: SmartLoad Guard

Project Title: Smart Energy Monitoring With IOT

Tashrif Ahmed, Indranil
Electrical and Electronic Engineering, Hajee Mohammad Danesh Science and Technology University

Abstract

The IoT-Based Smart Load Monitoring and Protection System is an embedded solution designed to track real-time electrical parameters and provide automated circuit protection. Powered by an ESP32 microcontroller, the system measures current draw using an ACS712 current sensor and calculates power and total energy consumption based on a 220V AC grid supply.

Real-Time Energy Metering: Continuously measures RMS current, power (Watts), and accumulated energy (kWh).

Automated Overload Protection: Features an automatic energy limit cut-off mechanism (e.g., 0.005 kWh) that trips a 2-channel relay module to prevent overload hazards.

Dual Monitoring & Control: Allows direct control and reset via local serial commands (ON, OFF, RESET) alongside real-time cloud visualization using the Blynk IoT mobile app.

Cloud Data Telemetry: Streams electrical metrics across four dedicated Blynk virtual pins: Voltage (V0), Energy (V1), Power (V2), and Current (V4).

By combining local physical protection with remote IoT data logging, the system prevents electrical overloading, improves power efficiency, and provides accessible telemetry from anywhere, serving as an adaptable prototype for smart grids and home automation.

Team Name: IUB-IRCPS

Project Title: PIKU 2.1 Rev A: Low-Cost Environmental Inspection Robot for Safer Indoor Hazard Assessment

Md Habibur Rahman Habib
Electrical and Electronic Engineering, Independent University, Bangladesh

Abstract

PIKU 2.1 Rev A is a low-cost environmental inspection robot designed for supervised indoor exploration. It combines temperature, humidity, gas/smoke, flame, ultrasonic, IR, and motion sensing with reactive autonomous navigation and browser-based manual control. The robot can move through an environment, avoid obstacles using local decisions, respond to configured hazards, and send live status to a human supervisor.

The system is intended for remote inspection of potentially hazardous indoor areas before human entry. In situations involving suspected gas leakage, fire risk, abnormal environmental conditions, or physical obstacles, PIKU can help observe the environment from a safer distance while keeping a human in the loop for supervision and control.

Team Name: IEEE_IU_HeXORcists

Project Title: JOINIK: A Proactive Edge-based modular IoT Sensing Network for pre-breach intrusion detection in retail security

Md Iftekhar Ahmed, Numaira Loqman, Imtiaz Mahmud Sobuj, Sakeef Mahmud Foysal, Jareen Tasnim
Information and Communication Technology, Islamic University, Bangladesh

Abstract

Traditional security systems are reactive- only actionable after the intruder has already breached the premises, unable to intervene until the crime has been completed. JOINIK shifts the paradigm from reactive to detective- aiming to catch the attempt of intrusion via shutter tampering, drilling, hammering, power line sabotage etc.

The system runs a three-stage ML pipeline: local signal fusion by combining frequency analysis, power status compared to neighbouring sources; image processed through analyzing motion data via CCTV footage; and a cloud-side risk-decision model combines both to decide whether the interruption classifies as a threat or false positives wind, traffic, and storms via live weather-API weighting. Future planning includes preventative measures such as de-escalating the attempt with non-lethal riot control agents.

Resilience is layered throughout, such as- a GSM failover chain lets backup nodes report if the primary fails. The mmWave radar detects even stationary intruders.

Owners monitor and control everything through a web dashboard and companion mobile app — live node status, event history, and remote alarm cancellations. The platform is designed to bring enterprise-level detection to small and medium retail shops.

Team Name: JSTU_R1

Project Title: Wi-Fi CSI-Based Non-Contact Respiration Rate Monitoring System

Abdullah Al Minhaz
Electrical and Electronic Engineering, Jamalpur Science and Technology University

Abstract

This project aims to develop a non-contact respiration rate monitoring system using Wi-Fi Channel State Information (CSI). The system will utilize multiple ESP32-S3 N16R8 devices as distributed sensing nodes and a TP-Link Archer C6 router as the Wi-Fi access point. Changes in the wireless signal caused by subtle chest movements during breathing will be captured through CSI data. The collected signals will be processed to extract respiration-related patterns, remove noise and interference, and estimate the subject's respiration rate in real time. Unlike conventional respiratory monitoring methods, the proposed approach does not require electrodes, wearable sensors, or physical contact with the subject. The multi-node configuration is intended to improve sensing reliability and provide more robust measurements under different subject positions and environmental conditions. The project will serve as a research prototype for contactless respiratory monitoring and may have future applications in remote healthcare, patient monitoring, sleep monitoring, and assisted living environments.

Team Name: JSTU_R2

Project Title: Low-Cost Real-Time ECG Monitoring System for Accessible Cardiac Health Screening

M. Miyad Islam Nion, Md. Umar Faruk, Md. Nirob Hossain, Abdullah Al Minhaz
Electrical and Electronic Engineering, Jamalpur Science and Technology University

Abstract

This project presents a low-cost, real-time single-lead ECG monitoring system designed for accessible cardiac health screening. The system uses an AD8232 ECG sensor module and Arduino Nano to acquire ECG signals from three surface electrodes and calculate heart rate in real time. The acquired ECG waveform and BPM data are transmitted to a computer through USB, where a custom Python-based GUI provides live waveform visualization, BPM monitoring, lead-off detection, session recording, and automated PDF report generation. The system is designed as a portable and affordable alternative for ECG signal acquisition and health-awareness applications, particularly in settings where conventional ECG equipment may be less accessible. A working prototype will be demonstrated through live ECG signal acquisition and monitoring. Future development can extend the system toward wireless data transmission, multi-lead ECG acquisition, cloud-based monitoring, and intelligent ECG analysis.

Team Name: EnviroGuard

Project Title: IoT-Based Intelligent Environmental Monitoring System

Md Fazla Bari Mithu, Arafat Islam, Md. Navid Hasan
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

This Project presents an IOT-Based Intelligent Environment monitoring system designed to provide real-time monitoring and automation control in different indoor environments. The system uses here ESP32 and multiple sensors to monitor important environmental parameters, including temperature, humidity, smoke, odour, dust concentration, flame detection, and human presence. For the purpose of remote monitoring, the system can be integrated with the Blynk IoT platform. In this case, a user will be able to collect all the data and check sensor values remotely from their smartphone or tablet. Furthermore, the data from sensors can be sent to Google Sheets for future analysis and visualization. Overall, the proposed system can be implemented in different environments in order to ensure safety and provide automatic control. The system can be utilized in agricultural facilities, food storage, laboratories, classrooms, offices, warehouses, or other enclosed places.

Team Name: FloodEye

Project Title: IoT-Based Flood Risk Monitoring and Emergency Response System

Salsabil Jannat, Maria Sultana Nova, Md. Mosharof Hossain Mashrafi, Md Fazla Bari Mithu
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Flood Early Warning, Environmental Monitoring and Rescue Support System is a project that gives warnings and help before during and after floods. Flood Early Warning uses an ESP32 and HC-SR04 ultrasonic sensor to check water levels and conditions into Normal Safe, Warning and Danger. Flood Early Warning shows alerts with yellow and red LEDs and a buzzer that makes a sound. Flood Early Warning has a DHT22 sensor that watches temperature and humidity and an MQ-2 gas sensor that spots smoke or gas in the area and raises a warning when the level goes above a set limit. Flood Early Warning has a display that shows water level and other environment data in real time. When water is dangerous Flood Early Warning uses an A7670C GSM module to send emergency SMS to people in danger. People can reply with their location and Flood Early Warning sends that location to rescuers with details. Flood Early Warning also has a soil moisture sensor to check soil after floods so farmers know how dry the land is and can recover. Flood Early Warning is an versatile tool for flood warnings emergency messages, environment safety and farming, after floods.

Team Name: Green Pulse

Project Title: Multi Crop Irrigation System

Jahidul Islam Shanto, Milton Mistry
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Bangladesh is an agriculture-based country where for sustainable production and proper utilization of water sources there is a crucial need for effective irrigation. This paper presents a smart irrigation system for multi-crop and crop-age aware system, which consists of the ESP32 microcontroller, DHT11 sensor, soil moisture sensor, relay module, water pump with machine learning controller that is low cost and affordable. Based on the crop type and age, alongside soil moisture, temperature, humidity, and an engineered dryness index, the proposed system is designed to predict crop irrigation needs. We tested Extra Trees, Random Forest and Gradient Boosting classifiers with a dataset of 501 samples from nine crop types. Extra Trees classifier with 93.07% maximum classification accuracy was able to identify all samples in the test set that needed irrigation, having no false negative. The expected Pump ON/OFF signal is transmitted to ESP32 which is responsible for the automatic operation of the relay and the water pump. The suggested framework can provide an implementable real-time decision-making for irrigation management with consideration to the specific crop and can help to ensure more effective use of water in the agriculture sector in Bangladesh and elsewhere.

Team Name: PowerCore

Project Title: Portable and Modular based DC Power System

Md. Ashiqur Rahman, Mohammad Anisul Mostafa Badsha, Mahjabin Rahman Orhi, Limon Haldar
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

This project presents a low-cost, standalone, and modular DC power system designed to provide flexible and reliable renewable energy access for off-grid and underserved communities. The proposed system integrates solar charging, battery storage, and a dedicated Solar-Battery Control System that manages load priority, protection, and power distribution. The control system is connected between the battery bus and load bus, creating a plug-and-play architecture that allows users to easily connect or disconnect power modules and loads without redesigning the entire system. This modular approach makes the system simple to install, maintain, and expand according to changing energy requirements. Essential DC loads can be prioritized to ensure better utilization of available stored energy while integrated protection improves operational safety. By combining renewable energy, modular design, intelligent load management, and affordable components, the proposed system provides a practical and scalable solution for rural households, remote locations, and emergency applications where reliable electricity access is limited.

Team Name: SustainaVolt

Project Title: Development of a Dual-Mode IoT-Based Water Quality Data Acquisition Platform for Machine Learning-Based Dissolved Oxygen Prediction

Anindhow Chowdhury, Woliar Rahman
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Maintaining optimal water quality—especially dissolved oxygen (DO)—is essential for aquaculture productivity, yet traditional manual monitoring and basic threshold systems fail to capture complex environmental dynamics or offer predictive insights. To address this, this project presents a dual-mode IoT-based data acquisition platform powered by an ESP32 microcontroller, designed to gather high-quality datasets for machine learning (ML)-based DO prediction. The platform continuously tracks key parameters—including water pH, turbidity, TDS, temperature, humidity, light intensity, and pressure—using two operational modes: offline logging to a MicroSD card synced via a DS3231 RTC module for remote stability, and real-time wireless telemetry via Wi-Fi for cloud integration. To ensure hardware reliability, all analog inputs are routed to ESP32 ADC1 channels to avoid Wi-Fi conflicts, featuring voltage-matched circuitry and a modular architecture for easy sensor expansion. Ultimately, this scalable system provides the foundational data required for ML models to enable predictive aeration control, lower energy costs, reduce fish mortality, and advance sustainable aquaculture practices.

Team Name: Team Infra-red

Project Title: Development of an Automated Smart Infrared Therapy Unit for Safe DeepTissue Pain Relief in Physiotherapy

Joyanto Bormon, Sajib Islam Khan, Md. Rafid Al Jami, Md. Naeem Howlader
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Infrared therapy tool is one of the tools in physiotherapy. Infrared therapy tools with temperature control are utilized as controllers to prevent the risk of burns due to excessive heat exposure. This infrared lamp exposure temperature controller is designed using a diode connected to a solid state relay (SSR), which is able to dim the infrared lamp if the temperature exceeds the set one, the temperature can be detected by the AMG8833 sensor. Arduino Nano as the brain of the system that controls input and output. This research uses the research and development method, which is a method used to make a product and test the effectiveness of the product results. Research and development on infrared therapy tools with AMG8833 sensors to detect temperature in patients during therapy. The results of this study are the highest measurement difference = 1.03° C, the lowest measurement difference = 0.05°C making comparisons with temperature measuring instruments. The average calculation on all respondents obtained a temperature of 41.46° C by measuring body temperature before and after therapy. The conclusion of the infrared therapy tool can be used to patients because it has been validated by comparing measurements with temperature measuring instruments and infrared therapy tools can be used safely to patients.

Team Name: AgroFly

Project Title: Pesticide Spraying Drone

Md. Al Muzahid Pranto, Estiak Hossen, Sifat Rahman, Mohammad Basel
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Small-scale farmers in Bangladesh currently rely on manual pesticide spraying, which is both time-consuming and exposes them to harmful chemicals. Commercial agricultural drones could help by reducing the amount of work needed, but their high cost and limited capacity often prove impractical for farmers with small plots of land. This project proposes a low-cost and compact agricultural spraying drone that can carry approximately 800 g of liquid nano-urea pesticide per flight to address the needs of these farmers. In order to increase the amount of spraying this project utilizes a fast battery replacement method so that the operator can carry 4-5 low-cost, easily accessible batteries in Bangladesh and perform multiple 800 g spraying cycles in one day. This project emphasizes to serve the small-scale agricultural plots, rather than large-scale commercial agriculture. The project then introduces the idea of a low-cost drone that can reduce the exposure of farmers to harmful chemicals and serve as an alternative to both manual pesticide spraying system and high cost commercial agricultural drones.

Team Name: AquaSync

Project Title: IoT-Based Flood Monitoring, Alarming and Automatic Sluice Gate Control System

Humayra Jahan, Nafis Abedin, Anindita Mazumder, S.M. Sawat
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Flooding is a common natural hazard that can cause serious damage to lives, agriculture, infrastructure, and local communities. This project presents an IoT-Based Flood Monitoring, Alarming and Automatic Sluice Gate Control System developed to monitor rising water levels and provide an automatic response to critical conditions. An ESP32 is used as the main controller to collect data from different sensors. The HC-SR04 ultrasonic sensor continuously measures the distance between the sensor and the water surface, while a water-level sensor provides an additional indication when the water reaches a critical level. A rain sensor is also included to detect rainfall and support the overall decision-making process. Based on the water-level condition, the system provides green, yellow, and red LED indications and activates a buzzer alarm when the situation becomes critical. An SG90 servo motor automatically operates the sluice gate to regulate water flow and reduce excessive water accumulation. A 16×2 I2C LCD displays important sensor readings and system status. The ESP32 also provides the foundation for IoT-based monitoring. Overall, the proposed system offers a simple, low-cost, and automated approach to flood monitoring, early warning, and water-flow control.

Team Name: Caremate

Project Title: Design and Development of a Low-Cost Smart Integrated Biomedical Monitoring System for Real-Time Monitoring of Vital Signs and Motion

Jannatun Nayeem, Yeasmin Mim
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

This project presents the design and development of a low-cost smart integrated biomedical monitoring system for real-time observation of essential physiological parameters and body motion. The proposed system utilizes an ESP32 microcontroller to integrate multiple sensors for measuring body temperature, heart rate, blood oxygen saturation (SpO₂), humidity, and acceleration. The MAX30102/MAX30105 sensor is used to monitor heart rate and SpO₂, while the LM35 and DHT11 sensors measure temperature and humidity, respectively. An MPU6050 accelerometer and gyroscope are incorporated to detect motion and changes in body orientation. The collected data are processed by the ESP32 and displayed on an OLED screen, enabling convenient and continuous monitoring. The system is designed with affordability, portability, and ease of use in mind, making it potentially suitable for preliminary health observation, home-based monitoring, and resource-limited healthcare environments. Although the system is not intended to replace clinical-grade medical equipment, it demonstrates the potential of embedded technology for accessible and integrated health monitoring.

Team Name: IMPULSE TECH

Project Title: NEURONEX – Intelligent Paralysis Patient Monitoring System Based on Hand Motion and Vital Signs

Nazim Ahmed, Shahriar Hosen, Sumon Hossain, Md Takib Hossain
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

NEURONEX is a smart healthcare monitoring system designed to support paralysis patients and their caregivers. It enables continuous observation of essential health conditions and helps identify abnormal changes at an early stage. The system aims to improve patient safety, reduce the need for constant manual supervision, and provide timely alerts during critical situations. It is designed as an affordable and accessible healthcare solution, particularly for patients with limited mobility. NEURONEX also aims to support caregivers by providing reliable health information and improving their ability to respond quickly when necessary. The project focuses on enhancing patient care through smart and practical healthcare technology.

Team Name: NextGen Biomed

Project Title: OtoNasoThroX–PharyScan: A Low-Cost IoT-Based Portable ENT Device with EfficientNetB0-Assisted Real-Time Pharyngitis Detection

Md. Azraful Islam Khan, Jasim Uddin, Trisha Barua
Biomedical Engineering, Jashore University of Science and Technology

Abstract

In rural and low-resource areas, specialist ear, nose, and throat (ENT) care is hard to reach, and common ENT problems often go undiagnosed for too long. This work presents a low-cost portable ENT system that pairs simple Internet of Things (IoT) hardware with a deep learning model for pharyngitis screening, named OtoNasoThroX–PharyScan. The device comprises an ESP32-CAM (ESP32 system-on-chip with an on-board camera) module inside a 3D-printed enclosure designed in SOLIDWORKS. An Android app, OtoNasoThroX, handles the live view and image capture, and a Flask web app, PharyScan, runs a fine-tuned EfficientNetB0 model to classify each throat image. We compared four pretrained convolutional neural networks (CNNs): ResNet18, ResNet50, MobileNetV3-Small, and EfficientNetB0, on a public dataset of pharyngitis and non-pharyngitis throat images. Model selection used stratified 5-fold cross-validation, class-weighted loss, and two-phase fine-tuning, under a leakage-free protocol in which an independent 60-image test set was frozen before any model selection and audited for exact-duplicate leakage. EfficientNetB0 gave the best cross-validation AUC-ROC (0.9717 ± 0.0069). On the frozen independent test set, averaged over five training seeds, it reached $97.33 \pm 1.70\%$ accuracy and an AUC-ROC of 0.9965 ± 0.0043 , with well-calibrated confidence (expected calibration error, ECE, ≈ 0.056). An ablation confirmed the contributions of two-phase fine-tuning, data augmentation, and class weighting. Grad-CAM visualizations indicate the model attends to the oropharyngeal region. However, a quantitative faithfulness test shows these maps are best read as qualitative support rather than proof of clinical reasoning. The system returns a classification within about 1–2 seconds. Taken together, the results suggest low-cost IoT imaging, a mobile app, and deep learning can combine into one practical tool for early ENT screening.

Team Name: PulseGuard**Project Title: Wrist Guardian: A Low-Cost, Real-Time IoT Wearable for Heatwave Protection in the Elderly**

Masud Rana, Md. Shoriful Islam, Md. Fazla Bari Mithu, Nafis Abedin
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

Most low-cost wearables are able to measure physiological or environmental parameter - but not both, which is significant when referring to elderly and their physiological risk due to the increased frequency of heat waves. This paper proposes the concept of Wrist Guardian, a wearable device with motion, environment and physiology sensing capabilities based on IoT, designed by use of the ESP32 microcontroller, the ADXL345 sensor, the DHT11 sensor, the MAX30102 sensor and the MLX90614 sensor. This deterministic rule-based algorithm checks 5 engineering thresholds per observation and sets the status into RISK or GOOD, sends local alerts to the OLED and to the vibration, sends remote alerts via Telegram and logs into MicroSD as CSV. In addition, a conceptual Heat-Stress Risk Score (HSRS) is also introduced for future adaptive modeling although results are presented here using only the rule-based detector. For three days, there were 560 outdoor observations collected from an engineering validation with 60 elderly volunteers (aged 60-70 years) without any personally identifiable data. Of these, 312 (55.7%) were classified as RISK; heart rate ranged 66–129 BPM (mean 96.35) and SpO₂ ranged 86–99% (mean 93.19%). The prototype demonstrates the feasibility of developing low-cost multi-sensor heat-stress wearables in resource-constrained environments, at a cost of BDT 3,760 (≈USD 30).

Team Name: Team Agantuk**Project Title: FireGuard AI: AI-Based Early Fire Detection and Warning System (Offline)**

Ragib Ishraq, Mirza Mahadir Hasan, Toufique Omar, Minhaz, Sourav Kamal
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

FireGuard AI is a low-cost offline AI system that can provide early fire/safety warning for home, school, workshop, market, community centers or highly compact settlements in places where traditional fire monitoring systems may be too expensive or unfeasible.

It turns any standard webcam or existing CCTV camera connected to a laptop into an intelligent fire monitoring system. An underlying fire and smoke specific YOLO deep learning model runs on the GPU (when available) or CPU to analyze the incoming video stream. There is no need for any additional computer hardware or constant internet connection.

To avoid false alarms, it does not consider individual frames but uses a combination of confidence, object persistence, area of the frame and growth to classify events into 'SAFE', 'SUSPICIOUS', 'WARNING', 'CONFIRMED' and 'CRITICAL' levels.

In case of a confirmed event, it displays information about the emergency on-screen and generates an incident pack with annotated images, video, metadata and a database record.

Processing and storage is done on device to ensure privacy and offline operation. Bangla documentation and a Windows friendly interface allows untrained volunteers or even non-technical people to setup and operate FireGuard.

Team Name: Team Habla

Project Title: Low-Cost IoT-Enabled Portable Smart Multi-Sport Match Management System with Wireless Substitution Control, Integrated Scoreboard, and Match Time Synchronization

Md. Al Muzahid Pranto, Estiak Hossen, Sifatur Rahman
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

This project describes the design and development of a low-cost, portable, IoT-enabled smart system for managing multi-sport matches in schools, universities, and sports clubs. The system combines wireless player substitutions, real-time score management, match timing, editing team names, and recording substitution history into one portable device. Using an ESP32 microcontroller, the system enables wireless communication and web-based control, while an Arduino Nano handles real-time processing and powers P10 LED matrix displays. Match officials are able to manage substitutions, scores, team details, and match timing via a smartphone or laptop using an easy-to-use web interface. A physical keypad has been included so that the system can continue to function when wireless control is unavailable. The prototype was tested during actual football matches and attained a wireless communication range of about 25 metres, with command response and display update latencies under 500 milliseconds, approximately two hours of continuous battery operation, and a 98% success rate in field tests. The P10 LED matrix display ensured good visibility both in daylight and at night. The system can also be set up for handball, basketball, volleyball, and other sports that involve substitutions, showing its versatility as a practical and inexpensive solution for match management.

Team Name: JUST LOOSERS

Project Title: Simulation-driven Patient-specific Hydrocephalus & Eloquent-cortex Risk Estimation

Waliullah, Mokshadul Momin Jim, Md. Samiur Rahman, Md. Redwan Ahmed
Electrical and Electronic Engineering, Jashore University of Science and Technology

Abstract

We develop a patient-specific finite element pipeline that transforms BraTS20 multi-modal MRI (T1, T1ce, T2, FLAIR) into a per-domain deficit-probability risk profile for eloquent cortex. The pipeline includes: (i) parenchymal mesh generation using Otsu thresholding, marching-cubes surface extraction, and iso2mesh Voronoi-Delaunay tetrahedralisation; (ii) solution of the post-resection SlicerCBM boundary-value problem under gravity with penalty contact against the inner skull using a compressible neo-Hookean constitutive model; (iii) per-domain scoring across motor, language, cognitive, sensory, and visual regions using MNI-atlas eloquence weights; and (iv) tertile classification using cohort-relative cutpoints. For the primary cohort ($n = 22$), p99 DPD (median 1.44 mm, range 0.13–3.12 mm) falls within the published cortical-shift (5–25 mm) band, while peak von-Mises stress correlates weakly with cavity volume ($r \leq 0.18$). Headless FEA takes ≈ 90 s per patient using a single CPU. Clinicians can drop a BraTS20 folder to visualize real-time parenchymal deformation or view the per-domain deficit map through a Flask + VTK web service with a client-side WebGL viewer (vtk.js). The code is open-source.

Team Name: 3D FAB KU

Project Title: Design and Development of a High-Voltage, Energy Efficient 3D Printer with TEC-Assisted Thermal Management

Mahathe Hasan Shamim, Prof. Dr. Md. Mahbub Hossain, Prof. Dr. Md. Shamim Ahsan, Ekbal Shamim Pritul
Electronics and Communication Engineering, Khulna University

Abstract

We have made a 3D printer. It can print any type of prototype if the Adobe Illustrator file can be designed. For different types of fracture cases, it can design customized bandages or similar products of customized size that may not be available in the market.

Team Name: Vertex Vangurd

Project Title: FLAMENOVA—Autonomous Fire Suppression Robot

Maysha Tasnim Mahi, Samania Jannati Ahana, Masrurul Hasan Taniym, Shoun Basak
Electronics and Communication Engineering, Khulna University

Abstract

FLAMENOVA is an autonomous fire detection and extinguishing robot designed to provide a rapid response to small-scale fire incidents. The robot uses multiple flame sensors to detect the presence and direction of a fire. Based on the sensor readings, it automatically moves toward the fire source using a motor-driven robotic platform. When the fire is detected, FLAMENOVA activates a water pump through a relay and uses a servo motor to direct the water toward the affected area. The system is controlled by an Arduino, which processes sensor inputs and manages the robot's movement, pump, and water direction. By combining real-time fire detection, autonomous movement, and automatic water-based suppression, FLAMENOVA aims to reduce the need for immediate human intervention in potentially hazardous environments. The project demonstrates how low-cost embedded systems and robotics can be applied to disaster response and fire-safety applications.

Team Name: Bumblebee

Project Title: Flood Early Warning & Emergency Network

Arik Hasan, Tanmoy Islam, Toufiqur Rahman Takdir, Md. Naim Rahman Arafat, Arastatalis Tasdik Khan
Electronics and Communication Engineering, Khulna University

Abstract

Communication Network is a low-cost disaster management system designed to improve flood preparedness and emergency response in vulnerable communities. The system will detect potential flooding by using water-level, weather, and ultrasonic sensors to monitor environmental conditions in real time. When dangerous conditions are detected, the system can provide early warnings to help people take necessary precautions. For emergency communication, ESP32 microcontrollers will use the built-in ESP-NOW wireless communication feature to transmit SOS or emergency signals quickly without relying on internet connectivity. The project is designed to be expandable, allowing additional sensors, warning mechanisms, and communication features to be integrated in the future. The overall goal is to develop an affordable, reliable, and community-oriented flood monitoring and emergency communication network.

Team Name: ElectroVive

Project Title: EdgeGuard V1: Wearable System for Fall Detection and Vital-Sign Monitoring

Md. Arman Sheikh, Srijon Saha, Tanzim Zaman, Afifah Sultana Orin, Sadman Rafid
Electronics and Communication Engineering, Khulna University

Abstract

Falls among elderly or at-risk individuals often go unnoticed, delaying critical emergency response. EdgeGuard addresses this problem through a low-cost, ESP32-based wearable prototype that performs fall detection and health monitoring using rule-based logic rather than machine learning, ensuring transparent, explainable, and computationally lightweight operation. A motion sensor continuously supplies acceleration and angular-velocity data, from which movement magnitudes, orientation estimates, configurable thresholds, and a finite-state machine are used to identify candidate fall events while minimizing false positives. Upon detection, the wearer is given a brief cancellation window via a local buzzer and display before an emergency alert is transmitted wirelessly for remote notification. The system further integrates a pulse-oximetry sensor for heart-rate and blood-oxygen monitoring, and is being extended with a dedicated cardiac sensing module to capture real-time electrical activity of the heart, enabling richer physiological context such as heart-rate variability and rhythm anomalies. All sensing modules are implemented as non-fatal additions, so the core fall-detection and alerting pipeline continues to function reliably even if an optional sensor is absent or fails, preserving the system's safety-first, offline-capable design.

Team Name: Team SHIELD

Project Title: Smart Air & Safety Monitor

Salman Ahmad, Mahmudul Haque, Fazle Azim, K.M Ahasan Abdullah
Electronics and Communication Engineering, Khulna University

Abstract

Smart Air & Safety Monitor, a Remote Health Monitoring System, is designed to monitor and assess the air quality and environmental conditions of a room continuously. The device measures temperature, humidity, CO₂, eCO₂, TVOC, smoke, gas leakage, and flame detection, presenting the results clearly.

The main purpose is to provide an effective way to understand indoor air quality and identify potentially unhealthy or unsafe conditions. Users can observe the current environmental status locally. If gas leakage or fire is detected, an alarm will be activated on both the device and web dashboard. The room owner will receive an alert message, while fire detection will additionally trigger an emergency call or SMS notification to the local fire service for rapid response.

The system sends collected data to a remote server for storage and analysis. A centralized dashboard displays the data clearly, enabling remote monitoring of the room. Both current and historical data can be reviewed to understand changes in environmental conditions over time. Overall, the system aims to support healthier and safer indoor environments through continuous monitoring, remote accessibility, data storage, and early hazard detection.

Team Name: The Annotators**Project Title: Dengue Monitoring System Using AI- based Mosquitos Detection**

Rana Ahmed, Biplob Das, Mostafa Mahmud Raiyan
Electronics and Communication Engineering, Khulna University

Abstract

Dengue fever is one of the most rapidly spreading mosquito-borne diseases in tropical and subtropical regions, transmitted through the bite of the Aedes mosquito. Traditional surveillance methods are largely manual, time-consuming, and prone to human error, making early outbreak detection difficult. To address this, we present an AI-based Dengue Monitoring System that uses Computer Vision and Internet of Things (IoT) technologies for automated, real-time mosquito detection and disease risk estimation.

The proposed system uses a camera to continuously capture images of insects, which are processed by an object detection model trained on the Roboflow platform. The model classifies detected insects into two categories: Aedes Mosquito and Other Insects. Based on classification results, the system calculates a dengue risk percentage using the ratio of detected Aedes mosquitoes to total detected insects.

Team Name: KUET Vanguard**Project Title: Smart Rescue Beacon: A Wearable Emergency Monitoring and Rescue Prioritization System**

Istiaq Ahmed Utsho, Zulkar Nayeem, Sawgata Roy
Electrical and Electronic Engineering, Khulna University of Engineering & Technology

Abstract

During a disaster or search-and-rescue operation, rescuers may have to deal with many victims at the same time, while having very limited information about who needs help most urgently. The Smart Rescue Beacon was developed to address this challenge by giving rescuers a continuous picture of a victim's condition and location.

The system uses a wearable beacon to monitor heart rate, SpO₂, body temperature, and GPS location in real time. The wearer can also manually send SOS or bleeding alerts using dedicated buttons. This information is wirelessly transmitted to a central Smart Rescue Command Center, where multiple victims can be monitored simultaneously.

The command-center dashboard shows each victim's vital signs, emergency status, and location on a live map. It then calculates a rescue priority based on the victim's health condition, emergency status, and distance from the rescue team. The system can also adjust its prioritization according to different disaster scenarios, such as flooding, cold weather, or building collapse.

The key idea is that the system does more than simply send an SOS. It organizes continuously updated information so that rescuers can quickly understand the situation and identify cases requiring greater attention. The prototype demonstrates a low-cost, end-to-end wearable rescue monitoring and triage system, with future scope for long-range communication, improved positioning, and enhanced multi-victim monitoring.

Team Name: Team Deadline

Project Title: AI-Based Smart Assistant for Visually Impaired People (Smartphone App)

Sanzida Alam Jerin, Fariha Tabassum
Computer Science and Engineering, Khulna University of Engineering & Technology

Abstract

This project is a mobile application that uses the smartphone's camera and Artificial Intelligence (AI) to help visually impaired users understand and interact with their surroundings. The app continuously captures images through the phone's camera, analyzes them using computer vision models, and provides voice guidance to the user.

Unlike expensive smart glasses, this solution only requires a smartphone, making it much more affordable and accessible.



ENGINEERING THE DEEP

Designing, Building, and Operating Underwater ROVs

A Workshop on Underwater Robotics

Co-Located with IEEE RAAICON 2026

IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things 2026



September 19, 2026
Start at 10:00 AM



**Jashore University of Science
and Technology**

Save Your Seat



WORKSHOP · CO-LOCATED WITH IEEE RAAICON 2026

Engineering the Deep

A Workshop on Underwater Robotics

Presented by Team MIST MAVIROV | IEEE RAS MIST Student Branch Chapter | 2-Hour Technical Session

Team MIST MAVIROV, the Military Institute of Science and Technology’s first underwater robotics team and members of the IEEE MIST Robotics and Automation Society (RAS) Student Branch Chapter, presents a 2-hour technical workshop on underwater robotics, co-located with IEEE RAAICON 2026. The team recently won the Pioneer Class of the MATE ROV World Championship 2026, recording the highest points ever achieved in that class, and this session distills that competition-tested engineering into a structured technical discussion.

WORLD CHAMPIONS
Pioneer Class, MATE ROV World Championship 2026 — highest points ever recorded in the class

Three Core ROV Subsystems

The workshop covers the three core subsystems of a remotely operated vehicle (ROV).

Mechanical	Electrical & Power	Software, Controls & Perception
Frame design and material selection for submerged operation, buoyancy and ballasting for neutral trim, gasket sealing and enclosure design for waterproofing, thruster configuration, and manipulator/gripper mechanisms.	Power distribution and integrity within a sealed enclosure, motor drivers and electronic speed controllers (ESCs), tether design for power delivery and signal integrity over distance, thermal management, and electromagnetic interference (EMI) mitigation in a confined underwater housing.	Ground control station (GCS) and topside telemetry architecture, closed-loop control for depth, heading, and station-keeping, and onboard computer vision for underwater object detection and navigation.

From Working Prototype to Field-Reliable System

A dedicated segment addresses the practical challenges of deploying embedded systems in real hardware applications — debugging sensor and actuator faults in a sealed, inaccessible enclosure, handling power brownouts and signal noise under real operating loads, managing firmware reliability over a tethered link, and the iterative testing-and-fixing process required to take an embedded system from a working prototype to a field-reliable one under competition conditions.

Each segment is grounded in actual design decisions and trade-offs made on the team's competition ROV, supported by recorded footage from pool testing and world championship runs. The session addresses engineering constraints specific to underwater systems — corrosion resistance, limited visibility, tethered communication, and pressure/buoyancy management — that distinguish ROV design from aerial or ground robotics.

WHO SHOULD ATTEND

Aimed at students, researchers, and young professionals in robotics, mechatronics, embedded systems, or marine/field robotics, the workshop offers a research-informed, competition-validated perspective on a still-nascent field in Bangladesh.

Instructors

Name	Role	Affiliation
Sheikh Tanvir Islam Uday	Chair <i>IEEE RAS MIST SB Chapter</i>	Level 4 EEE MIST
Md. Rokonzaman Shaad	General Secretary <i>IEEE RAS MIST SB Chapter</i>	Level 4 EEE MIST
Areeb Ayman Haq	Webmaster <i>IEEE RAS MIST SB Chapter</i>	Level 4 EEE MIST



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